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Different Approaches for Lift Coefficient Prediction on an Airfoil: Classical CFD and Deep Learning Techniques

Project for the course of Advanced Methods for Scientific Computing and Numerical Analysis for Machine Learning

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Abstract: This work presents an integrated computational pipeline for creating highly accurate, data-driven surrogates for rapid aerodynamic analysis. We first extend the **LifeX** C++ finite element library with a Reynolds-Averaged Navier-Stokes (RANS) solver using the Spalart-Allmaras turbulence model. This solver is deployed on the CINECA Leonardo HPC cluster to generate a reference-quality dataset. This dataset is then used not only to train but also to systematically optimize a Convolutional Neural Network (CNN), implemented from scratch in C++, which predicts lift coefficients from a Signed Distance Function (SDF) representation of the airfoil. Through a comprehensive tuning campaign, we identified the optimal network topology, optimizer, and training protocol. The final, optimized model achieves an exceptional test Mean Squared Error of 0.000487 while delivering predictions with a speedup of over four orders of magnitude compared to direct simulation. This work establishes a validated methodology for creating robust and computationally efficient surrogates for complex aerodynamic design.

1. Introduction

Aerodynamic design balances a trade-off between the demand for physical fidelity and the need for rapid design iteration. High-fidelity Computational Fluid Dynamics (CFD) simulations, such as those based on the Reynolds-Averaged Navier-Stokes (RANS) equations, offer unparalleled accuracy in predicting complex flow phenomena. However, their significant computational cost makes them impractical for design-space exploration or real-time analysis. Conversely, data-driven surrogate models, especially deep learning architectures like Convolutional Neural Networks (CNNs), can provide near-instantaneous predictions but are entirely dependent on the quality and availability of the high-fidelity data used for their training and the robustness of their design.

This project develops a pipeline that integrates both methods. Inspired by the framework of Cuozzo et al. (2026), we first build a high-performance CFD solver to generate reference-quality aerodynamic data. We then leverage this data not only to train a custom CNN surrogate model but also to perform a systematic investigation into its architecture and training protocol. By establishing this workflow for clean symmetric airfoils, we made a base idea for future extensions to more complex iced geometries.

The document is structured to follow this approach, beginning with the development of the high-fidelity simulation framework. Section 2 (**Navier-Stokes Simulation**) details the extension of the **LifeX** C++ finite element library to handle external aerodynamic flows. We describe the integration of the one-equation Spalart-Allmaras

RANS model, the adaptation of the solver to two-dimensional domains, the implementation of specialized boundary conditions and the development of utilities for computing aerodynamic forces. This section also provides a comprehensive overview of the High-Performance Computing (HPC) workflow established on the CINECA Leonardo cluster.

Then, Section 3 (**Convolutional Neural Network**), presents the design, implementation and training of our surrogate model. A key contribution of this work is the development of a CNN entirely from scratch in C++, designed to predict the lift coefficient from an airfoil’s geometric representation along with the angle of attack and Reynolds number. We elaborate on the mathematical foundations of the network architecture, the physics-informed loss function that regularizes training and the implementation of distributed-memory parallelism using MPI, enabling efficient execution on HPC hardware.

Building upon this foundation, Section 4 (**Tuning**) transitions from model implementation to rigorous optimization. This chapter details the comprehensive hyperparameter tuning campaign undertaken to identify the most accurate and stable model configuration. Using a five-fold cross-validation methodology, we systematically investigate the impact of network topology, optimizer selection, regularization techniques (including dropout and the physics-informed loss term), activation functions and learning rate schedules. This empirical analysis provides the evidence-based justification for the final production model’s design.

Finally, Section 5 (**Conclusions**) synthesizes the findings from all stages of the project, evaluates the performance of the final optimized model and discusses the implications and future directions of this integrated framework.

2. Navier-Stokes Simulation

The main goal of this section of the project is to extend the **LifeX** library with new features, making it a more general-purpose and high-performance simulation framework. Specifically, the work focuses on integrating one-equation **RANS turbulence model** (Spalart-Allmaras) into the existing incompressible **Navier-Stokes solver** (NS), enabling the simulation of turbulent flows in external aerodynamic configurations.

2.1. LifeX Implementation

As detailed by Africa et al. (2024), **LifeX** is an open-source C++ library built on top of **deal.II**, originally designed for life-science applications such as cardiac and cardiovascular fluid dynamics. While the library provides a robust and well-structured finite element infrastructure, its original focus on biomedical applications necessitated several non-trivial adaptations to extend it to external aerodynamic flows.

This work addressed four primary challenges:

- adapting the solver and utility functions, originally designed for three-dimensional problems, to two-dimensional domains (Section 2.1.1);
- integrating the one-equation Spalart-Allmaras (SA) RANS model to account for turbulent effects (Section 2.1.2);
- implementing specialized conditions for aerodynamic simulations, including wall functions and far-field boundaries (Section 2.1.3)
- developing a residual-based post-processing utility to compute lift and drag forces (Section 2.1.4).

The following sections detail the implementation of each of these components.

2.1.1. 2D Implementation

While the **LifeX FluidDynamics** solver is templated on the spatial dimension **dim** through the **deal.II** convention, its original test cases and utilities were primarily designed for three-dimensional domains. Extending the framework to 2D involved adapting three key components: the finite element function space, the calculation of vorticity and the computation of the wall distance field.

Function space and triangulation

The SA solver uses Q_1 **finite elements** (FE_Q of degree 1) on the same triangulation as the NS solver:

$$\tilde{v}_h \in V_h = \{ v \in H^1(\Omega) \mid v|_{\partial\Omega_D} = g_D \} \quad (1)$$

On the implementation side, the SA Degree of Freedom (DoF) handler (`dof_handler_sa_`) is independent from the NS one (`dof_handler`), in a way that the two systems have disjoint degrees of freedom without altering the block structure of the Navier-Stokes system. The shared triangulation object is the key architectural choice that allows both DoF handlers to be iterated in lockstep during matrix assembly.

Vorticity in 2D

The modified vorticity $\tilde{S}$ entering the SA production term requires the vorticity magnitude $S = |\boldsymbol{\omega}|$. In three dimensions, this is computed as $S = |\nabla \times \mathbf{u}|$.

In a 2D context, it simplifies to the absolute value of the scalar curl:

$$S = \left| \frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y} \right| \quad (2)$$

which is extracted from the velocity gradient tensor $\nabla \mathbf{u}^{n+1}$ interpolated at quadrature points via `fe_values_ns_.get_function_gradients`.

Wall distance in 2D

The wall distance $d(\mathbf{x})$ is computed once during `setup()` via a parallel algorithm. Each MPI rank collects the barycenters of boundary faces tagged as `prm_wall_tag_` (airfoil surface), serializes them into a flat `double` array of size $N_{\text{face,loc}} \times 2$, and participates in a global `MPI_Allgatherv` to build the complete set $\{\mathbf{c}_w\}_{w=1}^{N_{\text{face,tot}}}$.

The distance at every locally owned SA node i is then:

$$d_i = \max \left(\min_w \|\mathbf{x}_i - \mathbf{c}_w\|_2, 10^{-12} \right) \quad (3)$$

where the lower clamp prevents division by zero for nodes sitting exactly on the wall boundary.

The result is stored in `wall_distance_ghosted_`, a Trilinos ghosted vector indexed on `dof_handler_sa_`, following the standard pattern:

1. write into an owned vector and call `compress(VectorOperation::insert)`.
2. copy owned $\rightarrow$ ghosted to propagate ghost entries across MPI ranks.

The overall complexity per rank is $O(N_{\text{face,tot}} \cdot N_{\text{nodes,loc}})$, which is acceptable for the 2D meshes used in the airfoil example.

2.1.2. Turbulence Model

To enable the simulation of turbulent flows, the framework was extended with the one-equation SA RANS model, introduced by Spalart and Allmaras (1992). This model introduces a transport equation for a scalar variable known as the *modified kinematic viscosity*, $\tilde{\nu}$:

$$\frac{\partial \tilde{\nu}}{\partial t} + \mathbf{u} \cdot \nabla \tilde{\nu} = \underbrace{c_{b1} \tilde{S} \tilde{\nu}}_{\text{production}} - \underbrace{c_{w1} f_w \left(\frac{\tilde{\nu}}{d} \right)^2}_{\text{destruction}} + \underbrace{\frac{1}{\sigma} \left[\nabla \cdot ((\nu + \tilde{\nu}) \nabla \tilde{\nu}) + c_{b2} |\nabla \tilde{\nu}|^2 \right]}_{\text{diffusion + cross-diffusion}} \quad (4)$$

The model constants (Table 1) follow the standard set by **NASA Turbulence Modeling Resource (TMR)** without the f_{t2} trip term.

Table 1: Spalart-Allmaras model constants based on the NASA TMR standard.

Constant	Value	Role
c_{b1}	0.1355	production coefficient
c_{b2}	0.622	cross-diffusion coefficient
σ	2/3	turbulent Prandtl number
κ	0.41	von Kármán constant
c_{v1}	7.1	viscous damping
c_{w1}	≈ 3.2391	destruction (derived from log-law consistency)
c_{w2}	0.3	regularization in f_w
c_{w3}	2.0	regularization in f_w

Closure functions

The model relies on a set of non-linear closure functions to relate $\tilde{\nu}$ to the turbulent eddy viscosity. Defining the viscosity ratio $\chi = \tilde{\nu}/\nu$, these functions are:

$$f_{v1}(\chi) = \frac{\chi^3}{\chi^3 + c_{v1}^3}, \quad f_{v2}(\chi) = 1 - \frac{\chi}{1 + \chi f_{v1}(\chi)},$$

$$\tilde{S} = \max\left(S + \frac{\tilde{\nu} f_{v2}}{\kappa^2 d^2}, 10^{-10}\right),$$

$$r = \min\left(\frac{\tilde{\nu}}{\tilde{S} \kappa^2 d^2}, 10\right), \quad g = r + c_{w2}(r^6 - r), \quad f_w(g) = g \left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6},$$

The final turbulent eddy viscosity, ν_t , which modifies the momentum equations, is then computed as:

$$\nu_t = \tilde{\nu} f_{v1}(\chi).$$

Time discretization and operator splitting.

To solve the coupled Navier-Stokes and Spalart-Allmaras system, this is advanced at each time step using a **first-order Lie splitting** scheme (splitting error $O(\Delta t)$). This approach is consistent with the first-order BDF1 time integration used for both solvers.

Given the state $(\mathbf{u}^n, p^n, \tilde{\nu}^n)$ at time t^n , the solution at t^{n+1} is obtained in two steps:

1. the **NS step** solve the momentum and continuity equations with the with effective viscosity frozen at $\tilde{\nu}^n$:

$$\rho \frac{\mathbf{u}^{n+1} - \mathbf{u}^n}{\Delta t} + \rho(\mathbf{u}^* \cdot \nabla) \mathbf{u}^{n+1} - \nabla \cdot (2\mu_{\text{eff}}^n \nabla^S \mathbf{u}^{n+1}) + \nabla p^{n+1} = \mathbf{0},$$

where $\mu_{\text{eff}}^n = \mu + \nu_t(\tilde{\nu}^n)$ is *explicit* with respect to the SA variable;

2. the **SA step** solve the transport equation using the velocity field $\mathbf{u}^{n+1}$ just computed and a semi-implicit linearization:

$$\frac{\tilde{\nu}^{n+1} - \tilde{\nu}^n}{\Delta t} + \mathbf{u}^{n+1} \cdot \nabla \tilde{\nu}^{n+1} = c_{b1} \tilde{S}^n \tilde{\nu}^{n+1} - c_{w1} f_w^n \frac{\tilde{\nu}^n}{d^2} \tilde{\nu}^{n+1} + \frac{\nu + \tilde{\nu}^n}{\sigma} \Delta \tilde{\nu}^{n+1} + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}^n|^2.$$

The linearization strategy is the following:

- the **production** is treated implicitly using a Picard iteration. $\tilde{S}^n$ is frozen and $\tilde{\nu}^{n+1}$ appears linearly in the matrix (stabilizing for $\tilde{S}^n > 0$);
- the **destruction** is linearized around $\tilde{\nu}^n$. The coefficient $c_{w1} f_w^n \tilde{\nu}^n / d^2$ enters as a positive reaction term, ensuring coercivity;
- about the **Diffusion**, the coefficient $(\nu + \tilde{\nu}^n)/\sigma$ is explicit while $\Delta \tilde{\nu}^{n+1}$ is implicit;
- about the **Cross-diffusion**, $c_{b2}/\sigma |\nabla \tilde{\nu}^n|^2$ is fully explicit and moved to the right-hand side.

Weak formulation and SUPG stabilization

To handle the advection-dominated nature of the SA equation, the weak form is stabilized using the **Streamline-Upwind Petrov-Galerkin** (SUPG) method.

Multiplying by the SUPG test function $\psi_h = v_h + \tau_{\text{SUPG}} \mathbf{u} \cdot \nabla v_h$ and integrating over Ω , the discrete linear system components become:

$$A_{ij}^K = \int_K \left[\psi_i \frac{\phi_j}{\Delta t} + \psi_i (\mathbf{u} \cdot \nabla \phi_j) + D \nabla \phi_i \cdot \nabla \phi_j - \psi_i c_{b1} \tilde{S}^n \phi_j + \psi_i c_{w1} f_w^n \frac{\tilde{\nu}^n}{d^2} \phi_j \right] dK,$$

$$b_i^K = \int_K \psi_i \left[\frac{\tilde{\nu}^n}{\Delta t} + \frac{c_{b2}}{\sigma} |\nabla \tilde{\nu}^n|^2 \right] dK,$$

where $D = (\nu + \tilde{\nu}^n)/\sigma$.

The stabilization parameter τ_{SUPG} is defined locally for each cell K as:

$$\tau_{\text{SUPG}} = \frac{1}{\sqrt{\left(\frac{2}{\Delta t}\right)^2 + \left(\frac{2|\mathbf{u}|}{h_K}\right)^2 + \left(\frac{4D}{h_K^2}\right)^2}},$$

where h_K is the minimum vertex distance of cell K .

Physical bound enforcement.

After each linear solve, a physical bound is enforced on the owned vector before updating the ghosted copy (the ordering `compress(VectorOperation::insert)` $\rightarrow$ ghosted copy is mandatory):

- negative values ($\tilde{\nu} < 0$) are clamped to 0.0, since turbulent viscosity cannot be negative (SA physical bound);
- NaN values ($\tilde{\nu} = \text{NaN}$) are reset to $\tilde{\nu}_\infty$ (freestream value) rather than zero, so that the affected cell remains active and self-corrects over subsequent time steps, avoiding a permanently "dead" region in the domain.

Adaptive time-stepping and bisection.

The bisection mechanism of `FluidDynamics` is fully supported.

`try_timestep_with_bisection` is declared `virtual` in the base class, and `FluidDynamicsSA` overrides it by calling `sa_solver.take_snapshot()` before each attempt and `sa_solver.restore_snapshot()` on failure, ensuring that the SA state $\tilde{\nu}^n$ is rolled back consistently at every bisection level.

2.1.3. Boundary Conditions

The implementation handles three distinct boundary condition types (the airfoil wall, the farfield inlet and the outlet boundaries) imposed on the SA variable each with a different physical motivation.

Wall-resolved (low-Re regime).

A homogeneous Dirichlet condition is applied on the airfoil surface: $\partial\Omega_{\text{wall}}$:

$$\tilde{\nu} = 0 \quad \text{on } \partial\Omega_{\text{wall}}. \quad (5)$$

This is the physically correct condition in the viscous sublayer, as the viscosity ν_t must vanish as the distance to the wall $d \rightarrow 0$.

Wall functions (high-Re regime).

When the first mesh node is at $y^+ > 5$ (unresolved sublayer), a non-homogeneous Dirichlet condition is derived from the **log-law**:

1. the friction velocity is estimated via Schlichting's flat-plate formula:

$$C_f = \frac{0.026}{Re^{0.143}} \quad \text{and} \quad u_\tau = U_\infty \sqrt{C_f/2}$$

2. the wall-normal distance of the first node is converted to wall units:

$$y^+ = \frac{d_{\text{node}} u_\tau}{\nu}.$$

3. a switch criterion at $y_{\text{sw}}^+ = 11.06$ separates the sublayer from the log-layer:

$$\tilde{\nu}_{\text{wall}} = \begin{cases} 0 & \text{if } y^+ < y_{\text{sw}}^+, \\ \text{root of } \nu_t(\tilde{\nu}) = \nu \kappa y^+ & \text{otherwise.} \end{cases}$$

4. the inversion of $\nu_t(\tilde{\nu}) = \tilde{\nu} f_{v1}(\tilde{\nu}/\nu)$ is performed with a **fixed-point iteration**:

$$\tilde{\nu}^{(k+1)} = \frac{\nu \kappa y^+}{f_{v1}(\tilde{\nu}^{(k)}/\nu)}, \quad k = 0, \dots, 4,$$

which converges in 5 iterations due to the monotonicity of f_{v1} .

Inlet and farfield.

A Dirichlet condition with freestream value is imposed:

$$\tilde{\nu} = \tilde{\nu}_\infty \in [3\nu, 5\nu] \quad \text{on } \partial\Omega_{\text{inlet}} \cup \partial\Omega_{\text{farfield}}. \quad (6)$$

Outlet.

No condition is explicitly imposed: the natural Neumann condition $\partial_n \tilde{\nu} = 0$ arises automatically from the boundary term produced by integration by parts of the diffusion operator in the weak form.

2.1.4. Aerodynamic Coefficients (Residual-Based Approach)

Lift and drag coefficients are computed at the end of each time step via a **residual-based boundary force method**, using the `FluidDynamics::compute_force(tag)` method available in **LifeX**, which integrates the full stress tensor over the boundary face with the prescribed tag (airfoil wall):

$$\mathbf{F} = \int_{\partial\Omega_{\text{wall}}} \boldsymbol{\sigma} \cdot \mathbf{n} \, d\Gamma, \quad \boldsymbol{\sigma} = -p \mathbf{I} + 2\mu_{\text{eff}} \nabla^S \mathbf{u}. \quad (7)$$

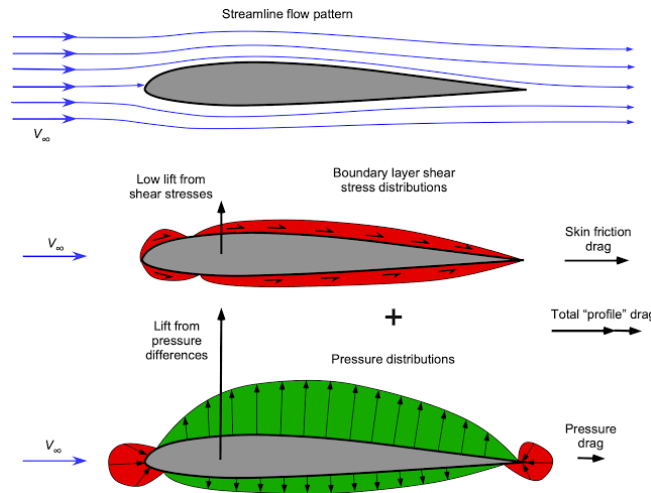


Figure 1: Aerodynamic forces on an airfoil. Source: Eagle Pubs

As illustrated in Figure 1, the Cartesian force components (F_x, F_y) are projected into the aerodynamic reference frame using the angle of attack α :

$$D = F_x \cos \alpha + F_y \sin \alpha, \quad L = -F_x \sin \alpha + F_y \cos \alpha. \quad (8)$$

The forces are normalized by the dynamic pressure q_∞ and the reference surface S_{ref} (which corresponds to the chord length in 2D simulations):

$$q_\infty = \frac{1}{2} \rho U_\infty^2, \quad C_L = \frac{L}{q_\infty S_{\text{ref}}}, \quad C_D = \frac{D}{q_\infty S_{\text{ref}}}. \quad (9)$$

Implementation details

This procedure is encapsulated in a header-only utility class `AerodynamicCoefficientsEvaluator` that exposes a single method:

```
1 compute_coefficients(tag, density, U_inf, S_ref, alpha)
```

Listing 1: Method from `AerodynamicCoefficientsEvaluator`.

It is registered as a callback via `FluidDynamics::set_end_timestep_callback`, so it executes automatically after each successful NS time step. Results are then appended to a CSV file (`force.csv`) with columns `time`, `F_lift`, `F_drag`, `C_L`, `C_D`.

2.1.5. Configuration

The turbulence model is selected via the parameter `"Turbulence model": "Spalart-Allmaras"` in the JSON configuration file.

For backward compatibility, `FluidDynamicsSA` also accepts the legacy value `"None"`, overriding it internally with `Turbulence::SpalartAllmaras` after `parse_parameters`.

The enum class `Turbulence` in `fluid_dynamics.hpp` was extended accordingly:

```
1 enum class Turbulence { None, VMS_LES, SpalartAllmaras };
```

Listing 2: Extension of the class `Turbulence`.

2.2. Computational framework and execution workflow

This section details the computational environment and the end-to-end workflow established for the simulation.

2.2.1. HPC platform and software stack

All simulations were executed on **Leonardo** at CINECA (Bologna), targeting the `dcgp_usr_prod` partition (Intel Sapphire Rapids nodes, 112 cores per node) under the EuroHPC allocation EUHPC_D35_025, with job submission managed by the SLURM scheduler.

The simulation framework is **LifeX** 2.1, introduced in Section 2.1, compiled against `deal.II` 9.7.1 with **Trilinos** as the linear-algebra backend, `MPICH` 4.0 for distributed-memory parallelism, and `HDF5` 1.14.5 for field output in XDMF format. Meshes are generated externally with `Gmsh` and read by **LifeX** in MSH 2.2 format.

The full software stack is delivered to the compute nodes through an Apptainer image (`lifex_env.sif`) maintained by the **LifeX** developers, which removes the need for any host-level `module load` of HPC libraries; the build procedure and the rationale behind this choice are detailed in Section 2.2.2.

The resources allocated for each run of the simulation campaign are summarized in Table 2.

Component	Specification
Cluster	Leonardo @ CINECA (EuroHPC)
Partition	dcgp_usr_prod
CPU	Intel Sapphire Rapids
Cores per node	112
MPI ranks per simulation	8
Wall-time per run	up to 24 h
Software stack	LifeX 2.1 / deal.II 9.7.1 / Trilinos / MPICH 4.0 / HDF5 1.14.5
Container runtime	Apptainer 1.4.5 (image lifex_env.sif)

Table 2: Computational resources for the simulation campaign on Leonardo.

2.2.2. Containerization and build process

The compute nodes of Leonardo do not expose a pre-configured HPC software stack through the `module` command, which only loads base system libraries. A native build of **LifeX** on the cluster would therefore require manually compiling `deal.II 9.7` together with all its required backends¹, an operation that is both fragile and not portable across nodes. The entire toolchain was therefore packaged into an Apptainer image and executed through the container runtime available on the compute nodes (Apptainer 1.4.5), so that the build environment is identical to the execution environment.

Identifying an image that simultaneously satisfies all **LifeX** dependencies required iterating over several candidates, summarized in Table 3, encountering a series of incompatibilities:

- the Spack stack pre-installed on Leonardo provides `dealii@9.6.2`, but only with the `~trilinos` variant, which is incompatible with **LifeX 2.1**;
- the official `dealii/dealii` Docker images expose `deal.II 9.7` with all required backends, but lack the `boost_filesystemConfig.cmake` component required by the **LifeX** CMake configuration;
- an earlier MOX `mk` container shipping `deal.II 9.5.1` fails to compile **LifeX 2.1** due to API changes in `get_mpi_communicator`, `CellStatus`, and `AffineConstraints` that occurred upstream between 9.5 and 9.7

The image finally adopted, `mk-dealiiimaster_latest.sif`, is the MOX MK container with the `deal.II` module tracking the upstream master branch (`deal.II 9.7`), together with matching `vtk`, `boost`, `trilinos`, `petsc`, `p4est`, and `hdf5` modules, activated inside the container by sourcing `/u/sw/etc/profile` and loading the relevant modules.

Stack	Components	Outcome
Spack <code>dealii@9.6.2</code> (Leonardo)	<code>deal.II ~trilinos</code> , PETSc, MPICH	Missing Trilinos/Sacado required by LifeX 2.1
<code>dealii/dealii</code> Docker images (<code>v9.7.1-jammy</code> , <code>master-jammy</code> , <code>v9.7.1-noble</code>)	<code>deal.II 9.7.1</code> with Trilinos, PETSc, <code>p4est</code> , VTK, HDF5	<code>boost_filesystemConfig.cmake</code> missing in the image
MOX <code>mk</code> container (<code>deal.II 9.5.1</code>)	<code>deal.II 9.5.1</code> , Boost 1.76, VTK, HDF5	API mismatch: LifeX 2.1 requires <code>deal.II ≥ 9.7</code>
<code>mk-dealiiimaster_latest.sif</code>	MK toolchain with <code>deal.II master (≥ 9.7)</code> , Boost, VTK, Trilinos, PETSc, <code>p4est</code> , HDF5	Adopted for the simulation campaign

Table 3: Container strategies evaluated for building LifeX on Leonardo.

The CMake configuration is invoked through `apptainer exec -cleanenv -bind`, which strips host environment contamination and mounts the workspace inside the container. Because the simulation campaign is purely two-dimensional, the build is configured with `-DLIFEX_DIM=2` and restricted to the two example targets actually used, `example_fluid_dynamics_airfoil` and `example_fluid_dynamics_external_flow`. A global `make all` cannot be used in 2D because some files in the test suite (e.g. `tests/multidomain_laplace/laplace.cpp`)

¹Trilinos (used by `LinearAlgebraTrilinos` and the `sacado_dfad` AD operators), PETSc, `p4est`, VTK, Boost, and HDF5

rely on overloads of `GridTools::rotate` that exist only for three-dimensional triangulations, and the resulting compilation error stops the whole build. The complete build script is reported in Listing 3.

```

1 #!/bin/bash
2 #SBATCH -A EUHPC_D35_025
3 #SBATCH -p dcgp_usr_prod
4 #SBATCH --time=00:24:00
5 #SBATCH -N 1
6 #SBATCH --ntasks-per-node=112
7 #SBATCH --job-name=lifex_build
8
9 set -euo pipefail
10
11 WORK_DIR="$HOME/Ice_acceleration_model_on_airplane_wing"
12 SIF="${WORK_DIR}/containers/mk-dealii-master_latest.sif"
13 SRC_SUBDIR="external/lifex-public"
14 BUILD_SUBDIR="external/lifex-public/build_2d"
15
16 rm -rf "${WORK_DIR}/${BUILD_SUBDIR}"
17
18 singularity exec --cleanenv \
19   --bind "${WORK_DIR}:/work" \
20   --bind /dev/shm:/dev/shm \
21   "${SIF}" bash -c "
22     source /u/sw/etc/profile
23     module load gcc-glibc dealii vtk
24
25     cmake -B /work/${BUILD_SUBDIR} -S /work/${SRC_SUBDIR} \
26       -DCMAKE_BUILD_TYPE=Release \
27       -DLIFEX_DIM=2 \
28       -DDEAL_II_DIR=\${mkDealiiPrefix} \
29       -Dlin_ALG=Trilinos
30
31     cmake --build /work/${BUILD_SUBDIR} \
32       --target example_fluid_dynamics_airfoil \
33         example_fluid_dynamics_external_flow \
34       -j 112
35   "
```

Listing 3: SLURM build script for LifeX on Leonardo (`job_build_leonardo.sh`). The Apptainer image is invoked through the legacy `singularity` command alias.

2.2.3. Mesh generation and pre-processing pipeline

The computational meshes are generated with Gmsh from parametric `.geo` scripts, driven by a set of Python generators (`generate_geo*.py`, `naca_generator_v*.py`) that expose the airfoil profile (NACA 0006 through NACA 0018), the domain extent, and the boundary-layer structure as input parameters.

The resulting mesh is hybrid: a structured layer of quadrilaterals wraps the airfoil surface to resolve the boundary layer, while the outer region is filled with unstructured triangles. Boundary entities follow a fixed tagging convention (1 = inlet, 2 = outlet, 3 = airfoil wall) matching the tags expected by the **LifeX** configuration (Section 2.1), and the mesh is exported in the MSH 2.2 format read natively by `deal.II`.

In accordance with the wall-resolved turbulence modeling approach described in Section 2.1.3, the near-wall region is carefully refined. The height of the first cell layer is adjusted until the estimated dimensionless wall distance satisfies the criterion $y^+ \lesssim 1$ at a Reynolds number of $Re = 10^6$. This ensures that the use of a homogeneous Dirichlet condition for $\tilde{\nu}$ is physically valid.

Because a single malformed mesh wastes hours of allocated CPU time, every mesh is screened by a pre-processing pipeline before submission. Four independent Python checks are run in sequence:

- `mesh_info.py` reports the node and element counts grouped by element type and physical tag and verifies that the inlet, outlet, and airfoil tags are all present;
- `check_extents.py` computes the bounding box of each tagged boundary, confirming that the inlet, outlet and airfoil entities are positioned where the geometry expects them;
- `check_jacobians.py` reads the mesh through `meshio` and computes the signed area of every quadrilateral via the cross product of its diagonals, flagging inverted or non-convex cells that would produce a negative Jacobian;
- `check_yplus.py` estimates the wall-normal y^+ at the first node by measuring the height of the cells adjacent to the airfoil edges and applying a flat-plate skin-friction correlation ($u_\tau/U_\infty \approx 0.036$), consistent with the estimate in Section 2.1.3.

Any mesh that fails one or more of these checks is automatically discarded and regenerated. An example of a mesh that has passed this quality control pipeline, used for the NACA 0012 simulations, is shown in Figure 2.

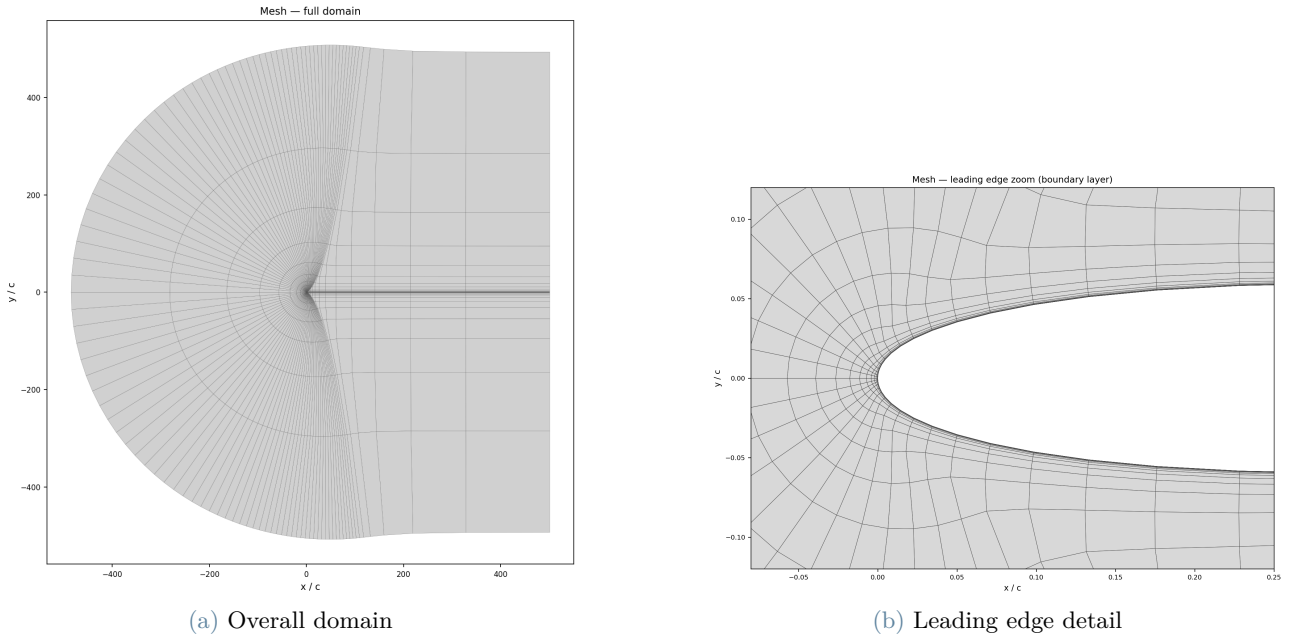


Figure 2: Computational mesh for the NACA 0012 airfoil. (a) Overall domain, with a structured quadrilateral boundary layer around the profile and an unstructured triangular outer region. (b) Detail of the wall-resolved boundary layer at the leading edge, refined to give $y^+ \lesssim 1$ at $\text{Re} = 10^6$.

2.2.4. Simulation execution and post-processing

Each simulation is launched as a single-node SLURM job that runs the **LifeX** executable inside the Apptainer container with `mpirun -n 8`, distributing the mesh across eight MPI ranks on the `dcgp_usr_prod` partition.

The workspace is bind-mounted into the container and the run-time configuration is supplied as a JSON file, following the parameter structure described in Section 2.1: the turbulence model is activated through `"Turbulence model": "Spalart-Allmaras"`, and the Reynolds number is set implicitly through the kinematic viscosity ($\nu = 10^{-6}$ with $U_\infty = 1$ and chord $c = 1$, giving $\text{Re} = 10^6$). Before execution, the configuration file is copied into the run directory `output/run_${SLURM_JOB_ID}/` as `config_used.json`, so that every result remains paired with the exact parameters that produced it.

During the run, the solver generates two primary forms of output: a time history of the aerodynamic coefficients, which is appended to `force.csv`, and the full velocity, pressure, and turbulent-viscosity fields, saved as HDF5 files with corresponding XDMF descriptors for visualization in ParaView. The complete SLURM submission script orchestrating this process is reported in Listing 5.

```
1 #!/bin/bash
```

```

2 #SBATCH -A EUHPC_D35_025
3 #SBATCH -p dcgp_usr_prod
4 #SBATCH --time=24:00:00
5 #SBATCH -N 1
6 #SBATCH --ntasks-per-node=8
7 #SBATCH --job-name=lifex_run
8 #SBATCH -o output/run_%j/solver.log
9
10 set -euo pipefail
11
12 WORK_DIR="$HOME/Ice_acceleration_model_on_airplane_wing"
13 SIF="${WORK_DIR}/containers/mk-dealiiimaster_latest.sif"
14 EXEC="/work/external/lifex-public/build_2d/examples/fluid_dynamics_airfoil
15     /lifex_example_fluid_dynamics_airfoil"
16 CONFIG="/work/cases/naca0012/config_new_method.json"
17 RUN_DIR="${WORK_DIR}/output/run_${SLURM_JOB_ID}"
18
19 mkdir -p "${RUN_DIR}"
20 cp "${WORK_DIR}/cases/naca0012/config_new_method.json" "${RUN_DIR}/
21     config_used.json"
22
23 singularity exec --cleanenv \
24     --bind "${WORK_DIR}:/work" \
25     --bind /dev/shm:/dev/shm \
26     "${SIF}" bash -c "
27     source /u/sw/etc/profile
28     module load gcc-glibc dealii vtk
29     export LD_LIBRARY_PATH=/work/external/lifex-public/build_2d/lib:\
30         $LD_LIBRARY_PATH
31
32     cd /work/output/run_${SLURM_JOB_ID}
33     mpirun -n 8 ${EXEC} ${CONFIG}
34 "

```

Listing 4: SLURM run script for a single simulation (job_run_leonardo.sh).

The field output is inspected in ParaView, while the coefficient time histories are plotted with the `plot_forces.py` script (matplotlib with the Agg backend, so that figures are rendered without a display on the compute nodes).

Figures 3 and 4b show representative output for the NACA 0012 at $\alpha = 4^\circ$ and $Re = 10^6$. The pressure coefficient field (Figure 3) exhibits the suction peak on the upper surface near the leading edge that generates the net lift; the velocity field (Figure 4a) shows the flow accelerating over the suction side and the thin viscous wake downstream of the trailing edge; the spanwise vorticity (Figure 4b) isolates the shed boundary layers in the wake.

2.2.5. Dataset generation for the surrogate model

The simulation is not an end in itself: its role is to build the training set for the convolutional surrogate model presented in Section 3.

To build a comprehensive dataset, the parameter space was systematically explored. For each airfoil profile in a family of symmetric NACA sections (NACA 0006 to NACA 0018), the angle of attack was swept over the range $\alpha \in \{-14^\circ, -12^\circ, \dots, +14^\circ\}$ using templated configuration files (`config_isa_aoa_*.json`).

For each configuration, the pipeline produces a single label: the converged lift coefficient, C_L . These labels populate the `Y_c1` array of the final dataset, which is stored in the compressed NumPy format (`.npz`). The corresponding inputs are the Signed Distance Function representation of each geometry and the scalar pair of physical parameters (Re, α).

Computing the lift labels with the turbulence-resolved solver, rather than with a low-fidelity panel method, is what allows the surrogate to be trained against high-fidelity data in the high-incidence regime where thin-airfoil

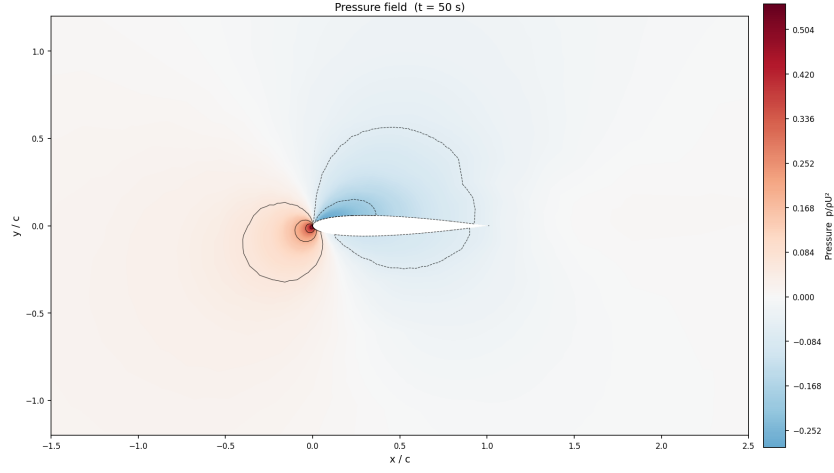
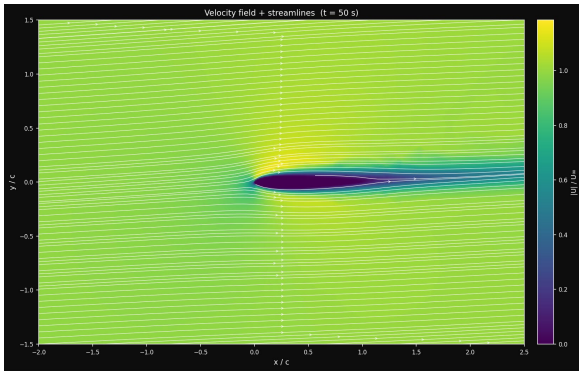
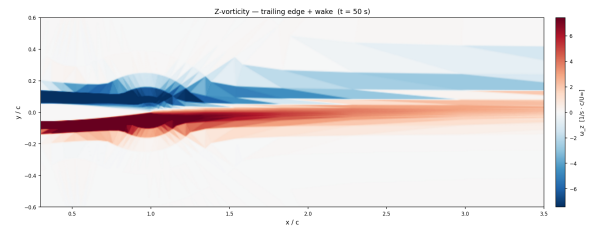


Figure 3: Pressure coefficient field around the NACA 0012 at $\alpha = 4^\circ$, $\text{Re} = 10^6$. The suction peak on the upper surface near the leading edge and the stagnation region on the lower side generate the net lift.



(a) Velocity magnitude with superimposed streamlines.



(b) Spanwise vorticity in the wake.

Figure 4: Flow field around the NACA 0012 at $\alpha = 4^\circ$, $\text{Re} = 10^6$. (a) The flow accelerates over the suction side and produces a thin viscous wake downstream of the trailing edge. (b) The spanwise vorticity isolates the shed boundary layers.

theory no longer holds.

The same workflow extends to the iced-airfoil geometries. A modified leading edge produced by an icing model could be created in future workflows and processed through the identical execution and post-processing chain, so that the dataset can later be augmented with iced profiles without any change to the simulation pipeline. The complete workflow is summarized in Figure 5.

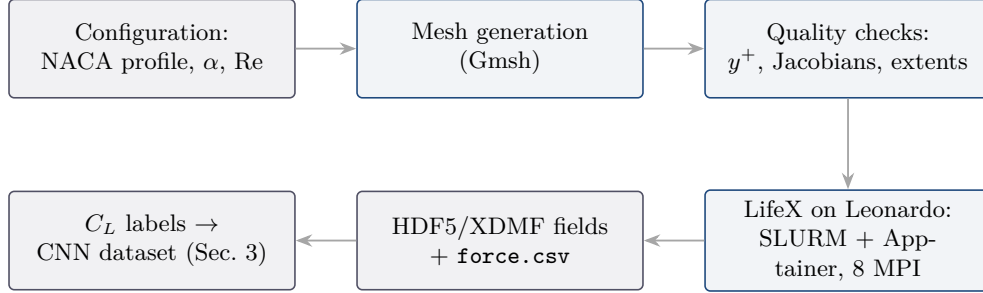


Figure 5: End-to-end simulation pipeline (top row left to right, then bottom row right to left). Mesh generation and quality control run interactively on the login node; the solver runs as a SLURM batch job inside the Apptainer container on the compute nodes; the resulting lift coefficients feed the convolutional surrogate model of Section 3.

3. Convolutional Neural Network

Building upon the dataset established in the preceding Section 2, this Section introduces a surrogate model designed to approximate the results of CFD simulations with a significant reduction in computational time.

Our approach is centered on a **Convolutional Neural Network (CNN)**, a deep learning architecture adept at learning the complex physical relationships that govern aerodynamic performance directly from simulation data.

Specifically, the network is designed to predict the lift coefficient (C_L) as a function of three key inputs:

1. the airfoil shape, represented as a **Signed Distance Function**;
2. the angle of attack (α);
3. the Reynolds number (Re).

By training on the generated simulation results, the CNN effectively learns to emulate the fluid dynamics solver, enabling near-instantaneous predictions.

The following sections will provide a comprehensive overview of this CNN model, from its underlying mathematical principles to its custom implementation designed for high-performance computing.

3.1. Mathematical Background

This section details the mathematical foundations of our surrogate modeling approach.

We first introduce the **Signed Distance Function (SDF)**, which transforms the airfoil’s complex geometry into a structured grid suitable for a neural network. We then provide a comprehensive breakdown of the **Convolutional Neural Network architecture**, explaining the mathematical operations within each layer and the physics-informed training algorithm used to learn the mapping from geometry and flow conditions to the aerodynamic lift coefficient.

3.1.1. Geometric Representation with Signed Distance Function

To construct a deep learning model capable of predicting aerodynamic coefficients, we need a robust method for representing the geometry of the airfoil $\Omega \subset \mathbb{R}^2$ and its boundary $\partial\Omega$ in a format suitable for Convolutional Neural Networks. Traditional boundary representations, such as coordinates of boundary points, are challenging to process with standard convolutional operators because they lack fixed spatial grids. To address this limitation, we represent the airfoil geometry using a **Signed Distance Function**, obtaining for each airfoil the result shown in Figure 6.

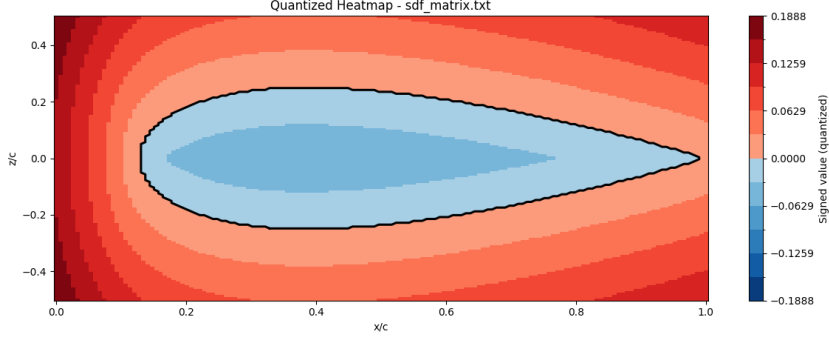


Figure 6: The Signed Distance Field representation of a NACA 0012 airfoil, discretized on a uniform grid. Negative values (blue) correspond to the airfoil interior, positive values (red) to the exterior while the boundary is represented by the zero-level contour (black line).

The SDF $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ maps any point $P = (x, z)$ in the spatial domain to its signed distance from the boundary $\partial\Omega$:

$$\Phi(P) = \begin{cases} -d(P, \partial\Omega) & \text{if } P \in \Omega \\ d(P, \partial\Omega) & \text{if } P \notin \Omega \end{cases} \quad (10)$$

where $d(P, \partial\Omega)$ represents the Euclidean distance from the point P to the boundary $\partial\Omega$:

$$d(P, \partial\Omega) = \inf_{Q \in \partial\Omega} \|P - Q\|_2 \quad (11)$$

Using this definition, the boundary of the airfoil is implicitly defined as the zero-level set of the function:

$$\partial\Omega = \{P \in \mathbb{R}^2 \mid \Phi(P) = 0\} \quad (12)$$

while the interior corresponds to negative values ($\Phi(P) < 0$) and the exterior to positive values ($\Phi(P) > 0$).

In our numerical implementation, the boundary $\partial\Omega$ is defined as a closed polygon composed of an ordered set of n vertices $\{V_0, V_1, \dots, V_{n-1}\}$.

To guarantee scale invariance, the coordinates of the airfoil profile are first normalized by the chord length c , defined as the Euclidean distance between the leading edge point $(x_{\min}, z|_{x_{\min}})$ and the trailing edge point $(x_{\max}, z|_{x_{\max}})$:

$$c = \sqrt{(x_{\max} - x_{\min})^2 + (z_{\text{at } x_{\max}} - z_{\text{at } x_{\min}})^2} \quad (13)$$

where $x_{\max}$, $x_{\min}$, $z_{\max}$, and $z_{\min}$ are the extreme coordinates computed over the boundary vertices. Each vertex is then scaled by dividing its coordinates by c :

$$V_i = \left(\frac{x_{V_i}}{c}, \frac{z_{V_i}}{c} \right) \quad (14)$$

The boundary $\partial\Omega$ is then represented as the union of n linear segments $S_i = [V_i, V_{i+1}]$ (with $V_n = V_0$). The minimum distance from a grid point P to the boundary $\partial\Omega$ is computed by finding the minimum distance to each individual segment S_i :

$$d(P, \partial\Omega) = \min_{i=0, \dots, n-1} d(P, S_i) \quad (15)$$

For a given segment S_i defined by the endpoints $A = V_i$ and $B = V_{i+1}$, any point on the infinite line passing through A and B can be parameterized as $P(t) = A + t(B - A)$ for $t \in \mathbb{R}$. The orthogonal projection of the point P onto this line is obtained by minimizing the distance $\|P - P(t)\|^2$, which yields the projection parameter:

$$t^* = \frac{(P - A) \cdot (B - A)}{\|B - A\|_2^2} \quad (16)$$

To restrict the projection to the finite line segment S_i , t^* is clamped to the interval $[0, 1]$:

$$t = \max(0, \min(1, t^*)) \quad (17)$$

The projection point on the segment is then $P_{\text{proj}} = A + t(B - A)$, and the distance to the segment is:

$$d(P, S_i) = \|P - P_{\text{proj}}\|_2 \quad (18)$$

To determine the sign of the distance field, we implement the **Winding Number Algorithm** to check whether a grid point P lies inside or outside the closed boundary. The winding number $wn(P)$ computes the number of times the boundary curve wraps around the point P . For a closed polygon, it is computed by casting a horizontal ray starting from P and pointing in the positive x -direction, and counting the signed crossings of this ray with the polygon edges:

- an **upward crossing** occurs if an edge $S_i = [V_i, V_{i+1}]$ crosses the horizontal ray from below, i.e., $V_{i,z} \leq z_P < V_{i+1,z}$ and P lies to the left of the directed line segment;
- a **downward crossing** occurs if an edge S_i crosses the horizontal ray from above, i.e., $V_{i+1,z} \leq z_P < V_{i,z}$ and P lies to the right of the directed line segment.

The relative position of the point P with respect to the directed segment $[A, B]$ is determined mathematically using the 2D cross product:

$$\text{cross}(P, A, B) = (A_x - x_P)(B_z - z_P) - (B_x - x_P)(A_z - z_P) \quad (19)$$

If $\text{cross}(P, A, B) > 0$, the point P lies to the left of the directed segment, whereas if $\text{cross}(P, A, B) < 0$, it lies to the right. The winding number is incremented by 1 for every upward crossing and decremented by 1 for every downward crossing. A point P is strictly inside the domain Ω if and only if the total winding number is non-zero:

$$P \in \Omega \iff wn(P) \neq 0 \quad (20)$$

Once the signed distance is computed for any point, we project it onto a discrete uniform grid. The spatial domain is discretized into a Cartesian grid of size $N \times N = 150 \times 150$ over the bounding box $[X_{\min}, X_{\max}] \times [Z_{\min}, Z_{\max}]$, with:

$$X_{\min} = -0.15, \quad X_{\max} = 1.01, \quad Z_{\min} = -0.12, \quad Z_{\max} = 0.12 \quad (21)$$

The grid spacing is defined by:

$$\Delta x = \frac{X_{\max} - X_{\min}}{N - 1}, \quad \Delta z = \frac{Z_{\max} - Z_{\min}}{N - 1} \quad (22)$$

For each grid node indexed by (row, col) for $row, col = 0, \dots, N - 1$, the corresponding spatial coordinates are given by:

$$x_{col} = X_{\min} + col \cdot \Delta x, \quad z_{row} = Z_{\max} - row \cdot \Delta z \quad (23)$$

This discretized representation yields a 2D matrix of shape 150×150 , where each entry contains the signed distance value. This matrix provides a spatially structured representation of the geometry that preserves boundary details, making it perfectly suited as an input to the convolutional branch of the neural network.

3.1.2. Convolutional Neural Networks for Aerodynamic Prediction

The architecture of our **Convolutional Neural Network** is designed as a multi-input model to effectively process both the spatial geometric data and the scalar physical parameters. A high-level schematic of this design, inspired by the model presented by Cuozzo et al. (2026), is depicted in Figure 7.

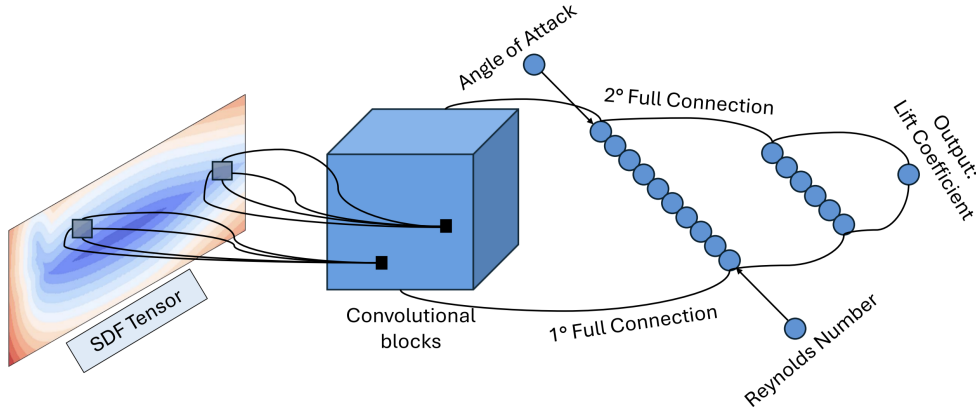


Figure 7: Architecture of the Convolutional Neural Network used for aerodynamic prediction.

The network is logically divided into three main components:

1. a convolutional feature extractor, responsible for interpreting the airfoil geometry. It accepts the 2D SDF matrix as input and processes it through a series of **Conv2DLayers**, each followed by a **LeakyReLU** activation. This stack of layers hierarchically extracts relevant spatial features, such as leading-edge curvature and surface details. The resulting multi-channel feature maps are then transformed into a one-dimensional vector by a **FlattenLayer**, producing a learned representation of the airfoil’s shape;
2. a feature fusion stage, in which, the flattened geometric feature vector from the convolutional branch is concatenated with the scalar physical parameters (the angle of attack and the Reynolds number) tensor using a **ConcatenateLayer**. This creates a single feature vector that contains comprehensive information about both the airfoil’s shape and the current flow conditions;
3. finally the unified vector is passed to the regression head, which consists of a sequence of **DenseLayers**. This fully-connected network maps the combined high-level features to a single scalar output, which represents the predicted lift coefficient, C_L .

Convolutional Layer

The **Convolutional Layer** (**Conv2DLayer**) is an operator specifically designed to process grid-like data such as the 2D SDF. Its primary function is to detect local spatial patterns within the input by applying a set of learnable filters. The difference with the dense layer, which treats all input pixels independently, is that a convolutional layer preserves the spatial relationship between pixels by operating on localized neighborhoods.

At its core, the operation is a discrete 2D convolution, which can be understood as a *sliding window dot product*. A small matrix of learnable weights, known as the kernel (K), slides over the input feature map (I). At each position, an element-wise multiplication between the kernel and the overlapping region of the input is performed, and the results are summed up to produce a single pixel in the output feature map (O).

For a single-channel input I of size $H \times W$ and a kernel of size $F \times F$, the value of the output O at position (i, j) is given by:

$$O(i, j) = \sum_{m=0}^{F-1} \sum_{n=0}^{F-1} I(i-m, j-n) \cdot K(m, n) + b \quad (24)$$

where b is a learnable scalar bias term applied to the entire output feature map.

The key principle here is parameter sharing: the *same* kernel K and bias b are used across all positions of the input, allowing the network to detect the same feature regardless of where it appears in the SDF².

In deep networks, layers typically process multi-channel inputs. If the input feature map has C_{in} channels³, the kernel must be three-dimensional, with a size of $F \times F \times C_{in}$. The convolution is then performed across all channels simultaneously. The results from each channel are then summed together to produce a single value for the 2D output map.

To generate an output with C_{out} feature maps, we simply apply C_{out} independent kernels. Each kernel is responsible for learning a different spatial feature, allowing the layer to build a complete representation of the input geometry.

The behavior of the convolution is controlled by two hyperparameters:

- the stride (S) defines the step size the kernel matrix as it traverses the input grid. While a unit stride ($S = 1$) evaluates adjacent localized neighborhoods sequentially, a stride of $S > 1$ skips spatial coordinates, downsampling the spatial resolution of the resulting feature map;
- padding (P)⁴ consists in appending a perimeter of zero-valued elements around the input grid boundaries. This can be used to control the output dimensions and mitigate the loss of information at the edges.

²In more details, if the kernel learns to recognize a leading edge curve of a profile, it will successfully detect the curve whether it appears at the top of the grid or in a bottom corner.

³This is not our case, since in the SDF we have only one input channel ($C_{in} = 1$).

⁴In our implementation, padding is set to 0.

The dimensions of the output feature map ($H_{out} \times W_{out}$) are determined by the input dimensions ($H_{in} \times W_{in}$), the kernel size (F), the stride (S) and the padding (P) using the following formula:

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1 \quad \text{and} \quad H_{out} = \frac{H_{in} - F + 2P}{S} + 1 \quad (25)$$

From an implementation perspective, this operation is typically realized as a set of nested loops that iterate over each position in the output map, each element of the kernel, and each input channel, performing the required multiply-accumulate operations⁵.

Non-Linear Activation (Leaky ReLU)

After each `Conv2DLayer` or `DenseLayer` performs its linear operation, an activation function applies a non-linear transformation element-wise to its output. Without this, a deep neural network, regardless of its number of layers, would behave as a single, large linear model. The introduction of non-linearity is what gives the network its power to approximate arbitrarily complex, non-linear relationships, such as the one between airfoil geometry and lift coefficient.

In this project, a **Leaky Rectified Linear Unit** (Leaky ReLU) is implemented as the `LeakyReLULayer`. This function, shown in Figure 8, is a popular and effective choice in modern deep learning due to its computational efficiency and its ability to mitigate some of the common problems encountered during training.

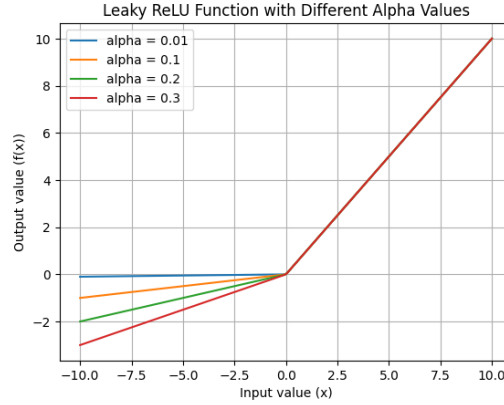


Figure 8: Example of Leaky ReLU function with different α .

The Leaky ReLU function is defined mathematically as:

$$\text{LeakyReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{if } x \leq 0 \end{cases} \quad (26)$$

where α is a small, positive constant (a hyperparameter, typically set to a value like 0.01).

The function behaves as a simple identity for all positive input values. For negative input values, instead of mapping them to zero like the standard ReLU function, it allows a small, non-zero gradient to pass through.

The primary advantage of the Leaky ReLU over the standard ReLU is its handling of negative inputs. In a standard ReLU, if a neuron's input is consistently negative, its gradient will always be zero. This means the neuron's weights will never be updated, and it can effectively "die", ceasing to contribute to the network's learning process. The Leaky ReLU prevents this "dying ReLU" problem by ensuring that a small gradient always exists, even for negative inputs.

From the backpropagation algorithm, the gradient of the Leaky ReLU is simple and computationally inexpensive to calculate:

$$\frac{d}{dx} \text{LeakyReLU}(x) = \begin{cases} 1 & \text{if } x > 0 \\ \alpha & \text{if } x \leq 0 \end{cases} \quad (27)$$

⁵We will explain in a better way highlighting our C++ implementation in Section 3.2.3 regarding our implementation in C++

During the backward pass of training, this straightforward derivative is used to scale the incoming gradient from the subsequent layer:

- if the input to the activation function was positive during the forward pass, the gradient passes through unchanged;
- if it was negative, the gradient is scaled down by the factor α .

This simple conditional check makes the backward pass for the **LeakyReLU** layer extremely fast, contributing to the overall efficiency of the training process.

Structural Layers

After the convolutional network has extracted a hierarchy of spatial features, the resulting multi-channel feature maps must be prepared for integration with the scalar parameters and for final processing by the regression head. This preparation is handled by two complementary **structural layers**, the **FlattenLayer** and the **ConcatenateLayer**, that are parameter-free: they do not contain any learnable weights or biases since their sole purpose is to rearrange data tensors.

Flatten Layer

The **FlattenLayer** links the convolutional layers and the fully-connected layers. Convolutional layers output a 4D tensor with a shape of [Batch Size, Channels, Height, Width], preserving spatial information. However, dense layers require their input to be a 2D tensor of shape [Batch Size, Features], where each row is a one-dimensional vector representing a single sample.

The flatten operation reshapes an input tensor T_{in} with dimensions $[B \times C_{out} \times H_{out} \times W_{out}]$ into an output tensor T_{out} with dimensions $[B \times D]$, where the new feature dimension D is the product of the input's channel, height and width dimensions:

$$D = C_{out} \times H_{out} \times W_{out} \quad (28)$$

During the forward pass, this is a straightforward memory rearrangement.

During the backward pass for backpropagation, the **FlattenLayer** performs the inverse operation: it receives an incoming gradient tensor of shape $[B \times D]$ and reshapes it back to the original 4D shape $[B \times C_{out} \times H_{out} \times W_{out}]$. This ensures that the gradients are correctly propagated back to the preceding convolutional layer in the required spatial format.

Concatenate Layer

Once the geometric features have been flattened into a vector, they must be combined with the scalar physical parameters by the **ConcatenateLayer**, that takes multiple input tensors and joins them together along a specified axis⁶.

Mathematically, given the flattened geometric feature tensor T_{flat} of shape $[B \times D_1]$ and the scalar input tensor T_{scalar} (containing angle of attack and Reynolds number) of shape $[B \times D_2]$, the Concatenate Layer produces an output tensor T_{concat} of shape $[B \times (D_1 + D_2)]$. This operation creates a unified feature vector that encodes both geometric and physical information for the network to make a prediction.

During backpropagation, the process is reversed. The incoming gradient tensor is split according to the original input sizes:

- the first D_1 elements of the gradient vector are passed back to the **FlattenLayer**'s branch;
- the D_2 elements corresponding to the scalar inputs are also computed, although they terminate at the input layer.

After this fusion step, the resulting unified tensor is ready to be processed by the final regression head of the network.

Dense Layer and final regression

The final computational stage of the network is handled by the **Dense Layer** (**DenseLayer**), known as a fully-connected layer. Its role is to learn the final mapping from the feature space to the target scalar value (the lift

⁶In our case, this is the feature axis (axis 1)

coefficient C_L).

A dense layer applies a linear transformation to its input vector. For an input vector $\mathbf{x} \in \mathbb{R}^{D_{in}}$, the layer computes an output vector $\mathbf{y} \in \mathbb{R}^{D_{out}}$ using a learnable weight matrix $W \in \mathbb{R}^{D_{out} \times D_{in}}$ and a learnable bias vector $\mathbf{b} \in \mathbb{R}^{D_{out}}$. The operation is defined as:

$$\mathbf{y} = W\mathbf{x} + \mathbf{b} \quad (29)$$

This operation performs a weighted sum of all input features to compute each output feature. Every neuron in the input layer is connected to every neuron in the output layer, allowing the network to model complex interactions between all elements of the input feature vector.

This layer is used in the regression head, which consists of a sequence of **DenseLayers**, each followed by a **LeakyReLU** activation:

- the initial dense layers take concatenated feature vector as input. Their purpose is to find and amplify the non-linear correlations between the learned geometric features from the SDF and the raw physical parameters (α and Re);
- the last layer is configured to have a single output neuron ($D_{out} = 1$) and does not have a subsequent activation function. This single neuron's output directly corresponds to the network's prediction for the lift coefficient (C_L).

During the backward pass, the dense layer must compute three distinct gradients based on the incoming gradient from the subsequent layer $\frac{\partial L}{\partial \mathbf{y}}$, where L is the loss:

- the gradient with respect to the weights (W) is calculated as the outer product of the incoming gradient and the layer's input vector from the forward pass:

$$\frac{\partial L}{\partial W} = \frac{\partial L}{\partial \mathbf{y}} \cdot \underbrace{\mathbf{x}^T}_{\frac{\partial \mathbf{y}}{\partial W}} \quad (30)$$

- the gradient with respect to the biases ($\mathbf{b}$) is equal to the incoming gradient:

$$\frac{\partial L}{\partial \mathbf{b}} = \frac{\partial L}{\partial \mathbf{y}} \quad (31)$$

- the gradient with respect to the input ($\mathbf{x}$) must be passed back to the preceding layer. It is computed by multiplying the transposed weight matrix by the incoming gradient:

$$\frac{\partial L}{\partial \mathbf{x}} = W^T \cdot \frac{\partial L}{\partial \mathbf{y}} \quad (32)$$

These calculations are the fundamental operations that enable the network's regression head to learn from the training data.

Network Training

The process of training the neural network involves iteratively adjusting its parameters to minimize a loss function, which quantifies the discrepancy between the model's predictions and the desired outcomes. Our approach embeds physical knowledge directly into the training process, making our model a type of **Physics-Informed Neural Network (PINN)**. This is achieved through a composite loss function that balances empirical data accuracy with a theoretical physical constraint.

The total loss, L , is a weighted sum of data-driven term and a physics-informed term:

$$L = L_{\text{data}} + \lambda L_{\text{physics}} \quad (33)$$

where λ is a hyperparameter that controls the strength of the physical regularization.

The first component is the standard **Mean Squared Error (MSE)**, which measures the difference between the network's predicted lift coefficients (P_i) and the ground-truth values from the CFD simulations (T_i):

$$L_{\text{data}} = \frac{1}{N} \sum_{i=1}^N (P_i - T_i)^2 \quad (34)$$

This term ensures the network learns to faithfully replicate the high-fidelity simulation data generated for this work.

The second component acts as a physical constraint, guiding the network’s predictions to be consistent with established aerodynamic theory. This term is based on **Thin-Airfoil Theory**, which provides an analytical approximation for the lift coefficient at small angles of attack. The general form of this relationship is $C_L = 2\pi(\alpha - \alpha_{L=0})$, where $\alpha_{L=0}$ is the zero-lift angle of attack.

For symmetric airfoils, such as those in our dataset, the zero-lift angle of attack is, by definition, $\alpha_{L=0} = 0$. This simplifies the theoretical relationship to the linear form:

$$C_L \approx 2\pi\alpha \quad (35)$$

The physical loss in the network leverages this simplified form, penalizing deviations from this theoretical relationship:

$$L_{\text{physics}} = \frac{1}{N} \sum_{i=1}^N M_i \cdot (P_i - 2\pi\alpha_i)^2 \quad (36)$$

where M_i is a mask function, introduced because Thin-Airfoil Theory is only valid in the linear flight regime. Therefore, the mask is defined as:

$$M_i = \begin{cases} 1 & \text{if } |\alpha_i| \leq 10^\circ \\ 0 & \text{if } |\alpha_i| > 10^\circ \end{cases} \quad (37)$$

This ensures the physical constraint is only active at low angles of attack.

In the non-linear, high-angle-of-attack regime, the mask deactivates this term ($M_i = 0$), forcing the network to rely entirely on the empirical CFD data, which accurately captures the complex flow separation phenomena.

By explicitly building in the assumption of symmetry, this physics-informed regularizer provides a robust theoretical anchor for the network’s predictions without being applied in regimes where its underlying assumptions would be violated.

Optimization with Adam

Once the backpropagation algorithm has computed the gradient of the loss function with respect to every trainable parameter in the network ($\nabla L(\theta)$, where θ represents all weights and biases), the final step in the learning process is to update these parameters. This is done using the **Adam (Adaptive Moment Estimation)** optimizer, which guides the parameters toward a minimum in the loss landscape.

Adam combines the advantages of the **RMSprop** and **Momentum** optimization methods. It maintains an exponentially decaying moving average of both the past gradients (first moment, the momentum term) and the past squared gradients (second moment, the adaptive learning rate term).

Let $g_t = \nabla L(\theta_{t-1})$ be the gradient at timestep t . The optimizer updates two moving averages:

- first moment (momentum):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (38)$$

- second moment (adaptive learning rate):

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (39)$$

where β_1 and β_2 are hyperparameters (typically close to 1, e.g., 0.9 and 0.999 respectively).

These raw moment estimates are then bias-corrected to account for their initialization at zero:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad \text{and} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (40)$$

Finally, the parameters are updated using these corrected estimates. The update rule for a parameter θ is:

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} \quad (41)$$

where η is the learning rate and ε is a small constant (e.g., 10^{-8}) to prevent division by zero.

The key intuition is that the update step is scaled by both a running average of the gradients (providing momentum to navigate flat regions) and an adaptive, per-parameter learning rate (scaling down updates for frequently changing parameters and scaling up updates for sparse ones). This adaptive nature makes Adam robust and generally leads to faster convergence compared to standard stochastic gradient descent.

3.2. Code Implementation

We implemented our CNN and SDF generator using an **object-oriented paradigm**. Figure 9 provides a comprehensive **UML class diagram** detailing the structural layout, member attributes and method signatures of the system.

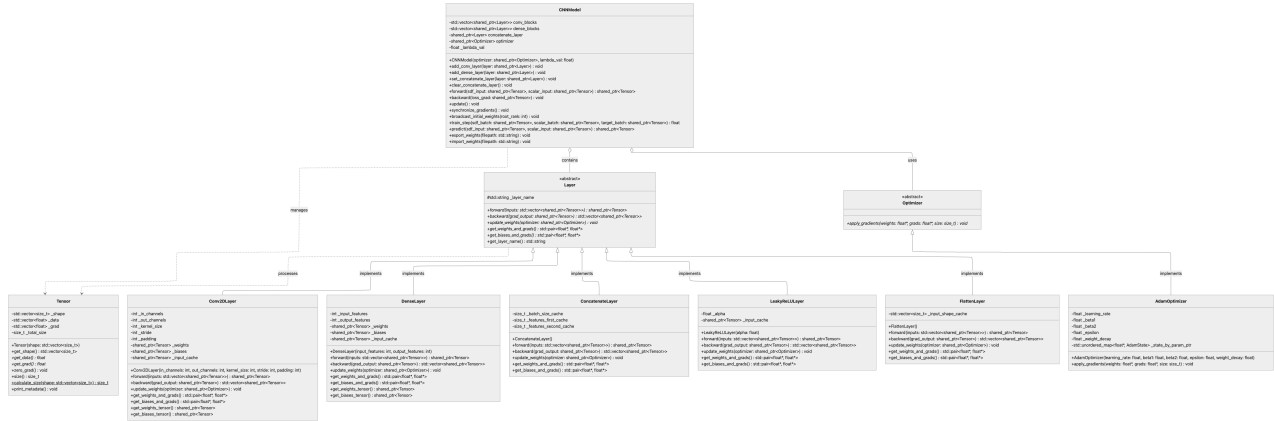


Figure 9: UML class diagram illustrating the object-oriented design of the C++ Convolutional Neural Network.

In the following sections, we decide to highlight the most interesting implementations of these components, detailing key choices and memory optimization layouts, as well as providing an explanation of the core functions implemented.

3.2.1. Geometric Preprocessing: SDF Generator

The generation of the Signed Distance Function is encapsulated within the `SDFGenerator` class, which maps airfoil geometries to structured 2D tensors suitable for convolutional processing.

To maximize CPU cache efficiency and minimize cache misses during spatial calculations, boundary vertices are stored in a `std::vector<Point>`, where `Point` is a simple structure holding `x` and `z` coordinates. This choice creates a single contiguous array for the boundary data.

Upon initialization, the constructor ensures scale invariance (as detailed in Section 3.1.1) by calling two private helper methods: `computeCord()` calculates the chord length c using `std::hypot` to prevent numerical overflow⁷ and `normalizePoint()` scales all vertex coordinates accordingly⁸.

With a normalized boundary in place, the `SDFGenerator` computes the signed distance for each point on the target grid. This is achieved through a set of optimized geometric kernel functions:

- **computeDistance(...)** **const** calculates the minimum Euclidean distance from a grid point to a boundary segment (as it is shown in Equations (16) and (17)). It includes a safety check for degenerate edges (where the two end-point of the segment coincide) to prevent division-by-zero errors;
- **getMinDistance(...)** **const** iterates through all boundary edges of the airfoil, calling for each one **computeDistance** and returning the absolute minimum distance found. Additionally, we used **static_cast<int>(airfoil.boundary.size())** to safely convert the unsigned vector size to a signed integer, preventing signed/unsigned comparison warnings;

⁷This function scales intermediate values before performing the sum of squares, avoiding the premature overflow or underflow that can occur when computing the norm directly as $\sqrt{\Delta x^2 + \Delta z^2}$

⁸This normalization step will be valuable for future implementations aimed at studying the effect of ice accumulation on the lift coefficient.

- finally, the `windingAlgorithm(...)` `const` implements the Winding Number Algorithm previously explained (Section 3.1.1) to verify whether a grid point p lies inside or outside the closed boundary. This is computed by analyzing edge crossings and using the 2D cross product (Equation (19))⁹, returning `true` if the point is inside the domain.

Once these values are computed for every grid point, the `mappingToMatrix` function converts the resulting 1D flattened SDF data into the required 150×150 2D matrix format and writes it to a file, utilizing the **RAII** (**R**esource **A**cquisition **I**s **I**nitialization) paradigm via `std::ofstream outfile(filename)` that automatically closes the file descriptor when `outfile` goes out of scope, in order to prevent system resource leaks in case of exceptions.

The most computationally intensive part of the process is executing these kernels for each one of the $150 \times 150 = 22,500$ points on the grid. This task is a classic example of an *embarrassingly* parallel problem and it is implemented in the `generateTensorMPI` method, the details of which are provided in Section 3.3.

3.2.2. Core abstractions and memory management

To achieve high computational throughput while preserving a flexible design, the system introduces two core abstractions: a unified multi-dimensional tensor representation and a polymorphic layer interface.

Tensor

The foundational data structure of our CNN implementation is the `Tensor` class, which manages both forward activation data (`_data`) and backward gradients (`_grad`) as multi-dimensional arrays of single-precision floating-point numbers.

First of all, the constructor `Tensor(std::vector<size_t> shape)` initializes the shape descriptor and sets the `_total_size` member through a member `initializer list`. It pre-allocates and zero-initializes the storage vectors using `std::vector::resize`, that guarantees contiguous memory allocation. Direct access to this underlying memory is provided via the `get_data()` and `get_grad()` methods, which return raw pointers (`float*`) using the `.data()` member function.

Beyond raw pointer access, the class provides safe, native array-like indexing for flattened 1D elements through overloaded `operator[]` methods, offering both mutable and read-only (`const`-qualified) references. Utility functions are also optimized: gradient values are efficiently reset before each backward pass using the `std::fill` algorithm within the `zero_grad()` method.

Furthermore, the total element count is computed by `calculate_size()`, which is declared as a `static` method¹⁰ and accepts a constant reference to avoid the overhead of copying the shape vector.

Finally, the class strictly adheres to `const` correctness, ensuring that utility methods like `print_metadata()` are executed safely without modifying the object's internal state.

Layer

The abstract base class `Layer`, defined in `Layer.hpp`, establishes a unified interface for all CNN components by leveraging runtime polymorphism to decouple the high-level network topology from the specific implementations of individual layers.

Runtime polymorphism is achieved through virtual functions and dynamic dispatch. When a class declares virtual methods, the compiler generates a virtual method table (`vtable`) containing pointers to these functions. Each object instance is implicitly equipped with a pointer (`vp_ptr`) to its class's `vtable`. During execution, when a virtual method is called via a base class pointer, the program resolves the call dynamically: it dereferences the object's `vp_ptr` to access the `vtable` and executes the overridden method corresponding to the actual derived type.

⁹Implemented through the `crossProduct(Point p, Point a, Point b)` `const` with each point passed by value, in order to improve performance for small `structs` by allowing them to be passed directly in CPU registers, avoiding memory indirection.

¹⁰This allows the method to be invoked without a class instance, for example, within a member initializer list.

This mechanism is fundamental to our design. Rather than relying on conditional logic to handle the different mathematical operations, polymorphism allows the model to store and manage all layers uniformly within a single container (`std::vector<std::shared_ptr<Layer>`). In addition, this design ensures high scalability and the effortless integration of new layer types simply by inheriting from the `Layer` base class.

Following the idea of implementing the polymorphism, we declared a virtual destructor `~Layer() = default`, ensuring that when a derived layer object is destroyed via a base class pointer, the derived class's destructor is executed, preventing resource leaks of dynamically allocated memory.

The core operations of the network are declared as pure virtual functions (indicated by the `= 0` suffix), which must be overridden by every concrete layer subclass. Specifically, we implemented:

- **forward**, that computes forward activations, and **backward**, that performs backpropagation, both accepting and returning `std::shared_ptr` in order to guarantee automatic memory deallocation when the tensors are no longer needed;
- **get_weights_and_grads()** and **get_biases_and_grads()** that returns `std::pair<float*, float*>` packaging the weight/bias arrays and their corresponding gradients into a single return value. Returning non-const pointers allows the optimizer to modify the parameter values directly in place.

Finally, architectural decoupling is achieved through a forward declaration of the `Optimizer` class; this resolves potential circular dependencies while allowing the `update_weights(std::shared_ptr<Optimizer>)` method to delegate parameter updates to external optimization algorithms seamlessly.

3.2.3. Computational and activation layers

The computational layers manage the network's mathematical transformations and contain all trainable parameters, while activation layers introduce the necessary non-linear activation boundaries.

DenseLayer

The concrete `DenseLayer` class implements the fully-connected linear transformation $\mathbf{y} = W\mathbf{x} + \mathbf{b}$, where W is the weight matrix and $\mathbf{b}$ is the bias vector, as described in details in the Section 3.1.2.

First of all, the constructor `DenseLayer(int input_features, int output_features)` allocates the weight and bias tensors utilizing `std::make_shared` rather than direct initialization via `std::shared_ptr(new Tensor(...))`. While this last one forces two separate heap allocations¹¹, `std::make_shared` performs a **single contiguous memory allocation** for both the reference-counting control block and the managed `Tensor` object, in order to reduce heap fragmentation and improve CPU cache locality by keeping the control metadata and the tensor data adjacent in memory.

To ensure gradient stability at the start of training, the weights are initialized according to the **Xavier/Glorot scheme**, drawing from a uniform distribution bounded by $\pm\sqrt{6/(N_{\text{in}} + N_{\text{out}})}$ using `std::default_random_engine` and `std::uniform_real_distribution<float>`.

During the **forward** pass, the input tensor is explicitly cached (`_input_cache`) to satisfy the mathematical requirements of backpropagation, as the input vector $\mathbf{x}$ must be retained to compute the weight updates as $\frac{\partial L}{\partial W} = \mathbf{x} \otimes \frac{\partial L}{\partial \mathbf{y}}$.

Moreover, the linear transformation $Y = XW^T + \mathbf{b}$ is evaluated through three nested loops:

- the outermost loop iterates over the batch size b ;
- the middle loop iterates over the output features j ;
- the innermost loop iterates over the input features i to evaluate the dot product $Y_{b,j} = \sum_i X_{b,i}W_{j,i} + b_j$.

The weight matrix W is stored in a transposed format of shape $(N_{\text{out}}, N_{\text{in}})$. This ensures that as the innermost loop increments its index over the input features, it accesses both the input tensor X and the weight tensor W with a unit stride. This design maximizes spatial locality, reduces cache misses, and enables the compiler to generate efficient SIMD vector instructions.

To conclude, also the **backward** method computes the gradients through three distinct sets of nested loops:

¹¹One for the object and one for its control block.

1. the gradient with respect to the inputs is calculated as $dX_{b,i} = \sum_j dY_{b,j} W_{j,i}$. In this loop structure, the weight matrix is traversed column-wise, resulting in non-contiguous memory access;
2. the gradient with respect to the weights is accumulated as $dW_{j,i} = \sum_b dY_{b,j} X_{b,i}$ using loops over j , i , and the batch dimension b . The innermost loop iterates over b , calculating the outer product of the output gradients and cached inputs;
3. the gradient with respect to the biases is accumulated as $dB_j = \sum_b dY_{b,j}$ by summing the output gradients across all batch elements for each output feature.

LeakyReLULayer

The concrete **LeakyReLULayer** class implements homonymous element-wise activation function shown in Equation (26). The negative slope parameter α is configured into the constructor **LeakyReLULayer(float alpha = 0.05f)**.

The interesting thing in this code is the use of the ternary operator:

- in the **forward** method, the element-wise activation is evaluated inside a contiguous loop using raw pointers `out_ptr[i] = (in_ptr[i] > 0.0f) ? in_ptr[i] : _alpha * in_ptr[i];`
- in the **backward** method, that is used to evaluate the gradient with respect to the input by scaling the output gradient as shown in Equation (27), implemented as `din_ptr[i] = dout_ptr[i] * ((cache_ptr[i] > 0.0f) ? 1.0f : _alpha).`

Because the derivative of the activation function is conditioned on the sign of the original input activations, caching the forward input tensor inside `_input_cache` is necessary to retrieve the values during the backward pass.

To conclude this class, because the activation layer contains no learnable parameters, the overridden method `update_weights` is implemented as a no-op. Moreover, the parameter accessors `get_weights_and_grads()` and `get_biases_and_grads()` return a pair of type-safe null pointers, `{nullptr, nullptr}`.

Conv2DLayer

The concrete **Conv2DLayer** class extends the abstract **Layer** interface to perform 2D spatial convolutions on four-dimensional tensors, which logically represent **batch samples, channels, height and width**.

Upon instantiation, the constructor initializes the convolutional kernels using a **Gaussian Xavier scheme** to ensure variance stability across deep network layers. The standard deviation for this distribution is computed as $\sigma = \sqrt{2/(F_{\text{in}} + F_{\text{out}})}$, where the fan-in and fan-out dimensions are defined as $F_{\text{in}} = C_{\text{in}} \times K^2$ and $F_{\text{out}} = C_{\text{out}} \times K^2$, with K denoting the kernel size.

During the **forward** pass, the spatial dimensions of the output tensor are explicitly calculated using Equation (25) and the resulting tensor is allocated contiguously via `std::make_shared`. Because the system flattens all multi-dimensional tensors into 1D contiguous vectors for optimal memory layout, the layer utilizes a private inline helper function, `idx4`, to resolve 4D indices to 1D offsets. This enforces a strict row-major mapping:

$$\text{Offset}(a, b, c, d) = [(a \times \text{dim}_b + b) \times \text{dim}_c + c] \times \text{dim}_d + d \quad (42)$$

where a, b, c, d represent the target indices across the batch, channel, and spatial dimensions.

The forward convolution operation is executed through seven-nested loop. The outer four loops iterate over the batch elements n , output channels oc and spatial output coordinates oh and ow , while the inner three loops traverse the input channels ic and kernel dimensions kh and kw . To handle boundaries without incurring the overhead of allocating memory for physical zero-padding, the implementation uses implicit padding. Source coordinates are dynamically evaluated at each step as $ih = oh \times S + kh - P$ and $iw = ow \times S + kw - P$:

- if these fall outside the input boundaries, the multiplication is safely bypassed using a **continue** statement;
- otherwise, the flattened indices are retrieved via `idx4`, and the kernel-input products are accumulated into a running sum initialized with the appropriate channel bias.

Finally, the **backward** method evaluates gradients using two distinct loop structures:

- the first structure computes the parameter gradients: the bias gradient (dB) aggregates the output gradients directly, while the weight gradient (dW) accumulates the product of the output gradients and the cached forward inputs;

- the second structure computes the gradient with respect to the input (dX) by performing a full convolution between the output gradients and the flipped kernel filters. To achieve this, the loop accesses the weights using inverted spatial coordinates:

$$kh_{\text{orig}} = K - 1 - kh, \quad kw_{\text{orig}} = K - 1 - kw \quad (43)$$

3.2.4. Structural Layers

The **structural layers** are parameter-free components designed to rearrange and route data tensors across the branches of the network.

FlattenLayer

The **FlattenLayer** class extends the base **Layer** interface to perform a logical reshaping of the 4D feature maps (generated by the convolutional branch) into a 2D tensor suitable for the subsequent **ConcatenateLayer**.

The **forward** method calculates the flattened size $F = C \times H \times W$ and allocates a 2D output tensor of shape $[B \times F]$ via `std::make_shared`. Since the flatten operation is basically a reshape, the data is copied from the input to the output tensor sequentially in a single loop, preserving the element order across contiguous memory arrays.

The **backward** method receives the 2D gradient tensor **grad_output** (of shape $[B \times F]$), validates the batch size consistency, allocates a 4D gradient tensor matching the cached shape `_input_shape_cache`¹² and performs the inverse sequential copy `grad_in_ptr[i] = grad_out_ptr[i]`, reconstructing the 4D gradient layout for the preceding convolutional layer.

ConcatenateLayer

The **ConcatenateLayer** class is used to obtain a single tensor that combines the information from the **Convolutional Layer** and the scalar values of the Reynolds number and angle of attack.

Beyond the mathematical formalism about the shape of the matrices described in a detailed way in Section 3.1.2, the merge operations are implemented with nested loops:

- the outer loop iterates over the batch size n , computing the row offsets for the first input (`n * _features_first_cache`), the second input (`n * _features_second_cache`) and the output (`n * (_features_first_cache + _features_second_cache)`);
- two inner loops then copy the data, ensuring row-wise contiguous layout.

During the **backward** method, this operation is inverted. The layer receives a combined gradient tensor of shape $[B \times (D_1 + D_2)]$ and must route the gradients back to their respective origins, reversing the copy logic: for each batch element, it copies the first D_1 components of the output gradient to the first input gradient buffer and the remaining D_2 components to the second input gradient buffer. Both elements are then returned in a standard vector `{grad_first, grad_second}`.

3.2.5. Training pipeline, loss and optimizers

The network's optimization and training pipeline is coordinated by mathematical loss functions and state-tracking optimization solvers.

Loss

Unlike the object-oriented structure of the computational layers, the **physics-informed loss function** and its corresponding gradient are implemented as non-member utilities within the **Loss** namespace. This design choice avoids the overhead of object instantiation and maintains a functional style for stateless operations.

¹²This member attribute of type `std::vector<size_t>` preserves the original spatial dimensions ($C \times H \times W$) to reconstruct and propagate the gradient back.

In the **forward** method¹³, the composite physics-informed loss, described in Section 3.1.2, is computed in a single loop using raw pointers retrieved via `get_data()` to ensure contiguous memory traversal. When computing the physics-informed penalty, the masking logic from Equation (37) is applied: the linear regime limit is saved in an anonymous namespace at the beginning of the file, in order to restrict the scope of this constant. The average data and physics terms are computed and normalized by the number of elements using `static_cast<float>(n)`

Finally, the **backward** method is implemented in a similar way.

Optimizer and Adam Optimizer

Similar to the layer architecture, the optimization logic is strictly decoupled from the neural network topology via an abstract **Optimizer** base class. This unified interface ensures that different numerical solvers can be swapped seamlessly, exposing a single pure virtual function (`apply_gradients(float* weights, float* grads, size_t size)`) to handle dynamic dispatch interface that must be implemented by concrete subclasses.

The only concrete implementation of this interface is the **AdamOptimizer**, which executes the **Adaptive Moment Estimation** algorithm detailed in Section 3.1.2.

To map specific weight and bias arrays to their corresponding optimization states without coupling the layers to the optimizer, the class utilizes a parameter-to-state lookup map as `std::unordered_map<float*, AdamState> _state_by_param_ptr`¹⁴, that provides average constant-time $\mathcal{O}(1)$ lookup complexity. Since weight and bias arrays are allocated once and reside in contiguous memory locations throughout the lifetime of the network, their raw memory addresses (`float*`) serve as unique keys. This pointer-based hashing avoids the overhead of index-based tracking or string-based lookups and enables on-the-fly initialization of the **AdamState** upon the first optimization step.

CNNModel

The concrete class **CNNModel** coordinates the execution, training and state management of the entire neural network. It aggregates the layers of the convolutional branch and the dense branch, manages feature fusion, interfaces with the optimizer and implements utilities for distributed data-parallel training and weight serialization.

The **forward** method propagates input tensors sequentially through the model. To maximize efficiency, the iteration over the layer containers (`conv_blocks` and `dense_blocks`) utilizes `const auto&`, thereby avoiding the performance overhead associated with incrementing reference counters on the underlying shared pointers. Conversely, the **backward** method employs reverse iterators to correctly propagate gradients upstream in strict accordance with the chain rule.

Another interesting detail is that in the **update** method gradient values are clipped to the interval $[-1.0, 1.0]$ to prevent gradient explosion¹⁵. This is executed via a local lambda function (`clip_layer`) containing a nested `clip_tensor` lambda, which directly modifies the elements in-place using standard pointer-based loops.

Whenever layer-specific parameters must be accessed for weight updates, gradient synchronization, or serialization, the model dynamically resolves the abstract base pointer types through runtime downcasting via `std::dynamic_pointer_cast` (e.g., casting from `std::shared_ptr<Layer>` to `std::shared_ptr<Conv2DLayer>`).

Finally, the model implements binary weight export and import methods using `std::ofstream` and `std::ifstream`, opened with the `std::ios::binary` flag. Binary data is written and read directly via type-unsafe low-level operations utilizing `reinterpret_cast<const char*>` and `reinterpret_cast<char*>` to map raw byte buffers, which minimizes I/O latency. During the import phase, after reading the shapes and element counts from the file, weight vectors are populated using `std::move` (from the `<utility>` header) to transfer the ownership of heap-allocated arrays into the target structures, bypassing expensive memory allocation and copying. To illustrate this efficient binary format, Table 4 details the layout of the file header and the metadata corresponding

¹³That accepts predictions, targets and angles of attack (**alphas**) tensors as `const std::shared_ptr<Tensor>&`.

¹⁴Where **AdamState** is a private structure that contains the first moment vector `std::vector<float> m`, the second moment vector `std::vector<float> v`, and an unsigned integer `timestep` to keep track of the parameter update steps.

¹⁵This choice has been made after training the model for the first time, where we have noticed very strong fluctuations of the loss value during the training process.

to the initial `Conv2DLayer`.

Table 4: Example layout of the binary weight file header, showing the global count of layers followed by the name and weight/bias tensor metadata of the first layer (`Conv2DLayer`).

Byte Offset	Field Name	Value / Example
0x00-0x07	Total layers in model (N_{layers})	8
0x08-0x0F	Layer name length (L_{name})	39
0x10-0x36	Layer name string	"Conv2DLayer(1,8,5, stride=5, padding=0)"
0x37	Weights presence flag	true (0x01)
0x38-0x3F	Weights tensor rank (R_W)	4
0x40-0x5F	Weights tensor shape	[8, 1, 5, 5]
0x60-0x67	Total weights count (N_W)	200
0x68-0x387	Weights raw data	[-0.0125, 0.0841, ...]
0x388	Biases presence flag	true (0x01)
0x389-0x390	Biases tensor rank (R_B)	1
0x391-0x398	Biases tensor shape	[8]
0x399-0x3A0	Total biases count (N_B)	8
0x3A1-0x3C0	Biases raw data	[0.0, 0.0, ...]

3.2.6. Data Pipelines: `NPZDataLoader`

The `NPZDataLoader` class manages the retrieval, normalization, and partitioning of datasets stored in the compressed NumPy (`.npz`) format. This format was explicitly selected because it packages multiple named multidimensional arrays within a single ZIP archive, acting as an efficient bridge between Python-based preprocessing scripts (such as `build_dataset.py`) and the C++ training pipeline without the severe parsing overhead of text-based formats like CSV.

Specifically, the dataset is structured into four core arrays:

- `X_sdf`, representing the 150×150 SDF of the airfoil geometries;
- `X_scalars`, storing the Reynolds number (Re) and the angle of attack (α);
- `Y_cl`, containing the ground-truth lift coefficient (C_L) labels computed from the Navier-Stokes simulations;
- `metadata`, containing the specific NACA airfoil profile labels.

This class provides a complete input pipeline, converting these raw data arrays into normalized, formatted tensor batches for network training.

To import the data without introducing heavy external C++ NumPy parsing libraries, the internal reading method establishes a read-only POSIX pipe to execute a brief Python script, that utilizes Python's native numpy library to load the specified array, converts it to raw single-precision bytes and streams the output directly to the standard output. The C++ program then intercepts this byte stream, reading it sequentially into a standard vector buffer before safely closing the process and verifying its exit status.

Once the arrays are loaded into memory, the constructor applies Z-score standardization across the SDF, scalar parameters and target lift coefficients, to prevent numerical instability during the training phase. Then, the class handles the sequential feeding of this data during training. The batching process partitions the normalized dataset into mini-batches by maintaining an internal cursor that tracks the current read position. For each requested batch, it allocates new tensor objects and utilizes standard library copying algorithms to transfer the exact slice of data from the internal buffers to the newly created tensors.

3.2.7. Main execution of the CNN

The main execution entry point of the neural network workflow is defined in `main.cpp`. It coordinates the loading of datasets, the structural composition of the CNN layers and the training and validation loops over successive epochs.

The program begins by instantiating the training and testing pipelines using `NPZDataLoader` to load the compressed NumPy datasets. It then creates a shared instance of the `AdamOptimizer` wrapped in a `std::shared_ptr` via `std::make_shared` with a designated learning rate of 10^{-5} . This optimizer is passed to initialize the `CNNModel` instance.

The architecture of the neural network is composed using the model’s builder methods:

1. the convolutional branch is constructed by adding a `Conv2DLayer` with 1 input channel, 8 output channels, a 5×5 kernel, and a stride of 5, followed by a `LeakyReLULayer` (with an activation slope $\alpha = 0.05$) and a `FlattenLayer` to reduce the spatial feature maps to a one-dimensional representation of size 7200;
2. the feature fusion stage is enabled by registering a `ConcatenateLayer` via `set_concatenate_layer()`, which merges the geometry features with the two physical flow parameters (Reynolds number and angle of attack);
3. the fully-connected regression head is built using three sequential `DenseLayers` of dimensions $7202 \rightarrow 128$, $128 \rightarrow 64$, and $64 \rightarrow 1$, separated by `LeakyReLULayer` activations, resulting in a single scalar prediction representing the lift coefficient (C_L).

The training process iterates for 100 epochs. Within each epoch, it resets the data loaders and processes the dataset in mini-batches. For each batch, it invokes `train_step` on the model instance to run the forward pass, evaluate the PINN loss function, compute parameter gradients via backpropagation and update the weights. After the training phase of each epoch, it runs a validation pass by propagating the validation samples through the network via `forward()` to monitor generalization and check for overfitting by computing the MSE against the validation targets.

3.3. Parallelism

To achieve high-performance capabilities and scale the computational workflow to large-scale grids and datasets, the codebase employs distributed-memory parallelism using the **Message Passing Interface (MPI)** standard. The parallelization strategy is implemented at two distinct stages: the geometric preprocessing phase (for parallel SDF generation) and the neural network training phase (for distributed data-parallel training).

3.3.1. Parallel SDF Generation

The computation of the SDF, as described in Section 3.1.1, is an embarrassingly parallel task, since the evaluation of the signed distance for each point on the 150×150 grid is independent of all other points.

To do this, upon initializing the MPI environment via `MPI_Init`, the total number of grid points ($N_{\text{total}} = 22,500$) is partitioned by the communicator size (S , retrieved using `MPI_Comm_size`). Because there could be some cases where N_{total} is not perfectly divisible by S , the remainder ($R = N_{\text{total}} \bmod S$) is distributed among the first R processes¹⁶. Each process (with rank ID r , retrieved via `MPI_Comm_rank`) computes its local count and offset as:

$$N_{\text{local}} = \lfloor N_{\text{total}}/S \rfloor + \mathbb{I}(r < R) \quad \text{and} \quad O_{\text{local}} = r \lfloor N_{\text{total}}/S \rfloor + \min(r, R) \quad (44)$$

where $\mathbb{I}$ is the indicator function.

Each process independently computes the SDF values for its assigned local grid points. To reconstruct the complete 2D matrix, the local chunks must be gathered into a global vector of size N_{total} , using `MPI_Allgatherv`, which performs a vector gather-to-all communication. The collective call requires `recvcounts`, which stores the number of elements expected from each rank, and `displs`, which specifies the starting byte displacement of each segment in the global buffer. `MPI_Allgatherv` ensures that all processes receive the full SDF image.

Finally, the root process (rank 0) writes the complete matrix to disk via the `mappingToMatrix` method, while other ranks proceed to the next file or terminate safely via `MPI_Finalize`.

¹⁶This partitioning guarantees perfect load balancing, as the workload of any two processes differs by at most a single grid point.

3.3.2. Distributed Data-Parallel Training

The training of the Convolutional Neural Network is parallelized in `CNN/main.cpp` and `CNNModel.cpp` using a data-parallel training strategy to scale to large datasets and decrease training epoch time.

In `CNN/main.cpp`, data parallelism is implemented by distributing the training batch across multiple processes. The global mini-batch size is scaled according to $B_{\text{global}} = B_{\text{local}} \times S$, with a local batch size of $B_{\text{local}} = 64$. To prevent redundant memory duplication, each process uses a custom `slice_batch_tensor` method to extract only its specific segment of the global dataset, beginning at the index offset $r \times B_{\text{local}}$.

To guarantee consistent optimization across the distributed environment, the training pipeline enforces MPI synchronization at three critical stages:

1. initially, the root process generates the network weights and distributes them to all other ranks using `MPI_Bcast` on the raw tensor buffers, ensuring an identical starting state¹⁷ across the entire communicator;
2. during the backpropagation phase, each process computes gradients based on its local batch slice. Before applying the Adam optimization step, these local gradients are aggregated via an in-place `MPI_Allreduce` operation. By utilizing the `MPI_SUM` operator directly on the existing memory buffers, the system efficiently avoids redundant data allocations. The accumulated gradients are then divided by the total number of processes S to compute a global average, which guarantees that every rank applies the exact same mathematical update to its local parameters;
3. finally, at the conclusion of each epoch, the local training losses, validation losses, and processed sample counts are gathered back to the root process via `MPI_Reduce`, allowing it to calculate and report the consolidated global statistics.

3.4. Results

Given the significant model size of 930,513 trainable parameters, performing the training process on standard personal computers was computationally impractical within a reasonable timeframe. To overcome this limitation, the training was conducted on the high-performance computing (HPC) resources of the **CINECA Leonardo cluster**. Specifically, we utilized a full single node configured with two Intel Xeon Platinum 8480+ (Sapphire Rapids) processors, providing a total of 112 physical cores, backed by 512 GiB of DDR5 RAM operating at 4800 MHz. This hardware acceleration reduced the total execution time from several hours to approximately five minutes.

```
1 #!/bin/bash
2 #SBATCH --job-name=cnn_training
3 #SBATCH --account=EUHPC_D35_025
4 #SBATCH --partition=dcgp_usr_prod
5 #SBATCH --nodes=1
6 #SBATCH --ntasks-per-node=1
7 #SBATCH --cpus-per-task=112
8 #SBATCH --time=01:00:00
9 #SBATCH --output=cnn_%j.out
10 #SBATCH --error=cnn_%j.err
11
12
13 cd /leonardo/home/userexternal/gdonnine/CNN/
14     Ice_acceleration_model_on_airplane_wing/CNN
15
16 module purge
17 module load python/3.11.7
18 module load gcc/11.3.0
19 module load intel-oneapi-compilers
20 module load intel-oneapi-mpi
21
22 srun ./cnn_mpi_executable
```

¹⁷With all the weights initialized to the same value.

Listing 5: SLURM run script for CNN training (`run_cnn.sh`).

The network was trained for 100 epochs on a synthetically generated dataset of 2D SDF, representing five **symmetric NACA airfoil profiles** (NACA 0006, 0008, 0010, 0012 and 0018) across varying angles of attack (α) and Reynolds numbers (Re). Partitioned via an 80/20 split, the dataset yielded 1,713 training samples and 429 validation samples¹⁸.

Initial Unstable Training Run

In our first training attempt, we adopted a standard learning rate of 10^{-3} , without applying input feature normalization or gradient clipping. This configuration resulted in severe numerical instability during the optimization process. As illustrated in Figure 10, the training and validation losses exhibit chaotic, erratic oscillations, fluctuating wildly by over seven orders of magnitude (ranging from 10^{-1} to 10^6 in log scale). The presence of massive loss spikes¹⁹ is a clear indicator of exploding gradients. Without gradient clipping, large backpropagated gradients led to excessive parameter updates, pushing the weights into unstable regions of the loss landscape and preventing the optimizer from settling into a local minimum.

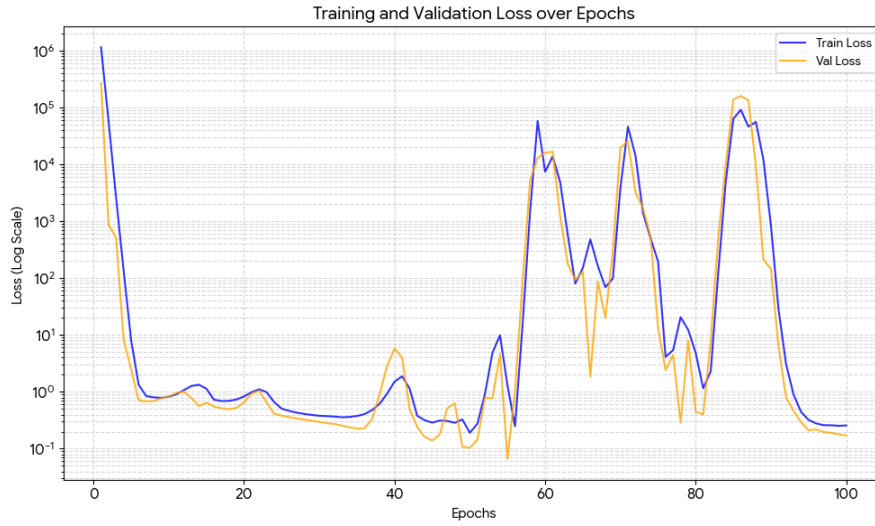


Figure 10: Log-scale evolution of the training and validation losses over 100 epochs during the initial unstable training attempt. The extreme oscillations and sudden spikes up to 10^6 highlight the numerical instability caused by an excessively high learning rate and the absence of gradient clipping or feature normalization.

Optimized Successful Training Run

To resolve the numerical instability and ensure smooth convergence, three key modifications were implemented in the training pipeline:

1. the Adam optimizer’s learning rate (η) was reduced from 10^{-3} to 10^{-5} ;
2. standard Z-score normalization was applied during the data-loading phase to scale the inputs (SDF and physical scalars) and targets (lift coefficients) to zero-mean and unit-variance;
3. gradient clipping was enabled in the update phase to clamp all gradient components strictly within the interval $[-1.0, 1.0]$ (as detailed in Section 3.2.5).

These modifications successfully stabilized the training trajectory.

As shown in Figure 11, the loss curves exhibit highly stable, monotonic convergence. The left plot (Full View) shows the rapid decay of the loss during the initial epochs, indicating that the network quickly learned the dominant physical relationships. The right plot (Zoomed View, Epochs 5-100) highlights the smooth, continuous reduction of both training and validation losses, with no signs of overfitting. The validation loss consistently tracks slightly lower than the training loss; this is a common characteristic when model regularization (such as

¹⁸All of which have been made publicly available on [Kaggle](#)

¹⁹Such as the prominent peaks around epochs 40, 60, 70 and 85.

L2 weight decay) is active during training or when the z-score normalization effectively centers the distribution of test cases.

This optimized training process completed in likely 5 minutes and 11 seconds, achieving a final training loss of 0.0104216 and a validation loss of 0.00784775, demonstrating both the efficiency and accuracy of our implementation.

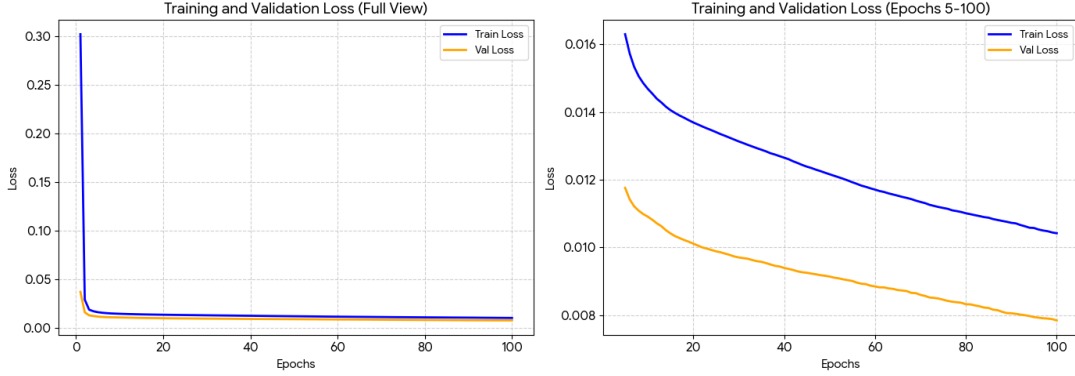


Figure 11: Evolution of training and validation losses for the optimized training run. The full view (left) illustrates the rapid initial convergence, while the zoomed-in view from epoch 5 to 100 (right) highlights the smooth, monotonic decrease of both metrics, confirming the effectiveness of Z-score normalization and gradient clipping.

4. Tuning

The Convolutional Neural Network, described and implemented in Chapter 3, provides the starting point for the numerical study presented in this Chapter. Although the initial model is already able to approximate the lift coefficient from the airfoil geometry and the flow parameters, we decided to understand which choice of architecture, optimization algorithm, regularization, activation function and learning rate allow the surrogate model to obtain more accurate and more reliable predictions. For this reason, we performed a sequence of controlled tuning experiments. In each experiment one group of quantities was varied, while the other settings were kept fixed. The results were then analysed not only through the final validation error, but also through the evolution of the training objective, the physical validation error, the gradient norms, the actual parameter updates and the activation distributions.

The order of the study follows the development of the model:

- first, we investigated the topology of the network, because the feature representation and the capacity of the regression head influence every subsequent experiment;
- then, we considered dropout, the optimizer, the weight of the SIMM physical term, and L1 and L2 regularization;
- next, we compared the available activation functions;
- as last thing, we studied the magnitude and the schedule of the learning rate.

The last part of the Chapter (Section 4.9) describes the recorded ordinary training sequence and its final production configuration, and collects the conclusions that can be drawn from the complete tuning.

4.1. Cross-validation and diagnostic methodology

All the tuning experiments use the training dataset stored in `dataset/cnn_dataset_train.npz`²⁰, which contains 1713 samples. Each sample consists of a 150×150 Signed Distance Function²¹, the scalar pair (Re, α) and the corresponding lift coefficient C_L . The separate file `dataset/cnn_dataset_test.npz`, containing 429 samples, is not used to compare candidates. In the intended procedure, it is loaded only after a candidate has been selected and the selected model has been refitted on the complete training dataset. This separation is important because using test data during the search would turn the final error into a selection quantity rather

²⁰Both the training and test datasets are generated according to the procedure detailed in Section 2.

²¹See Section 3.1.1 for details about the implementation.

than an independent estimate.

The comparisons are performed with the project implementation of **RandomKFold**. The samples are shuffled with a fixed seed and divided into five folds. An important detail is that the data is split on a per-sample basis, meaning individual simulation results are distributed randomly rather than grouping all data from a single airfoil into one fold. This means that for any given fold, data from the same airfoil (e.g., NACA 0012) can appear in both the training and validation sets. In this way, the tuning experiments measure the model’s ability to interpolate within the dataset, rather than its capacity for generalization to entirely new airfoil families. This per-sample choice is intentional: the project track defines the surrogate task as prediction over (Re, α) for the airfoils included in the dataset, not zero-shot transfer to unseen geometries. In every fold, approximately eighty percent of the samples are used for training and the remaining twenty percent for validation. In the recorded runs, the validation sets contain 342 (or 343 samples) and the training sets contain 1370 (or 1371 samples). The fold plan is generated once and reused for every candidate within a tuning experiment, guaranteeing that every candidate model is benchmarked on the exact same data for a direct comparison.

The experimental procedure enforces the following key principles, in order to ensure reproducible comparisons.

First of all, the normalization statistics (such as mean and standard deviation) are calculated exclusively from the training data. These same statistics are then used to scale the validation data. We also added a minor preprocessing step to convert the angle of attack from degrees to radians, which is necessary for the physics-informed loss function. Next, every trial is completely independent. For each candidate model and each fold, a new instance of the network and optimizer is created from scratch, in order to not carry over any learned parameters, optimizer moments or any other state from previous runs. Finally, the distributed training framework is designed to be consistent. The global batch size is fixed for all experiments, and increasing the number of MPI processes simply divides this global batch into smaller sub-batches for each worker. This ensures that the results and metrics are independent of the number of parallel ranks used.

To execute the extensive hyperparameter sweeps in an automated, scalable and reproducible manner, we engineered a modular, object-oriented C++ framework. The core of this framework is a "recipe-and-blueprint" design pattern, which decouples the abstract specification of a model from its physical runtime instantiation.

This process begins with **LayerRecipe** instances, which define each layer not as an object, but as a set of instructions. Each recipe encapsulates two core functionals:

- a shape inference mapping to dynamically compute output tensor dimensions;
- a deferred builder to allocate and initialize the layer’s parameters from a deterministic seed.

A **ModelBlueprint** then assembles these recipes into a complete network structure, logically separated into a convolutional feature extraction trunk and a fully-connected regression head.

To systematically generate the wide range of candidates for our sweeps, a parametric factory function, **make_blueprint**, constructs these blueprints on the fly. It accepts a full set of architectural parameters (such as kernel sizes, dense head widths and dropout rates) and chains together the appropriate recipes to produce a complete and custom **ModelBlueprint**.

A complete experimental candidate is then defined by a **TrialConfig** structure. This object bundles the **ModelBlueprint** with an **OptimizerRecipe** (e.g., Adam, SGD), a learning rate scheduler and all other loss and training settings. A **SearchSpace** utility then generates a full list of these **TrialConfig** objects, defining the entire grid for a given tuning experiment.

Orchestration is handled by the **CrossValidator** object. For each candidate **TrialConfig**, it executes the complete K-fold scheme. Crucially, for every fold, it uses the blueprint and recipes to instantiate a completely new, isolated model and optimizer. This deferred, just-in-time instantiation guarantees strict isolation and ensures zero state carryover between folds or candidates. Finally, the training loop itself is managed by the **Trainer**, which enforces our balanced batching and "20/20" early stopping rules, while a concurrent **TrainingDiagnostics** module captures the detailed stability metrics analyzed throughout this chapter.

With this experimental framework and software infrastructure established, we evaluate each candidate using a primary selection metric, supplemented by a series of diagnostic quantities to analyze model behavior.

The primary selection quantity is the physical-unit mean squared error on the validation samples. If $\hat{C}_{L,i}$ is the

predicted coefficient and $C_{L,i}$ is the CFD value, the physical MSE of fold k is computed as:

$$\text{MSE}_{\text{val}}^{(k)} = \frac{1}{n_k} \sum_{i \in \mathcal{V}_k} \left(\hat{C}_{L,i} - C_{L,i} \right)^2 \quad (45)$$

where $\mathcal{V}_k$ is the validation subset and n_k is its number of samples.

The final score for a candidate is the sample-weighted average of this MSE across all five folds, as returned by **CrossValidator**. This distinction matters because the training objective contains a data term and, in some experiments, a physics term expressed in fold-specific standardized target units, while Equation (45) is expressed in the squared physical lift-coefficient units. To assess the sensitivity of a model to the specific data split, the standard deviation of the MSE across folds is also reported alongside the mean.

Beyond the primary selection metric, we monitor several diagnostic quantities to ensure numerical stability and understand the training dynamics. The training objective is monitored through **train_objective**, while the physical training and validation errors are read from **epoch_metrics.csv**. The cross-validation history is normally stored every ten epochs, although diagnostics contain a validation value at every epoch.

The following diagnostic quantities are used throughout the Chapter:

- the gradient norm is read from rows with **parameter_scope=all** and it is measured after the MPI weighted gradient aggregation and before clipping and the optimizer update. This metric indicates whether gradients are naturally vanishing or exploding;
- the parameter update ratio is the actual movement of the parameters relative to their norm, computed as:

$$r_\theta = \frac{\|\theta_{\text{after}} - \theta_{\text{before}}\|_2}{\max(\|\theta_{\text{before}}\|_2, 10^{-12})}, \quad (46)$$

where θ_{before} and θ_{after} are the parameter vectors before and after the optimizer update, respectively.

A healthy value typically hovers around 10^{-3} . Excessively high values point to numerical instability or an overly aggressive learning rate, whereas excessively low values indicate training stagnation. The 10^{-12} term serves as a safety epsilon to prevent division by zero.

- the pre-activation and post-activation statistics are computed for every layer, including the mean, variance, minimum, maximum, and fixed-bin histograms of the activation tensors;
- the activation statistics are distinguished between the pre-activation tensor entering a non-linearity and the post-activation tensor returned by it. Their mean, variance, minimum, maximum and fixed-bin histograms are used to identify signal attenuation, saturation, or growth of the activation scale.

In all cases, we checked the raw CSV files for non-finite values and for late-epoch increases before describing a run as numerically stable.

4.2. Topology tuning

The first tuning activity concerned the topology of the network, as it dictates the model’s fundamental capacity to learn from the data. The architecture has two key components:

- a convolutional feature extractor responsible for extracting meaningful geometric features from the 150×150 SDF input;
- a dense head that fuses these geometric features with the scalar flow parameters (Reynolds number and angle of attack) to predict the final lift coefficient.

The main focus is to find the right balance. A model with insufficient capacity (too few layers or neurons) may fail to capture the non-linear relationship between an airfoil’s shape and its aerodynamic performance. Conversely, an overly complex model risks poor generalization, fitting noise in the training data rather than the underlying physical principles, while also increasing computational cost.

This type of structural comparison is consistent with the surrogate model motivation introduced in Chapter 3. The reference to the role of activation and gradient propagation in deep networks discussed by Glorot and Bengio (2010) is kept here only as motivation for studying signal propagation in principle, since that work considers saturating nonlinearities and initialization and does not predict a kernel size or a head width for this SDF trunk.

This experiment systematically explores different architectural choices, such as the convolutional kernel size, the width and depth of the dense head, to identify a topology that optimally balances representational power with generalization performance. All other hyperparameters, such as the optimizer and loss function, were held

constant to isolate the impact of the architecture itself.

For the convolutional trunk, we evaluated different kernel sizes, which directly govern the spatial resolution of the resulting feature maps and, consequently, the input dimensionality to the dense head. For a two-dimensional convolution with input size $H_{\text{in}} \times W_{\text{in}}$, kernel size K , stride S , and zero padding P , the output dimensions are

$$H_{\text{out}} = \frac{H_{\text{in}} - K + 2P}{S} + 1, \quad W_{\text{out}} = \frac{W_{\text{in}} - K + 2P}{S} + 1. \quad (47)$$

The topology experiment used $P = 0$ and set the stride equal to the kernel size, with eight output channels in every candidate. The two main alternatives for the trunk were a 5×5 kernel (producing a 30×30 feature map, leading to 7202 inputs for the dense head) and a 3×3 kernel (producing a 50×50 feature map, leading to 20002 inputs for the dense head).

For the dense head, we explored variations in both width (the number of neurons in each layer) and depth (the number of hidden layers). A head with widths $(w_1, \dots, w_m)$ then applies successive transformations $\mathbf{h}_{j+1} = \sigma(W_j \mathbf{h}_j + \mathbf{b}_j)$ with LeakyReLU nonlinearity σ and ends with a linear scalar output. Increasing the widths mainly affects the capacity of the first dense transformation, while adding hidden layers changes the depth of the nonlinear mapping. The parameter count makes this distinction concrete. With the 5×5 :

- the baseline head (128, 64) holds 930305 parameters;
- the intermediate head (512, 256) holds 3819521 parameters;
- the selected head (1024, 512, 256, 128) holds 8065025 parameters;
- the widest depth two head (2048, 1024) holds 16850945 parameters.

Each dense layer contributes $(D_{\text{in}} + 1)D_{\text{out}}$ weights and biases and the convolution contributes $(K^2 + 1) \times 8$ parameters.

To systemically evaluate these architectural trade-offs, we designed a comprehensive grid search. The search contained 18 candidates and therefore 90 fold evaluations. Each fold was trained following the same recipe:

- 100 epochs with global batch size 64;
- Adam optimizer with learning rate 10^{-5} ;
- a SIMM physics weight of 0.25;
- LeakyReLU activation with negative slope 0.05.

The baseline was **conv5x5-dense-128-64**. The entire experiment was conducted following the rigorous cross-validation framework, evaluation metrics and rules detailed in Section 4.1.

The official results and logs for this specific search are committed in the CINECA Leonardo log `run_leonardo_51807287.log` and its corresponding summary files. It should be noted that detailed per-epoch diagnostic logging (e.g. gradient norms, activation statistics) was disabled for this particular grid search. The only convergence histories are the public ten epoch checkpoints printed to the log.

The results of the 90-fold evaluation, summarized in Table 5, reveal the trends regarding the impact of network architecture on prediction accuracy. The ranked performance of all candidates is shown in Figure 12.

Candidate	Mean validation MSE	Fold standard deviation
conv5x5-dense-1024-512-256-128	0.004560	0.000946
conv5x5-dense-1024-512-256-128-64	0.004626	0.000918
conv5x5-dense-1024-512	0.004851	0.000911
conv5x5-dense-512-256-128-64	0.004872	0.001125
conv5x5-dense-512-256	0.005052	0.001067
conv5x5-dense-1536-768	0.005307	0.001177
conv5x5-dense-2048-1024	0.005673	0.001466
conv5x5-dense-128-64 (baseline)	0.006010	0.000823
conv3x3-dense-128-64	0.006330	0.001099
conv5x5-dense-64-32	0.006425	0.001292
conv3x3-dense-512-256	0.006459	0.001318

Table 5: Representative topology candidates in ranked order. The ranking was performed over the complete 18 candidate grid by sample-weighted mean physical validation MSE. The omitted intermediate candidates fall between the rows shown and do not change the conclusions.

The lowest observed validation MSE was obtained by `conv5x5-dense-1024-512-256-128`, with 0.004560 ± 0.000946 . This represents a substantial improvement over the baseline `conv5x5-dense-128-64`, which achieved 0.006010 ± 0.000823 , outperforming it consistently across all five cross-validation folds. The mean paired difference between the winner and the baseline is -0.001450 , indicating an advantage of the selected topology. After being refit on the entire training dataset, this winning topology achieved an MSE of 0.002805 on the untouched test dataset.

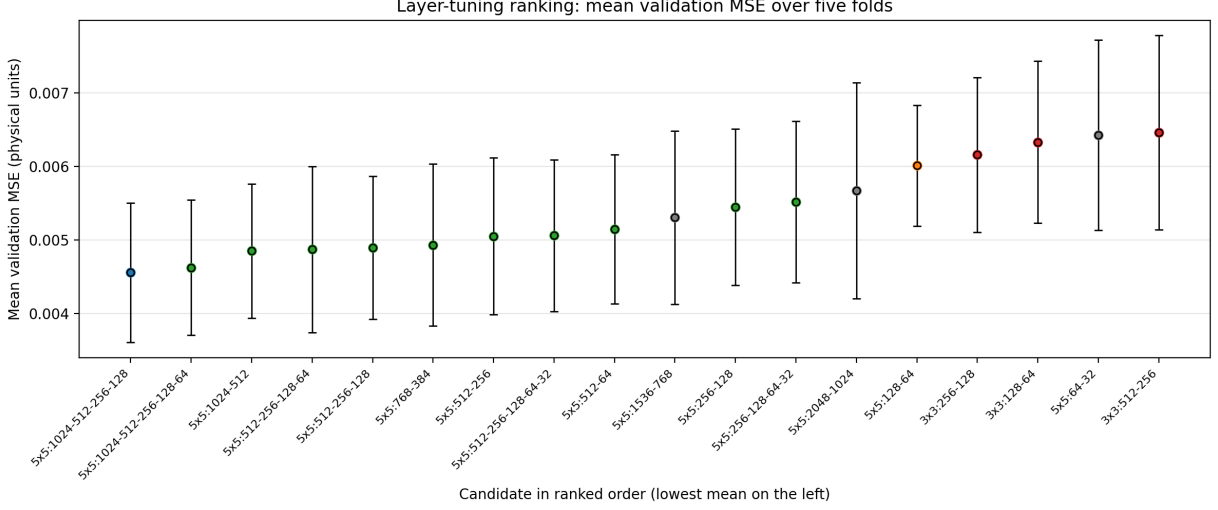


Figure 12: Layer tuning ranking by sample-weighted mean physical validation MSE over five folds. Error bars show the fold standard deviation. The leftmost candidate is the selected `conv5x5-dense-1024-512-256-128`. The plot summarizes `summary.csv` and does not imply a fitted scaling law.

The analysis shows that both network width and depth were crucial for achieving this performance, but with diminishing returns.

Deeper networks consistently proved more effective at learning the complex relationship between airfoil geometry and lift coefficient. We can see that at entry width (512, 256), increasing the head from two to four hidden layers reduced the mean error from 0.005052 to 0.004872, while at entry width (1024, 512) the corresponding increase in depth reduced it from 0.004851 to 0.004560. Increasing width also helped at fixed depth, but the gains became smaller and then reversed. At a fixed depth of two layers, the MSE decreased steadily as the head widened from (64, 32) (with an MSE of 0.006425) to (1024, 512) (with an MSE of 0.004851). However, this trend then reversed. Increasing the width further to (1536, 768) caused the error to rise again to 0.005307, and further to (2048, 1024) with an MSE of 0.005673. This indicates that the model was becoming too large and was beginning to lose its generalization capability.

For this reason, the best performance is not achieved by maximizing either width or depth in isolation, but by finding an effective interaction between the two.

The kernel comparison gives a second clear result. Across all paired comparison, the 5×5 convolution consistently outperformed the 3×3 convolution. This trend is clearly visible in the paired comparisons shown in Figure 13. For instance, when paired with the (512, 256) dense head, the 5×5 kernel achieved a mean MSE that was lower by approximately 0.001407 compared to the 3×3 kernel. This outcome is likely due to the larger receptive field of the 5×5 kernel. With a larger window, the model can capture broader spatial patterns and more complex geometric details from the SDF input in a single step, useful for the prediction of the lift coefficient.

The convergence history, shown in Figure 14, further supports this conclusion. Models using the 5×5 kernel consistently achieved lower validation MSEs throughout the training process, indicating that they were able to learn more effectively from the data.

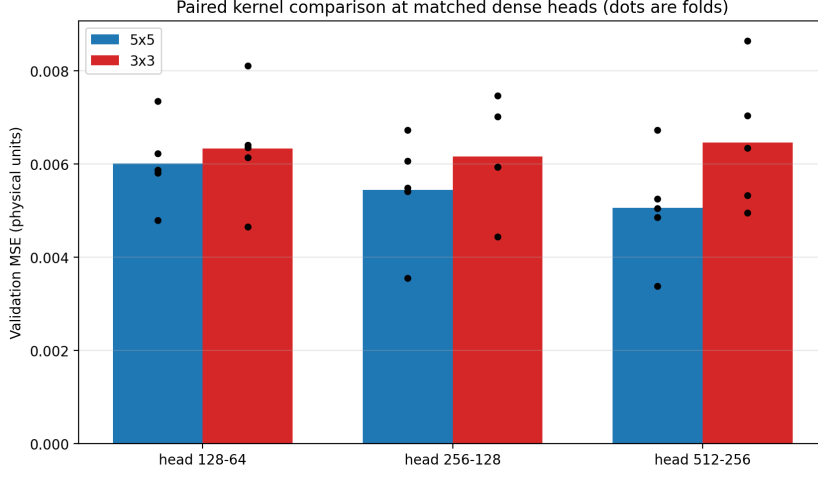


Figure 13: Paired kernel comparison at three matched dense heads. Bars show the sample-weighted mean physical validation MSE and dots show the five per fold values from `aggregated.csv`. All three pairs favour 5×5 with stride five over 3×3 with stride three at zero padding.

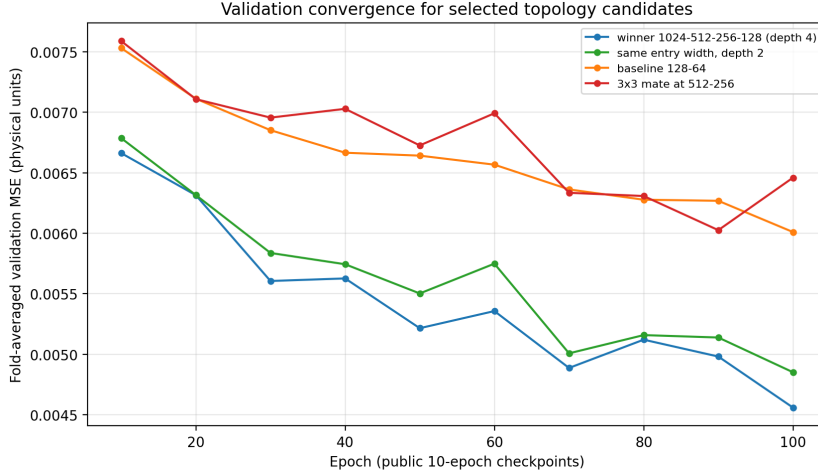


Figure 14: Fold averaged physical validation MSE at public checkpoints every ten epochs. Curves are shown for the winner `conv5x5-dense-1024-512-256-128`, the depth two candidate `conv5x5-dense-1024-512`, the baseline `conv5x5-dense-128-64`, and the 3×3 mate `conv3x3-dense-512-256`. The training objective is omitted because it uses different standardized units.

However, it is important to acknowledge several limitations inherent to this specific experimental setup:

- as already said in Section 4.1, this grid search was executed without recording detailed per-epoch diagnostics. While all runs completed successfully without signs of divergence, this prevents a deeper analysis of phenomena like vanishing or exploding gradients for these specific candidates;
- each candidate was trained only once per fold with a fixed random seed. Consequently, architecture difference cannot be separated from training seed noise;
- as established in our methodology (Section 4.1), the training and validation sets are split at the sample level, meaning that data from the same airfoil can appear in both sets. This setup measures interpolation within the dataset rather than generalization to unseen airfoil families;
- the grid search was limited to a specific range of kernel sizes, head widths, and depths. While the selected candidate performed best within this grid, it is possible that other configurations outside this range could yield even better results.

The cross-validation winner is `conv5x5-dense-1024-512-256-128`, with physical validation MSE 0.004560 ± 0.000946 and untouched test MSE 0.002805 after refitting on the complete training set.

4.3. Dropout tuning

Dropout is a regularization technique designed to prevent overfitting in large networks by randomly deactivating neurons during training, which discourages complex co-adaptations between them, as shown by Srivastava et al. (2014). Typically, it is most effective when there is a clear gap between training and validation performance. The preceding regularization experiment, however, showed only a small training to validation gap relative to the fold variability. The dropout experiment therefore tests a specific question: can stochastic masking improve validation accuracy when the available evidence does not indicate a large deterministic overfitting gap?

During training, a random subset of activations in a layer is set to zero. The remaining are scaled up by the reciprocal of the survival probability to maintain the expected magnitude of the layer’s output. This process, which effectively trains a large ensemble of thinned subnetworks, can be expressed mathematically as:

$$\tilde{z}_j = \frac{m_j}{1-p} z_j, \quad m_j \sim \text{Bernoulli}(1-p) \quad (48)$$

where p is the probability of dropping an element and $1-p$ is the survival probability. The same scaled mask is used during backpropagation, so that:

$$\frac{\partial L}{\partial z_j} = \frac{m_j}{1-p} \frac{\partial L}{\partial \tilde{z}_j} \quad (49)$$

At inference, the mask is disabled and the layer is the identity, so no additional rescaling is needed.

In the implementation, a **DropoutLayer** is inserted after every hidden **LeakyReLU** layer within the dense regression head. It is not applied to the final scalar output. The tested grid was $p \in \{0, 0.1, 0.2, 0.3, 0.5\}$, including the baseline case of no dropout.

To isolate the effect of dropout, all other hyperparameters were held constant, retaining the winning topology from Section 4.2. The experiment strictly followed the established cross-validation protocol detailed in Section 4.1, using the Adam optimizer (learning rate of 10^{-5}), a SIMM physics weight of 0.25, batch size 64 and a 100-epoch budget across identical data folds.

For rigorous reproducibility, the dropout masks are generated deterministically from a combination of the fold seed, epoch, sample row, and feature index. This design ensures that the stochastic mask remains identical regardless of the number of parallel MPI ranks employed. Crucially, the baseline candidate ($p = 0.0$) retains the dropout layer as an identity mapping rather than omitting it. This guarantees that all candidates share identical parameter initializations and random number sequences. Consequently, the resulting baseline performance is strictly paired within this sweep and should not be compared directly against standalone runs configured without the identity layer.

Candidate selection used the sample-weighted physical-unit validation MSE returned by **CrossValidator**. The official aggregate file is **cv_summary.csv**; the paired differences, final training MSE, and training to validation gaps are read from **sweep_dropout.csv**. The two files differ slightly in their arithmetic means because the official score weights the 342 and 343 sample validation folds, while the sweep CSV is also used to preserve the fold-by-fold paired inputs. Table 6 reports the complete grid. The last three columns are descriptive quantities and do not replace the physical validation MSE as the selection metric.

p	Mean val. MSE	Fold std.	Paired Δ	Mean train MSE	Val. – train
0.0	0.005834	0.001007	0.000000	0.005421	+0.000413
0.1	0.006686	0.001060	+0.000852	0.006390	+0.000296
0.2	0.007630	0.000861	+0.001796	0.007236	+0.000395
0.3	0.008467	0.001141	+0.002632	0.008098	+0.000369
0.5	0.011764	0.002603	+0.005930	0.011354	+0.000411

Table 6: Complete recorded dropout sweep. Mean validation MSE and fold standard deviation are the sample-weighted values from **cv_summary.csv**. Paired differences, training MSE, and validation minus training gaps are calculated from **sweep_dropout.csv**; a positive paired difference means that the rate is worse than the $p = 0$ reference.

The lowest validation error was achieved with dropout disabled ($p = 0$), yielding a mean MSE of $0.005833893 \pm 0.001006787$. As shown in Figure 15, performance systematically degraded as the dropout rate increased. Even

a small rate of $p = 0.1$ increased the error by nearly 15%, while the strongest rate of $p = 0.5$ more than doubled the validation MSE.

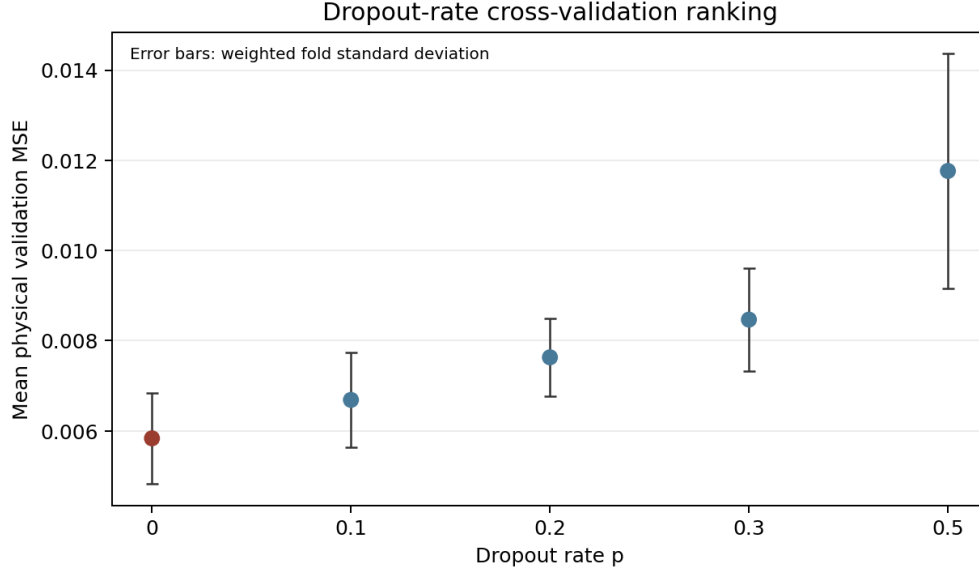


Figure 15: Dropout-rate ranking from the official `cv_summary.csv`. Points are sample-weighted mean physical validation MSE and error bars are the weighted standard deviations over the five folds. The $p = 0$ reference is highlighted in red.

This conclusion was remarkably consistent across all data splits. As visualized in the paired difference plot (Figure 16), no non-zero dropout rate improved performance in any of the five cross-validation folds.

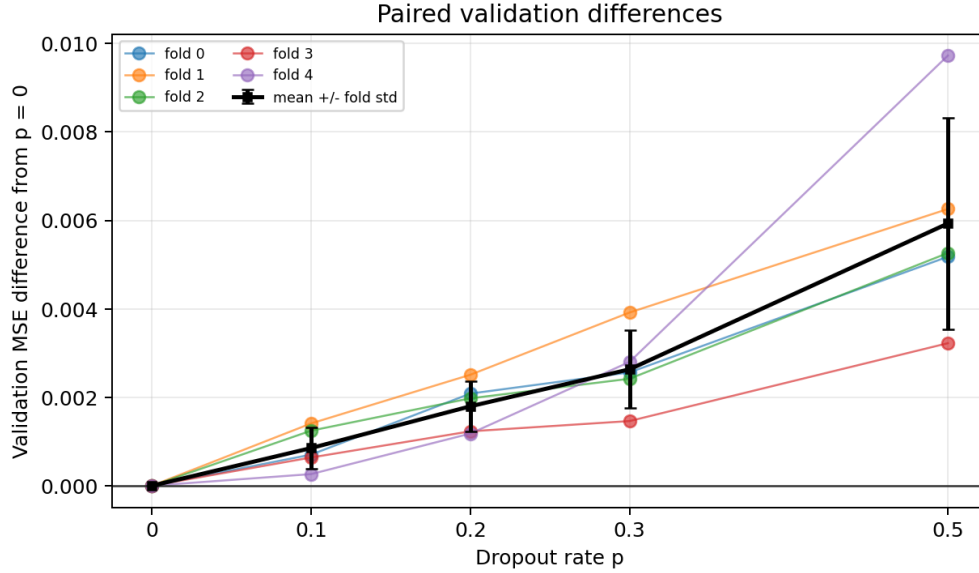


Figure 16: Paired physical validation MSE differences relative to the $p = 0$ reference, computed from `sweep_dropout.csv`. Coloured curves show the five zero-based folds, while black markers and error bars show the mean paired difference and its sample standard deviation. Positive values indicate deterioration relative to the reference.

This outcome directly answers the experimental question posed in our motivation. For the baseline model ($p = 0.0$), the gap between training and validation error was already very small (+0.000413), indicating no significant overfitting problem. While a small dropout rate ($p = 0.1$) did slightly reduce this gap (to +0.000296), this minor improvement came at the cost of a much larger increase in the overall validation error.

This pattern strongly suggests that, for this well-specified architecture and dataset, dropout did not act as a useful regularizer. Instead, it functioned primarily as optimization noise, hindering the model’s ability to converge to an accurate solution within the fixed 100-epoch training budget. Therefore, the evidence from this experiment supports disabling dropout entirely for the production model.

A closer look at the training dynamics, presented in Figure 17, reinforces this conclusion. The training history reveals that all candidates converge stably, but to different levels: the final mean is approximately 0.0123 for $p = 0$ and 0.1142 for $p = 0.5$. These objective values are in standardized SIMM units and are not numerical alternatives to the physical validation MSE.

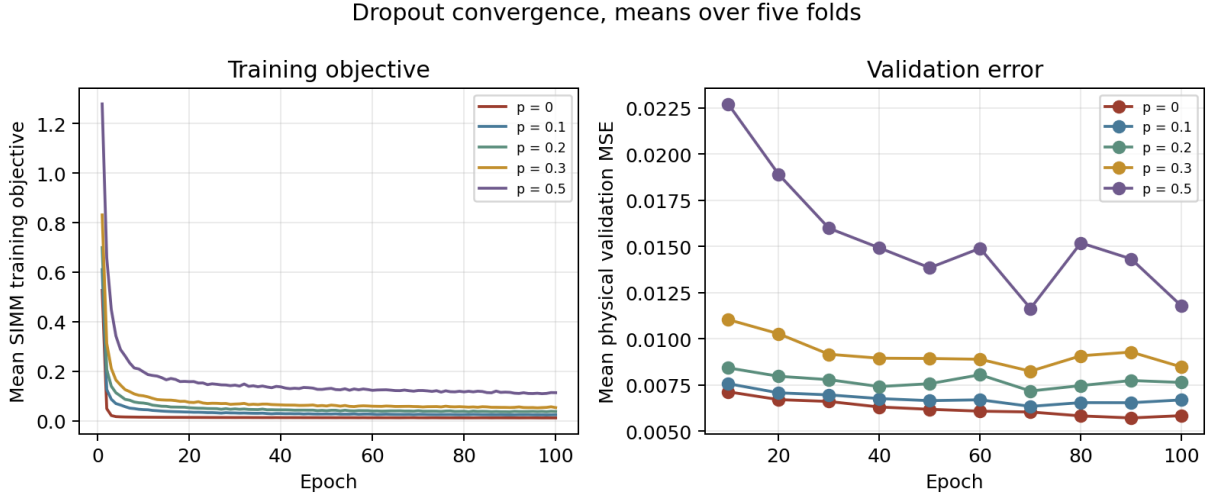


Figure 17: Means over the five recorded folds for the training SIMM objective and the physical validation MSE. The training objective is recorded at every epoch, whereas the validation values in the committed diagnostic files occur at ten-epoch checkpoints. The two panels use separate units, and the curves describe the recorded dropout sweep.

Next, as seen in the right panel, the validation error for the baseline $p = 0$ model consistently remains the lowest throughout the 100 epochs (from 0.0071253 at epoch 10 to 0.0058346 at epoch 100). Conversely, increasing the dropout rate raises the final error value, with the $p = 0.5$ candidate converging to a solution approximately twice as poor as the baseline (the MSE during validation ranges from 0.0227140 at epoch 10 to 0.0117650 at epoch 100). The curves contain small fold-averaged fluctuations, particularly for the nonzero rates, but none shows a late unbounded increase.

To understand why dropout degraded performance, we analyzed the detailed training diagnostics, summarized in Figure 18. These diagnostics confirm that the poorer performance was not caused by numerical instability but by the disruptive effect of dropout on the optimization process.

We can see that no run was numerically unstable. Looking at the bottom right plot, while stronger dropout rates led to larger initial gradient transients (peaking around 47.444652 for $p = 0.5$, compared to 37.841272 for $p = 0.0$), these were controlled, early-epoch events. All gradients remained finite and decreased substantially as training progressed, confirming that the high validation error was not the result of a failed or divergent run.

Looking at the other diagnostic panels, late in training, both the average gradient norm and the parameter update ratio systematically increase with the dropout rate. This indicates that dropout forces the optimizer to take larger, more aggressive steps even late in training. However, the post-activation variance shows the opposite trend: at epoch 100, the signal variance decreases as the dropout rate increases. This suggests that while dropout forces larger parameter updates, it may also lead to weaker overall signal propagation through the network.

Taken together, these diagnostics paint a clear picture. Dropout did not act as a helpful regularizer; instead, it introduced optimization noise that led to larger, less efficient parameter updates while simultaneously attenuating the network’s internal signal. This combination fully explains the degraded validation performance.

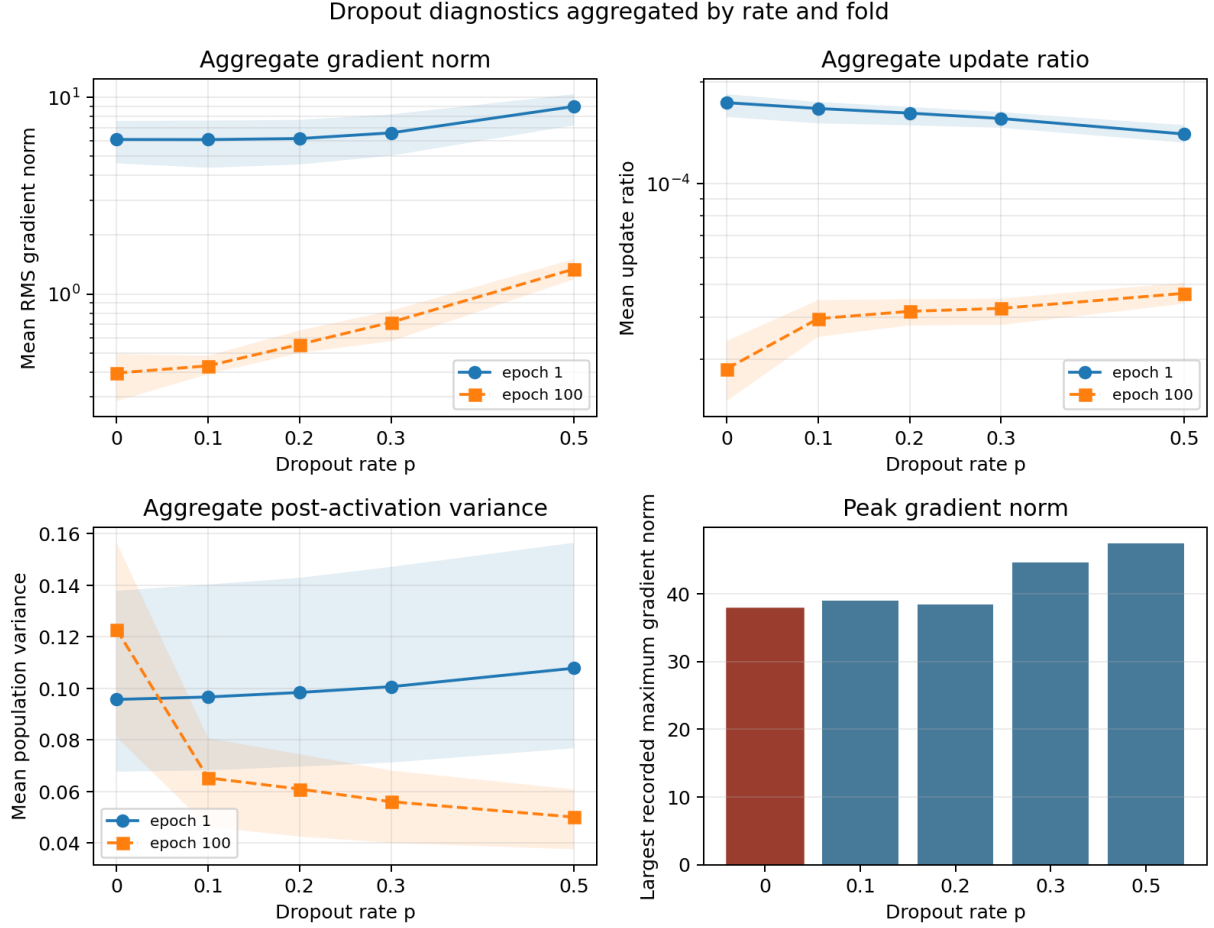


Figure 18: Diagnostics for the recorded dropout sweep. The aggregate gradient norm, actual update ratio, and post-activation variance panels show means with minimum to maximum bands over five folds at epochs 1 and 100; the horizontal axis is dropout rate and the first two vertical axes are logarithmic. The final panel shows the largest recorded `maximum_norm` with `parameter_scope=all` over all folds and epochs for each dropout rate. Diagnostic rows are aggregated and individual layers are not distinguished.

Therefore, the cross-validation winner is unequivocally `dropout_rate = 0.0`, with a final validation MSE of $0.005833893 \pm 0.001006787$. This choice aligns with the default CNN production configuration, which also disables dropout. The evidence from this experiment strongly supports retaining $p = 0$ for the final model, as every non-zero rate worsened the paired validation result without providing a meaningful reduction in the already-small overfitting gap.

4.4. Optimizer tuning

The optimizer determines how the gradient produced by the SIMM loss is converted into a change of the trainable parameters. For this CNN architecture, this choice is particularly critical: the convolutional feature extractor and the dense regression head differ substantially in parameter dimensions and gradient dynamics. Consequently, a fixed learning rate may lead to different effective progress depending on whether the method retains a direction from previous batches or scales each coordinate using past squared gradients. We decided to test different methods, including vanilla SGD, SGD with momentum, AdaGrad²², RMSprop, and Adam²³, to evaluate their impact on the physical validation error within the same finite training budget. The result is a comparison at one learning rate and is not a claim that an optimizer is universally preferable.

Before any parameter updates are made, we first apply gradient clipping to prevent numerical instability. In distributed execution, backpropagated gradients are first averaged across all MPI processes via collective reduction (`MPI_Allreduce`) to ensure mathematical equivalence regardless of the number of ranks. During

²²That formalizes cumulative coordinate adaptation, as explained by Duchi et al. (2011).

²³That combines first and second moment estimates with bias correction, as seen by Kingma and Ba (2015).

early epochs, gradients can become excessively large, especially across wide dense layers. To mitigate this, every individual component of the aggregated gradient vector is constrained to a fixed range. In our implementation, we apply coordinate-wise (component-wise) clipping to each gradient entry $g_{t,j}$ within the interval $[-1, 1]$:

$$\tilde{g}_{t,j} = \text{clip}(g_{t,j}, -1, 1) \quad (50)$$

This coordinate-wise clipping ensures that no single gradient entry can become disproportionately large, without altering the relative direction of smaller coordinates as a global Euclidean norm clipping ($\tilde{\mathbf{g}} = \mathbf{g} \cdot \min(1, \tau/\|\mathbf{g}\|_2)$) would. This clipped gradient, $\tilde{g}_{t,j}$, is then handed to the optimizer. No other modifications, such as explicit L1/L2 penalties, are applied to the gradients in this baseline experiment.

The five optimizers evaluated in this experiment are defined as follows:

- for Vanilla SGD, the update is

$$\theta_{t+1,j} = \theta_{t,j} - \eta \tilde{g}_{t,j}$$

- for SGD with momentum, the update is defined as

$$u_{t,j} = \mu u_{t-1,j} + \tilde{g}_{t,j}, \quad \theta_{t+1,j} = \theta_{t,j} - \eta u_{t,j}, \quad \mu = 0.9$$

- for AdaGrad, that stores the cumulative squared gradient, the update is defined as

$$G_{t,j} = G_{t-1,j} + \tilde{g}_{t,j}^2, \quad \theta_{t+1,j} = \theta_{t,j} - \frac{\eta \tilde{g}_{t,j}}{\sqrt{G_{t,j} + \varepsilon}}, \quad \varepsilon = 10^{-8}$$

- for RMSprop, that replaces the cumulative sum with an exponential average, the update is defined as

$$s_{t,j} = \rho s_{t-1,j} + (1 - \rho) \tilde{g}_{t,j}^2, \quad \theta_{t+1,j} = \theta_{t,j} - \frac{\eta \tilde{g}_{t,j}}{\sqrt{s_{t,j} + \varepsilon}}, \quad \rho = 0.9, \quad \varepsilon = 10^{-8}$$

- for Adam, that stores both a first moment $m_{t,j}$ and a second moment $v_{t,j}$, the update is defined as

$$\begin{aligned} m_{t,j} &= \beta_1 m_{t-1,j} + (1 - \beta_1) \tilde{g}_{t,j}, & v_{t,j} &= \beta_2 v_{t-1,j} + (1 - \beta_2) \tilde{g}_{t,j}^2, \\ \hat{m}_{t,j} &= \frac{m_{t,j}}{1 - \beta_1^t}, & \hat{v}_{t,j} &= \frac{v_{t,j}}{1 - \beta_2^t}, \\ \theta_{t+1,j} &= \theta_{t,j} - \eta \frac{\hat{m}_{t,j}}{\sqrt{\hat{v}_{t,j} + \varepsilon}}, & \beta_1 &= 0.9, \quad \beta_2 = 0.999, \quad \varepsilon = 10^{-8} \end{aligned}$$

In all five algorithms, the baseline learning rate is set to $\eta = 10^{-5}$ without an attached decay scheduler.

In our C++ framework, each candidate optimizer is encapsulated as an `OptimizerRecipe` (e.g., `Recipes::adam`, `Recipes::rmsprop`). When `CrossValidator` evaluates a fold, a fresh optimizer instance is constructed. For stateful optimizers, parameter state vectors (m_t, v_t, s_t, G_t) are zero-initialized and bound to weight arrays via an internal pointer hash map (`std::unordered_map<float*, State>`), ensuring strict fold isolation with $\mathcal{O}(1)$ lookup complexity and zero cross-fold state leakage. Although Adam triples the parameter memory requirement (storing two additional 32-bit floats per parameter, approximately 96 MB for our $\approx 8.065 \times 10^6$ parameter network), this overhead is easily accommodated by system memory.

The optimizer comparison was performed exactly in the same way that the dropout one was done, using the winning topology from Section 4.2 and the same cross-validation protocol. All other hyperparameters were held constant: SIMM physics weight of 0.25, batch size of 64, and a 100-epoch training budget.

The winning optimizer was selected based on the lowest physical-unit validation MSE (Equation (45)). The final scores reported in Table 7 are the sample-weighted averages across the five folds, ensuring that folds with slightly different numbers of validation samples are handled correctly. The results, as shown in Figure 19, demonstrate a clear performance hierarchy among the tested optimizers.

Adam was the cross-validation winner with $0.004426734 \pm 0.000793966$ physical validation MSE. It was the lowest-scoring candidate in every one of the five cross-validation folds. This represents a significant 15% error reduction compared to its closest competitor, RMSprop (MSE of 0.005216), and a 38% reduction compared to SGD with momentum. The other optimizers ranked as expected, with AdaGrad and momentum-based SGD delivering intermediate performance. Vanilla SGD, without any adaptive or momentum mechanisms, struggled significantly at this low learning rate, finishing with a validation error nearly four times higher than Adam's

Optimizer	Mean validation MSE	Fold standard deviation	Minimum fold	Maximum fold
Adam	0.004427	0.000794	0.002879	0.005044
RMSprop	0.005216	0.000585	0.004518	0.006173
AdaGrad	0.006184	0.001108	0.004239	0.007439
SGD + momentum	0.007150	0.001264	0.004791	0.008473
SGD	0.017924	0.005568	0.013063	0.028647

Table 7: Complete optimizer comparison at learning rate 10^{-5} . The means and fold deviations are sample-weighted physical validation MSE values, with the deviation defined as the weighted population standard deviation over the five folds. The minimum and maximum columns show the corresponding unweighted extrema of the fold scores.

(0.017924). While all runs completed successfully without diverging, the performance of plain SGD was substantially less accurate and consistent across the folds.

The physical training MSE recorded after the final fold epoch was also lower for the better validation candidates, but it must be interpreted separately from the SIMM objective: the methods that achieved better validation scores also reached a lower physical training MSE within the fixed 100-epoch training budget.

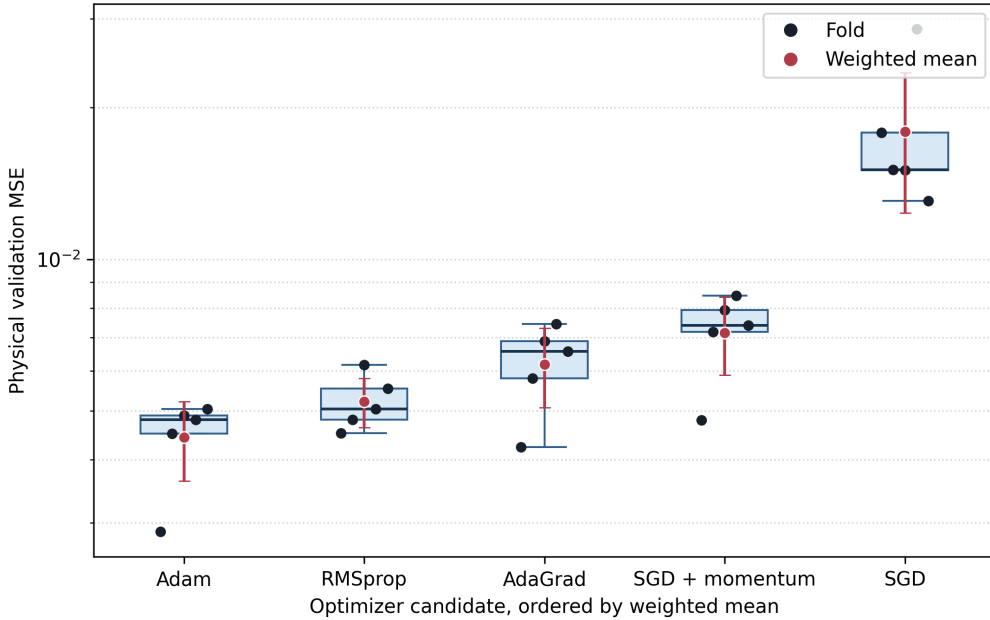


Figure 19: Optimizer ranking from the five-fold comparison at learning rate 10^{-5} . Dark points are individual physical validation MSE values, boxes summarize the five folds, and red points with bars show the sample-weighted mean and weighted population standard deviation. The vertical axis is logarithmic.

The training histories, visualized in Figure 20, provide a clear explanation for the final performance ranking by showing how each optimizer converged over the 100-epoch budget.

The superiority of the adaptive coordinate methods is evident from the very beginning. Adam and RMSprop immediately establish a lead, starting with and maintaining a significantly lower validation error than the other candidates. However, the plot also highlights key algorithmic distinctions:

- AdaGrad accumulates the sum of all past squared gradients $G_{t,j} = \sum_{\tau=1}^t \tilde{g}_{\tau,j}^2$. Because this denominator grows monotonically, the effective learning rate diminishes as $\mathcal{O}(1/\sqrt{t})$. Consequently, AdaGrad prematurely stalls in later epochs and cannot escape shallow local sub-optima;
- RMSprop resolves AdaGrad’s diminishing step size by replacing the monotonic sum with an exponentially decaying average ($\rho = 0.9$). However, it lacks first-moment momentum and initialization bias correction. Without momentum to filter high-frequency gradient jitter, RMSprop exhibits notable instability late in training, with its error spiking noticeably around epoch 90 before recovering;
- Plain SGD and SGD with momentum scale parameter updates directly by the unnormalized gradient

magnitude ($\|\Delta\theta\| \approx \eta\|\tilde{g}\|$). As backpropagated gradients naturally decay near flatter loss regions, the effective step size becomes vanishingly small ($\approx 10^{-7}$ – 10^{-8} for $\eta = 10^{-5}$), preventing SGD from covering sufficient distance across the loss landscape within 100 epochs. In contrast, Adam’s coordinate normalization by $\sqrt{\hat{v}_{t,j}}$ maintains active effective steps of scale η across all coordinates regardless of gradient magnitude decay.

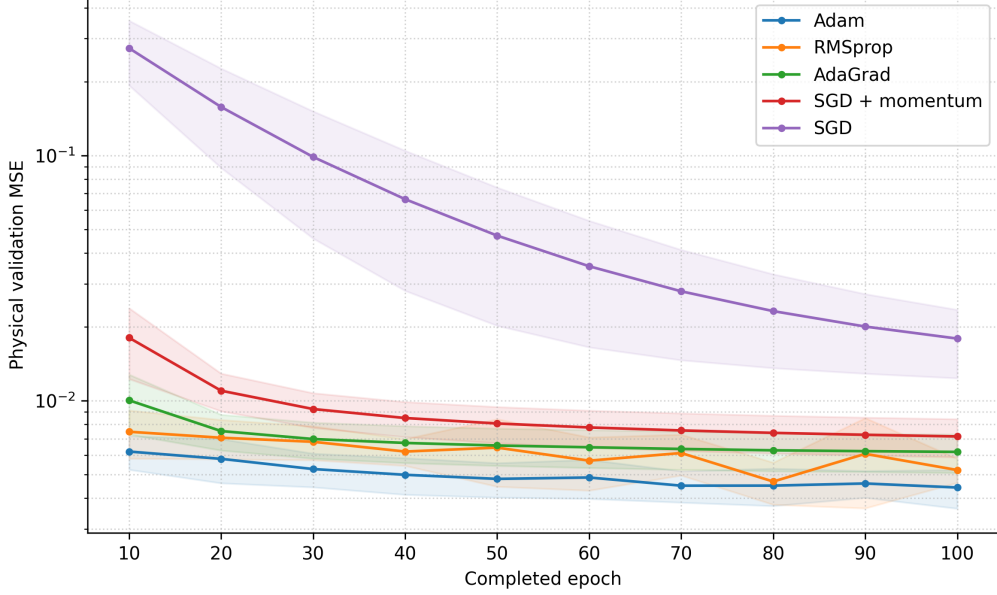


Figure 20: Physical validation MSE at the ten-epoch history checkpoints for the five optimizer candidates. Lines and shaded regions are the sample-weighted means and weighted population standard deviations over the five folds. The vertical axis is logarithmic; the SIMM training objective and the test set are not shown.

Overall, the convergence plot visually confirms the final rankings. It demonstrates that Adam’s victory is due not only to a lower final error but also to its combination of fast, efficient, and stable convergence throughout the entire training process.

The diagnostic plots for the winning Adam optimizer reveal why it was so much more effective and stable than the alternatives. The key architectural challenge is visualized in Figure 21, which plots the gradient magnitudes for each trainable layer at different points in training.

The plot shows a massive initial gradient peak at layer index 4, which corresponds to the first dense layer (`DenseLayer(7202, 1024)`). This single layer contains 7,375,872 parameters (weights and biases), accounting for over 91.4% of the total 8,065,025 parameters in the network. At epoch 1, the gradients in this layer are an order of magnitude larger than in any other layer. This transient gradient concentration occurs when the spatial features from the convolutional trunk ($30 \times 30 \times 8 = 7200$ features plus two flow scalars) are projected into the wide dense head. Adam is uniquely suited to handle this dimension bottleneck: coordinate-adaptive scaling ($\sqrt{\hat{v}_{t,j}} + \epsilon$) normalizes the effective step size per coordinate, preventing the massive dense layer from overpowering the convolutional kernels or destabilizing the network. In contrast, unscaled methods like SGD apply the same uniform scaling to both small convolutional kernels and massive dense layers, leading to severe under-stepping in some layers and oscillation in others.

The plot also shows that by epoch 10, Adam has successfully navigated this transient phase. The gradients across all layers have settled into a much more stable and well-behaved pattern, allowing for smooth convergence for the rest of training. This successful management of the network’s challenging gradient landscape is a primary reason for Adam’s superior efficiency and stability.

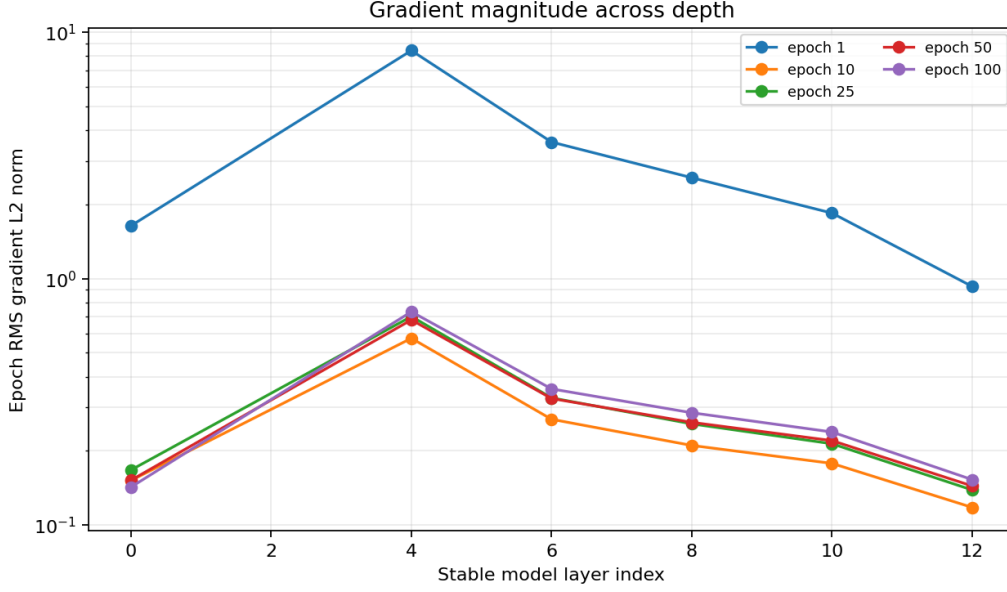


Figure 21: The RMS gradient norm for each trainable layer, plotted at different training epochs. The vertical axis is logarithmic. The plot reveals a massive gradient peak in the first dense layer (index 4) at the start of training (epoch 1). This initial transient is rapidly stabilized by the Adam optimizer, with gradient magnitudes becoming substantially smaller and more uniform across the network’s depth by epoch 10.

The cross-validation experiment identified Adam as the decisively superior optimizer for this task, outperforming all competitors in every fold. The diagnostic analysis confirms that its stability and efficiency stem from its ability to handle the large and varied gradient magnitudes that arise naturally from the network’s convolutional-to-dense architecture. The final refit of this experimental winner on the full training set yielded a test MSE of 0.003655.

4.5. Physics-weight tuning

The SIMM term was tuned because it is the only part of the objective that encodes an aerodynamic relation not learned directly from the CFD targets. A positive physics weight can improve the low-angle regime if the thin-airfoil relation supplies useful information, but it can also pull the prediction away from the simulation data when the relation is only approximate. The experiment therefore asks whether the prior improves the physical-unit validation error, rather than assuming that a larger physics contribution is beneficial. The search was performed over the training dataset only and the untouched test dataset was not used to choose the weight.

The implementation evaluates the loss in the fold-specific standardized target space. Let $C_{L,i}$ be the physical target, let μ_y and s_y be the target mean and standard deviation fitted on the training part of a fold, and let a_i be the angle of attack in radians. The standardized target and prediction are $y_i = (C_{L,i} - \mu_y)/s_y$ and $\hat{y}_i$, while the standardized physical target is:

$$y_i^{\text{phys}} = \frac{2\pi a_i - \mu_y}{s_y} \quad (51)$$

The mask is evaluated on the unstandardized angle,

$$M_i = \begin{cases} 1, & |a_i| \leq 0.174533 \text{ rad} \\ 0, & |a_i| > 0.174533 \text{ rad} \end{cases} \quad (52)$$

which corresponds to $|\alpha| \leq 10^\circ$. The actual batch objective is:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 + \lambda \frac{1}{N} \sum_{i=1}^N M_i (\hat{y}_i - y_i^{\text{phys}})^2, \quad (53)$$

where the physics sum is divided by the full batch size, not by the number of masked samples. The prediction gradient used by backpropagation is consequently

$$\frac{\partial \mathcal{L}}{\partial \hat{y}_i} = \frac{2}{N} \left[(\hat{y}_i - y_i) + \lambda M_i (\hat{y}_i - y_i^{\text{phys}}) \right]. \quad (54)$$

The validation score is converted back to physical units by multiplying the standardized prediction residual by s_y before squaring. Therefore, the SIMM objective and the selection metric have different units and cannot be ranked by their numerical magnitudes.

To determine the optimal strength of the physical term, we performed a sweep over seven different values for the physics weight $\lambda \in \{0, 0.05, 0.1, 0.25, 0.5, 1, 2\}$. The baseline case, $\lambda = 0.0$, corresponds to a purely data-driven model without any physics-informed loss.

As in the other tuning, all other hyperparameters were held constant, including the winning topology from Section 4.2, the Adam optimizer with a learning rate of 10^{-5} , a batch size of 64, and a 100-epoch training budget. In the codebase, the physics loss strength λ is configured through the `LossConfig` structure and `Trainer` computes the masked analytical thin-airfoil penalty on the standardized predictions at each forward pass. The cross-validation protocol was identical to the previous experiments, ensuring that the only variable was the physics weight λ .

The results of the sweep, summarized in Table 8, lead to a clear conclusion: the physics-informed loss term did not improve the model’s predictive accuracy.

λ	Mean validation MSE	Fold standard deviation	Δ_λ	Improved folds
0.00	0.005600634	0.001099091	0	–
0.05	0.005637647	0.001136246	+0.000037038	1/5
0.10	0.005629100	0.001116111	+0.000028457	1/5
0.25	0.005706661	0.001195944	+0.000106089	1/5
0.50	0.006025918	0.001193569	+0.000425322	0/5
1.00	0.006581512	0.001350790	+0.000981093	0/5
2.00	0.007406998	0.001519330	+0.001806727	0/5

Table 8: Recorded physics-weight sweep. The mean and deviation are the sample-weighted physical validation values in `cv_summary.csv`. Δ_λ is the paired candidate minus reference validation MSE, so positive values are worse than $\lambda = 0$.

The best performance was achieved by the purely data-driven model ($\lambda = 0$), which recorded a mean validation MSE of $0.005600634 \pm 0.001099091$. As shown in the performance plot (Figure 22, left panel), every non-zero value of λ resulted in a higher validation error. While the degradation was minimal for small weights like $\lambda = 0.05$, performance grew progressively worse as the physics term was given more influence.

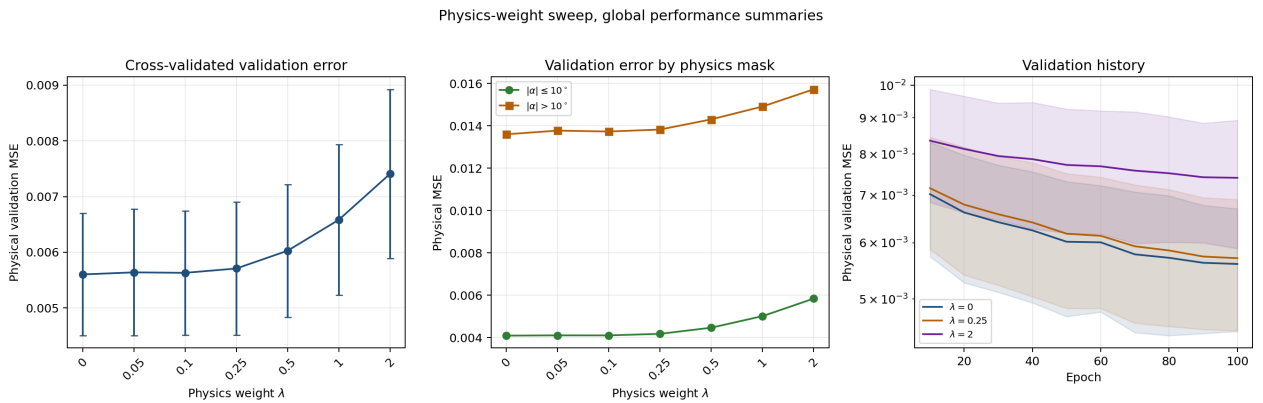


Figure 22: Global summaries of the recorded physics-weight sweep. The first panel shows sample-weighted mean physical validation MSE with fold deviations, the second separates the count-weighted low-angle and high-angle validation errors, and the third shows physical validation histories for $\lambda = 0, 0.25$, and 2 with fold-spread bands. The training SIMM objective is not placed on the validation-MSE axis.

To investigate this further, we tested whether the physics term provided a localized benefit in the low-angle-of-attack regime where its underlying theory is valid. We split the validation data into low-angle ($|\alpha| \leq 10^\circ$) and high-angle ($|\alpha| > 10^\circ$) groups, respectively with a count of 1442 and 271 samples across the five folds. The results of this split are shown in the second panel of Figure 22 (middle panel), are decisive. For $\lambda = 0$,

the count-weighted MSE was 0.004096627 in the low-angle group and 0.013603506 in the high-angle group. At $\lambda = 2$, the corresponding values were 0.005843330 and 0.015727326. Both groups therefore worsened as the prior became strong; the effect was not a selective improvement in the masked regime.

The validation history is consistent with the final ranking. The fold-averaged validation MSE for $\lambda = 0$ decreased from approximately 0.007025 at epoch 10 to 0.005602 at epoch 100, whereas the corresponding values for $\lambda = 2$ were 0.008356 and 0.007407. Every candidate completed 100 epochs with finite values, converging to a stable solution. The training objective started and ended higher as λ increased, from approximately 0.599 to 0.0104 for $\lambda = 0$ and from 1.393 to 0.0168 for $\lambda = 2$ when averaged over folds. These are standardized SIMM objectives, so their difference from the physical validation MSE is not a generalization gap.

The diagnostic plots in Figure 23 show that the optimizer successfully managed the training process for all candidates. As seen in the left panel, strengthening the physics prior led to a more pronounced initial gradient transient. At epoch 1, the aggregate RMS gradient norm for the $\lambda = 2$ model (mean of 45.36) was more than double that of the purely data-driven model ($\lambda = 0$, mean of 18.76). However, this was a controlled, early-epoch event. The optimizer rapidly stabilized all runs, and by epoch 100, the aggregate gradient norms had settled to a much smaller and more comparable level (0.842 for $\lambda = 0$ vs. 1.071 for $\lambda = 2$). This pattern confirms that the training was numerically stable and did not suffer from late-stage exploding gradients.

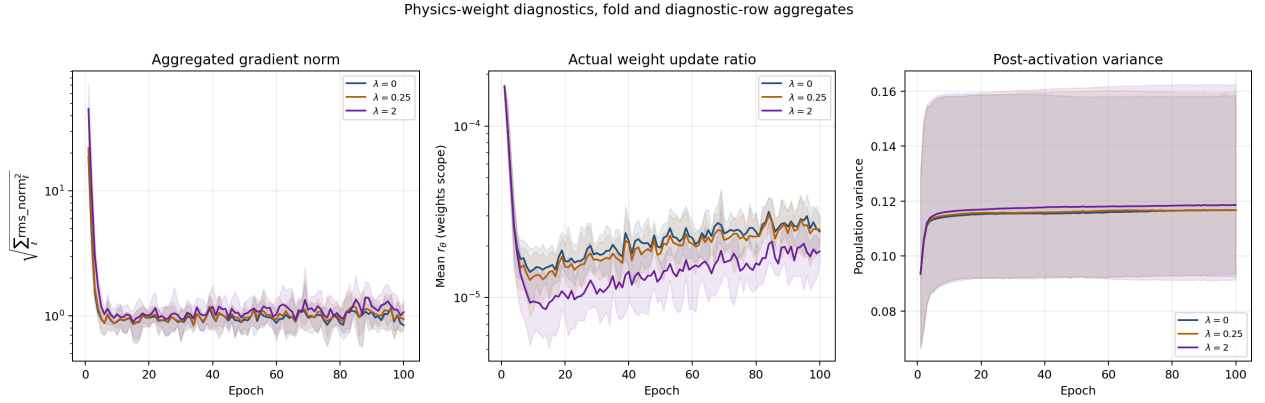


Figure 23: Layer-agnostic diagnostics for the recorded physics-weight sweep. Curves are means over folds and comparable diagnostic rows, with minimum to maximum fold bands. The gradient panel uses the square root of the sum of squared per-layer RMS norms for the all-parameter scope, the update panel uses the actual weights-scope update ratio, and the variance panel uses post-activation population variance. No layer index is shown.

The diagnostics also reveal that the physics term had a minimal effect on the network's internal dynamics:

- the middle panel shows that the "actual update ratio" for the model weights follows a very similar trajectory for all values of λ , decreasing steadily as the models converge. For the baseline model, this ratio decreased from an initial value of approximately 1.69×10^{-4} to 2.43×10^{-5} by the final epoch, indicating that the parameter updates became smaller and more refined over time;
- the right panel demonstrates that the post-activation variance was almost completely unaffected by the physics term. For both $\lambda = 0$ and $\lambda = 2$, the variance remained around 0.1092 after an initial transient. This is a crucial finding, as it rules out "vanishing" or "exploding" activations as a cause for the degraded performance.

The cross-validation experiment identifies the purely data-driven model with $\lambda = 0$ as the unconstrained winner, achieving the lowest validation MSE of $0.005600634 \pm 0.001099091$. The diagnostic analysis confirms that the physics-informed loss term did not cause numerical instability or signal attenuation, but rather introduced a slight optimization compromise against the high-fidelity CFD ground truth.

Nevertheless, the final production model adopts a physics weight of $\lambda = 0.10$. This decision is motivated by the project requirement of developing a Physics-Informed Neural Network surrogate model, which necessitates embedding theoretical domain knowledge ($\lambda > 0$). Under this operational constraint, $\lambda = 0.10$ emerges as the optimal admissible choice: its validation error increase relative to the unconstrained baseline is merely +0.000028, a magnitude more than an order of magnitude smaller than the cross-validation fold standard deviation (≈ 0.0011), rendering its empirical impact statistically indistinguishable from zero. Furthermore,

retaining this mild physical weight provides valuable structural grounding to classical Thin-Airfoil Theory ($C_L \approx 2\pi\alpha$) in the linear pre-stall regime ($|\alpha| \leq 10^\circ$), while completely avoiding the severe gradient inflation and loss landscape distortion observed at larger values such as $\lambda = 2.0$. Consequently, $\lambda = 0.10$ represents the most robust and sound compromise between physical consistency and empirical predictive accuracy.

4.6. Weight regularization

Weight regularization was tested because the dense head contains millions of trainable weights, so reducing unnecessary parameter magnitude could in principle improve validation performance. As we can see in Tibshirani (1996) and Hoerl and Kennard (1970), L1 regularization is associated with sparsity and L2 regularization with continuous shrinkage. The central question was whether these penalties could provide a benefit, or if they would instead hinder optimization or introduce unnecessary bias, particularly if the unregularized model was not significantly overfitting.

For every convolutional and dense weight tensor W_ℓ , the implementation defines the two penalties as:

$$R_{L1} = \lambda_1 \sum_{\ell} \sum_{w \in W_\ell} |w|, \quad R_{L2} = \lambda_2 \sum_{\ell} \sum_{w \in W_\ell} w^2. \quad (55)$$

The regularization is applied to the weights only, without considering the biases. The coefficients λ_1 and λ_2 are passed via `LossConfig`, allowing `Trainer` to compute the regularization gradients directly on the weights after distributed MPI synchronization and prior to gradient clipping. The corresponding gradient term, $\lambda_1 \text{sign}(w) + 2\lambda_2 w$, was added to the main loss gradient before the Adam update step. This implementation is distinct from the weight decay mechanism built into some optimizers.

To isolate the effects of regularization, we performed two independent, one-dimensional sweeps for the L1 and L2 coefficients (λ_1 and λ_2), starting from a baseline of zero for both:

- for L1, we evaluated eight values from 6.75×10^{-7} to 6.75×10^{-4} , with zero as the reference;
- for L2, we evaluated eight values from 10^{-4} to 10^{-1} , with zero as the reference.

All other hyperparameters were held constant. The full comparison followed the rigorous cross-validation framework detailed in Section 4.1.

The results from both the L1 and L2 sweeps, presented in Tables 9 and 10, point to the same clear conclusion: weight regularization did not improve model performance.

λ_1	Mean validation MSE	Fold standard deviation	Δ_{λ_1}	Improved folds
0	0.004419	0.000794	0	–
6.75×10^{-7}	0.004424	0.000788	+0.0000053	2/5
2.13×10^{-6}	0.004436	0.000800	+0.0000178	2/5
6.75×10^{-6}	0.004478	0.000811	+0.0000588	0/5
2.13×10^{-5}	0.004540	0.000832	+0.0001209	0/5
6.75×10^{-5}	0.004684	0.000902	+0.0002656	0/5
2.13×10^{-4}	0.005100	0.000950	+0.0006810	0/5
6.75×10^{-4}	0.006462	0.001189	+0.0020431	0/5

Table 9: Documented L1 reference sweep on the large topology. Values are transcribed from the experiment README because the corresponding raw result tree is absent. Δ_{λ_1} is the documented candidate minus reference mean validation MSE.

In both experiments, the baseline model with zero penalty ($\lambda = 0$) achieved the lowest mean validation MSE of 0.004419 ± 0.000794 ;

- the nearest L1 candidates were only 5.3×10^{-6} and 1.78×10^{-5} worse in the reported mean and improved in two folds, so neither meets the four-fold paired criterion or exceeds its own paired spread;
- the nearest L2 candidate, 10^{-4} , improved in one fold and was worse by 6.44×10^{-5} in the reported mean.

The degradation then increased across the tested grids.

At the same time, the validation minus training gap decreased, but the training MSE rose faster, indicating that shrinkage restricted the fit more than it corrected overfitting in this fixed 100-epoch regime.

λ_2	Mean validation MSE	Fold standard deviation	Δ_{λ_2}	Improved folds
0	0.004419	0.000794	0	–
10^{-4}	0.004483	0.000819	+0.0000644	1/5
3.16×10^{-4}	0.004580	0.000859	+0.0001613	0/5
10^{-3}	0.004768	0.000932	+0.0003489	0/5
3.16×10^{-3}	0.005092	0.000992	+0.0006730	0/5
10^{-2}	0.005902	0.001100	+0.0014836	0/5
3.16×10^{-2}	0.006994	0.001217	+0.0025749	0/5
10^{-1}	0.008453	0.001080	+0.0040345	0/5

Table 10: Documented L2 reference sweep on the large topology. Values are transcribed from the experiment README because the corresponding raw result tree is absent. Δ_{λ_2} is the documented candidate minus reference mean validation MSE.

The documented cross-validation winner is therefore $\lambda_1 = 0$ and $\lambda_2 = 0$, with reported mean physical validation MSE 0.004419 ± 0.000794 . The current production source also sets both penalties to zero, so the direction of this choice agrees with `CNN/main.cpp`.

4.7. Activation-function tuning

The activation function determines the nonlinear transformation applied after an affine layer and therefore affects both the signal passed to the next layer and the gradient returned during backpropagation. This choice is particularly relevant for the selected topology, which contains one activation after the convolution and four more in the dense head.

This experiment compares several common activation functions to determine which is most effective for our specific architecture and aerodynamic prediction task. The purpose was not to assume that one family is universally better, but to measure which transformation gives the lowest physical validation error under the fixed architecture and training budget. Saturating nonlinearities can reduce the derivative when their input has large magnitude, whereas a rectifier retains a unit derivative on its positive branch; these are established mechanisms for changes in signal and gradient propagation, as stated by Glorot and Bengio (2010).

From the rectifier family, we tested the standard ReLU, which deactivates for negative inputs, and several variants of LeakyReLU, which allow a small non-zero gradient to pass through, preventing the "dying ReLU" problem:

$$\begin{aligned}
 f_{\text{ReLU}}(x) &= \begin{cases} x, & x > 0, \\ 0, & x \leq 0, \end{cases} & f'_{\text{ReLU}}(x) &= \begin{cases} 1, & x > 0, \\ 0, & x \leq 0, \end{cases} \\
 f_{\alpha}(x) &= \begin{cases} x, & x > 0, \\ \alpha x, & x \leq 0, \end{cases} & f'_{\alpha}(x) &= \begin{cases} 1, & x > 0, \\ \alpha, & x \leq 0. \end{cases}
 \end{aligned} \tag{56}$$

The experiment tested three different negative slopes for LeakyReLU: $\alpha \in \{0.01, 0.05, 0.1\}$.

The smooth alternatives were implemented as:

$$\begin{aligned}
 f_{\text{Tanh}}(x) &= \tanh(x), & f'_{\text{Tanh}}(x) &= 1 - \tanh^2(x), \\
 f_{\text{Sigmoid}}(x) &= \frac{1}{1 + \exp(-x)}, & f'_{\text{Sigmoid}}(x) &= f_{\text{Sigmoid}}(x) (1 - f_{\text{Sigmoid}}(x)).
 \end{aligned} \tag{57}$$

Their outputs are bounded, and the derivatives become small in the corresponding saturation regions. The final dense layer remains linear, so these transformations are used only in the convolutional branch and in the four hidden dense layers.

In this case, all runs used the same best-performing topology (from Section 4.2), Adam optimizer with learning rate 10^{-5} , batch size 64, and 100 epochs. It also used a physics weight of 0.1, as stated at the end of Section 4.5. The entire experiment followed the rigorous cross-validation framework and evaluation metrics detailed in Section 4.1.

The results of the activation function sweep, summarized in Table 11, reveal a clear performance divide between the two families of functions tested.

Activation	Mean validation MSE	Fold standard deviation
LeakyReLU, $\alpha = 0.05$	0.004427	0.000865
LeakyReLU, $\alpha = 0.01$	0.004448	0.000876
ReLU	0.004454	0.000876
LeakyReLU, $\alpha = 0.1$	0.004545	0.000966
Tanh	0.005466	0.001045
Sigmoid	0.008971	0.001250

Table 11: Complete activation-function comparison on the fixed large topology. Means and deviations are the sample-weighted physical validation MSE and weighted population standard deviation from `cv_summary.csv`; the ranking was performed over all six candidates.

The ReLU family of activations (ReLU and Leaky ReLU) decisively outperformed the classic bounded functions. The best overall performance was achieved by LeakyReLU with $\alpha = 0.05$, which recorded a mean physical validation MSE of 0.0044274 ± 0.0008653 . However, the performance difference among the top three candidates was extremely small. As the error bars in Figure 24 show, the differences between the top three candidates are within the range of their fold-to-fold variability, indicating that while LeakyReLU with $\alpha = 0.05$ is the best performer, the other two rectifiers are statistically very close.

In contrast, the Tanh and Sigmoid functions performed significantly worse. Tanh’s validation error was over 23% higher than the winner’s, while Sigmoid’s error was more than double. This performance gap suggests that these functions suffered from the "vanishing gradient" problem, where the gradients become near-zero in the saturation regions, effectively stalling the learning process in the network’s deep layers.

The separation from the bounded alternatives was larger. Tanh reached $0.005465620520 \pm 0.001045446677$, about 23.4% above the selected mean, while Sigmoid reached $0.008971404356 \pm 0.001250088488$, about twice the selected mean. Figure 24 displays the official weighted means and deviations. All 30 candidate-fold records completed with finite metrics, and no candidate was removed from the ranking because of a failed or divergent run.

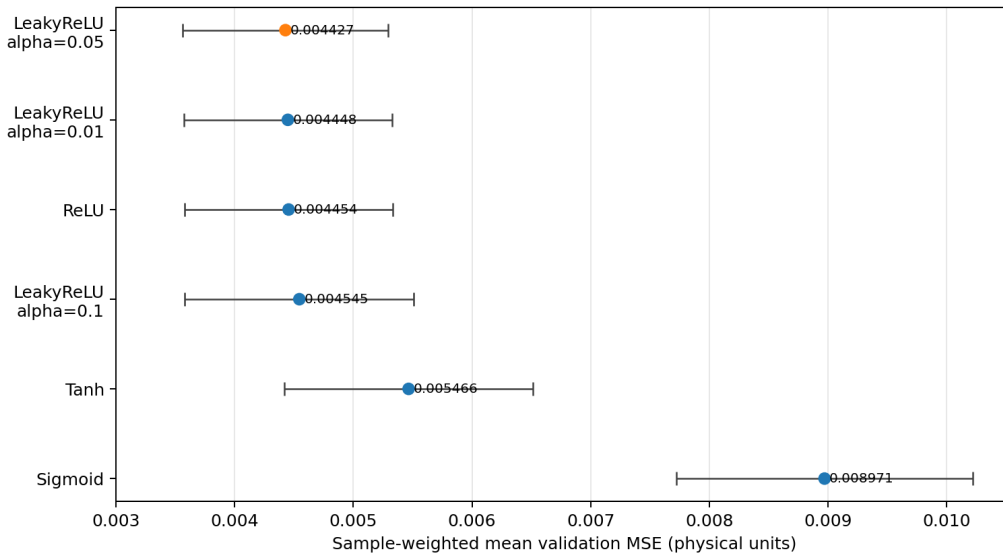


Figure 24: Complete activation-function ranking from the official `cv_summary.csv`. Points are sample-weighted mean physical validation MSE values and horizontal error bars are the corresponding weighted population standard deviations over the five folds. The orange point identifies the lowest observed candidate, LeakyReLU with $\alpha = 0.05$.

The validation histories, presented in Figure 25, provide a view of the training process. The ReLU family

candidates demonstrate fast and stable convergence, smoothly decreasing the validation error throughout the 100-epoch budget. In contrast, the saturating functions have a poorer convergence dynamics:

- Tanh converges to a solution that is consistently worse than any of the ReLU family functions;
- Sigmoid performs by far the worst. It starts with a validation error more than an order of magnitude higher than the others. While it does improve over time, its convergence is extremely slow, and it never comes close to matching the performance of the non-saturating functions.

This convergence behavior strongly supports the hypothesis that the saturating functions are suffering from vanishing gradients.

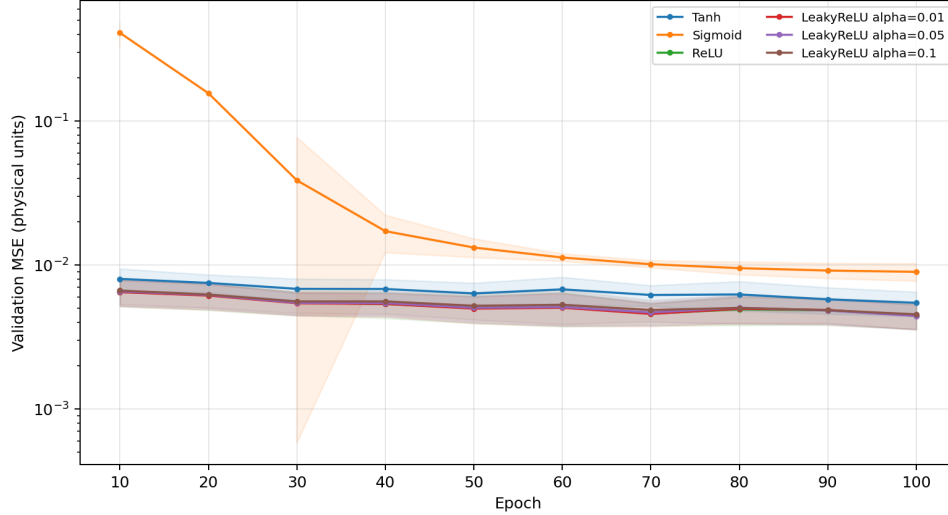


Figure 25: Fold-weighted physical validation MSE at the ten-epoch checkpoints for the six activation candidates. Shaded regions show the positive part of the weighted population standard deviation band over the five folds, and the vertical axis is logarithmic. The standardized SIMM training objective is not plotted on this axis.

The recorded diagnostics provide a numerical stability check, but they do not change the selection rule. For gradient flow, we use rows with `parameter_scope=all` and the stored `maximum_norm` for peak checks. As we can see in the top panels of Figure 26, the largest value for the selected candidate was 20.1379 at epoch 1, with similar peaks for ReLU and other LeakyReLU variants. Tanh was the numerical outlier with a peak of 111.508: this early-epoch instability likely hindered its ability to find a good solution.

Sigmoid had a lower peak of 18.2200, but this does not imply better optimization because its early validation curve was much worse. Indeed, the right panel confirms this instability, showing that Sigmoid resulted in the largest and most chaotic parameter updates, with a peak update ratio of 6.15050×10^{-4} . In contrast, the entire ReLU family, including the selected one, exhibited much smaller and more controlled updates, peaking around 4.5×10^{-4} .

In contrast, the bottom two panels show the detailed diagnostics for the winning LeakyReLU candidate. The gradients are well-behaved across the network’s depth. After an initial transient at epoch 1, the gradients at epoch 100 are small and stable, indicating smooth convergence without evidence of exploding gradients. The post-activation variance decreases toward the deepest hidden layers, but this reduction should not be interpreted by itself as signal degradation. Since the network ultimately performs scalar regression, the deeper layers progressively compress the high-dimensional representation into features that are directly useful for predicting C_L . Crucially, the variance remains clearly non-zero and stabilizes during training, providing no evidence of representational collapse.

Activation diagnostics for the recorded five-fold search

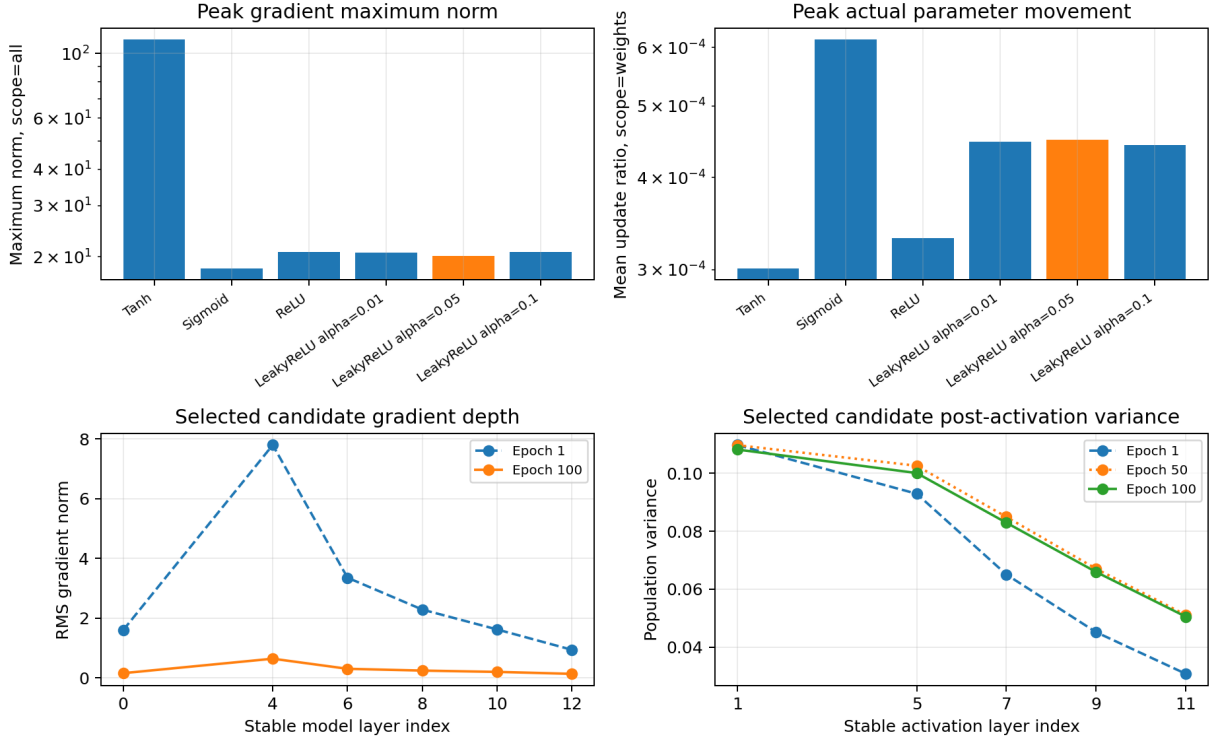


Figure 26: Diagnostics for the activation-function search. The upper panels show, for each candidate, the maximum gradient `maximum_norm` with `parameter_scope=all` and the maximum actual weight update ratio with `parameter_scope=weights`, taken over all five folds, layers, and epochs. The lower left panel shows fold-averaged RMS gradient norms for the selected candidate at epochs 1 and 100 over stable trainable layer indices. The lower right panel shows post-activation population variances for the selected candidate at epochs 1, 50, and 100, pooled over folds from the recorded population moments.

More in details, the activation statistics show the transformation expected from the equations. For the selected LeakyReLU, the post-activation population variances were pooled over folds from the recorded population moments as follows:

epoch	layer 1	layer 5	layer 7	layer 9	layer 11
1	0.109805	0.092960	0.065048	0.045128	0.030745
50	0.109755	0.102640	0.084950	0.067045	0.051060
100	0.108244	0.100116	0.082939	0.065922	0.050486

(58)

The variance decreases through the dense head, but it does not collapse to zero and changes little between epochs 50 and 100. The full selected-candidate diagnostic range was approximately $[-4.003, 4.035]$ before activation and $[-0.200, 4.035]$ after activation, which is consistent with retaining a scaled negative branch. The corresponding ReLU post-activation minimum was exactly zero. These observations support controlled negative attenuation and a stable signal scale for the selected run, although they do not establish that this mechanism alone caused its small validation advantage.

Conversely, the diagnostics provide clear evidence that the bounded functions failed due to neuron saturation. For Tanh, post-activation values were recorded as high as 0.9994, extremely close to the function’s limit where the derivative is nearly zero. The issue was even more pronounced for Sigmoid, where pre-activation values late in training reached as high as 10.67, deep into the saturation region. This behavior is perfectly consistent with the vanishing gradient hypothesis and directly explains the slow, ineffective convergence observed in the main results.

The vanishing-gradient risk is most directly assessed from the distribution of the pre-activation value z , rather than from the post-activation value alone. If $a_j = f_j(z_j)$, the chain rule inserts one derivative factor at every nonlinear layer. Ignoring the separate linear weight transformations, the activation contribution along a path

through k nonlinearities is:

$$\frac{\partial \mathcal{L}}{\partial z_j} = \frac{\partial \mathcal{L}}{\partial a_j} f'_j(z_j) \quad G_{\text{act}}(k) = \prod_{j=1}^k |f'_j(z_j)| \quad (59)$$

When the actual values z_j fall in a plateau, G_{act} becomes small even when the upstream loss gradient is non-zero. A broad or heavy-tailed pre-activation distribution therefore raises the fraction of samples and paths that receive weak local gradients; it does not imply that every path vanishes, because values near the central high-slope region can still carry a useful signal.

The scale of this effect is substantial. For Sigmoid, the derivative at an input of 5 is already tiny ($|f'(5)| \simeq 6.65 \times 10^{-3}$). To illustrate the multiplicative impact, if a gradient passed through just five such saturated neurons, it would be attenuated by a factor of approximately 10^{-11} . For Tanh, the effect is even more extreme. Our diagnostic data confirms this was happening, with pre-activation values for Sigmoid reaching as high as 10.67 late in training, well into the saturation region. This mechanism provides a definitive explanation for the slow and ineffective convergence of these functions.

The rectifiers have a different but equally important failure mode. ReLU has an exact zero derivative for $z \leq 0$, so a unit whose inputs remain on the negative branch contributes no local gradient through that activation. LeakyReLU replaces this strict plateau with a constant negative-branch slope α . This avoids an exactly dead path, but it does not eliminate attenuation: k consecutive negative-branch traversals contribute α^k , so the tested $\alpha = 0.05$ can still multiply the signal by 0.05^k , while $\alpha = 0.01$ attenuates it more strongly and $\alpha = 0.1$ less strongly. Thus, "non-saturating" should not be interpreted as "immune to vanishing gradients"; the relevant question is how much of the observed pre-activation distribution lies in each branch and how those branches are arranged across depth.

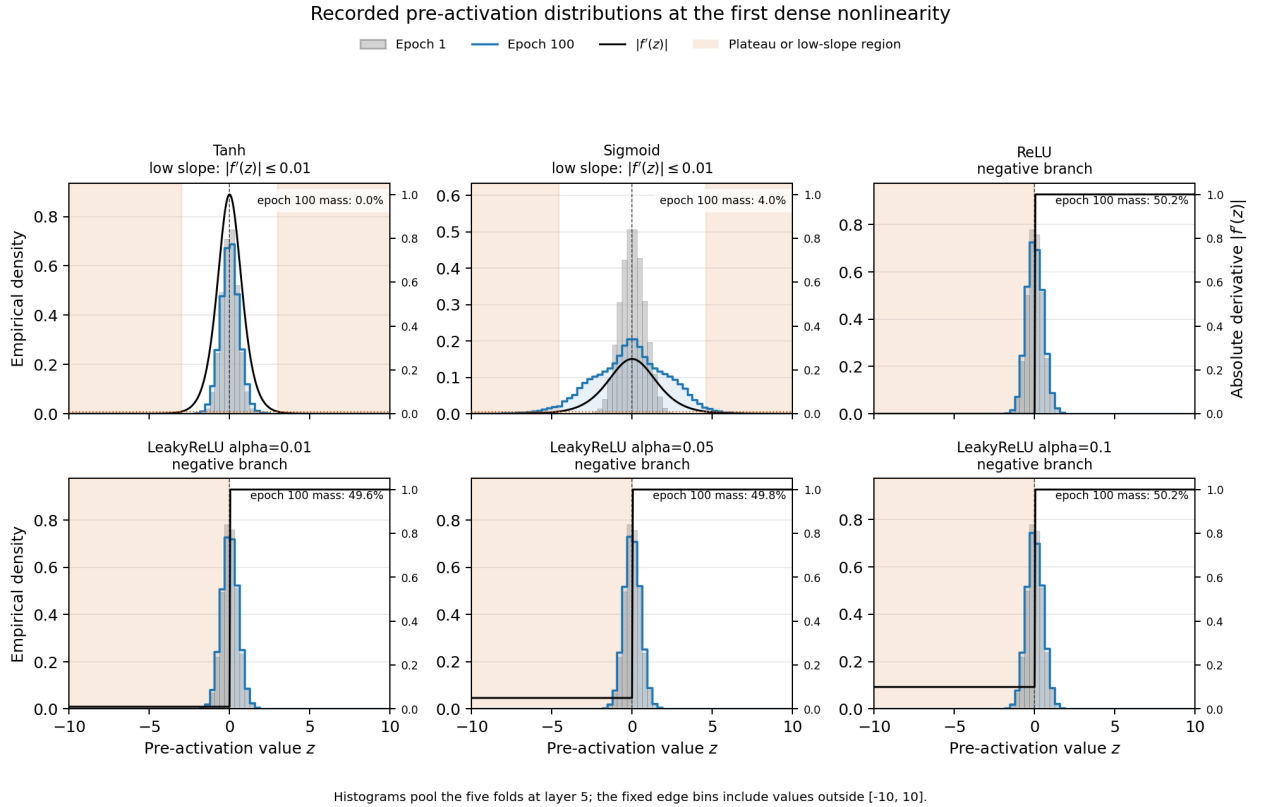


Figure 27: Empirical distributions of the actual pre-activation values z at the first dense nonlinearity (stable model layer 5) for the six activation candidates. Gray bars show epoch 1 and blue step curves show epoch 100, with histograms pooled over the five folds. The black curves show the exact absolute derivative $|f'(z)|$ on the right-hand scale. Orange shading identifies the Tanh and Sigmoid regions with $|f'(z)| \leq 0.01$ and the negative branch for ReLU and LeakyReLU. The epoch-100 mass annotations are approximate because the committed diagnostics store fixed-bin counts; the edge bins include values outside the displayed $[-10, 10]$ range.

This theoretical risk of vanishing gradients is confirmed by the empirical data. Figure 27 visualizes the distribu-

tion of the actual pre-activation values (z) at the first dense layer, providing a clear picture of why the different activation functions performed so differently.

The plot for Sigmoid is the most revealing. By epoch 100, the distribution of its pre-activation values has broadened significantly. Approximately 4.0% of its inputs now fall into the low-derivative "saturation" region (shaded in orange), where the gradient is less than 1% of its maximum. This provides direct, visual proof that a substantial portion of the neurons are receiving near-zero gradients, which explains the function's extremely slow convergence. While Tanh does not show a significant mass in the saturation region in this specific layer, other diagnostics confirmed that saturation was occurring elsewhere in the network.

In contrast, the plots for the ReLU family show that roughly half of their inputs fall into the negative branch (also shaded). For standard ReLU, this means that 50.2% of its neurons are "off" and passing a zero gradient. For the winning Leaky ReLU ($\alpha = 0.05$), approximately 49.8% of its inputs are in the negative branch, but they are passing a small, non-zero gradient. This visualization illustrates the key trade-off. While nearly half the neurons are in a low-gradient state for the ReLU family, this state is not catastrophic. The non-saturating positive branch allows strong gradients to flow, and the "leaky" negative branch prevents the complete death of neurons. This is a far healthier dynamic than the severe, widespread gradient attenuation caused by saturation in the bounded functions.

The final refit of the selected activation used all 1713 training samples and retained the large search topology. Its training SIMM objective decreased from 0.286768 at epoch 1 to 0.00865917 at epoch 100. The final MSE on the 429-sample test set was 0.00273892. This is a post-selection estimate for the refitted activation-search model and must not be presented as the test performance of the ordinary production model.

The cross-validation winner is therefore **LeakyReLU** with $\alpha = 0.05$, with physical validation MSE $0.00442742 \pm 0.00086526$ on the fixed large topology.

4.8. Learning-rate and schedule tuning

After choosing the topology, optimizer, loss settings and activation, we studied the learning rate because it controls the scale and duration of the parameter updates. In Adam, the learning rate multiplies the bias-corrected adaptive direction; it is therefore not removed by the coordinate-wise denominator, even though that denominator adapts to the history of each parameter. A rate that is too small may lead to slow convergence, failing to reach an optimal solution within the fixed epoch budget. Conversely, a rate that is too large can cause the optimization to become unstable and diverge.

Beyond the learning rate's magnitude, its schedule, that represents how it changes over time, is also crucial. In addition to a constant rate, we tested step decay, cosine decay²⁴ and warm-up, that was introduced to address early optimization difficulties in large-minibatch stochastic-gradient training (Goyal et al., 2017). Those references motivate the schedule families tested here, but they do not predict the behaviour of this Adam run: the present experiment uses no warm restarts, a different optimizer and a different data set.

The goal was to identify the combination of learning rate and schedule that leads to the fastest and most stable convergence for our specific aerodynamic modeling task.

Let $e \in \{0, \dots, E - 1\}$ denote the zero-based epoch index and let E be the total number of epochs. The experiment evaluated four distinct types of learning rate schedules:

- in the constant rate, the learning rate remains fixed at its initial value η_0 throughout the training process:

$$\eta_e = \eta_0$$

- in the step decay, the learning rate is multiplied by a decay factor γ at fixed step intervals S :

$$\eta_e = \eta_0 \gamma^{\lfloor e/S \rfloor}$$

The step schedules change at zero-based indices $e = 30, 60, 90$ for $S = 30$, and at $e = 15, 30, 45, 60, 75, 90$ for $S = 15$, which correspond to completed epochs 31, 61, 91 and 16, 31, 46, 61, 76, 91, respectively;

- in the cosine decay, the learning rate smoothly anneals from a maximum value $\eta_{\max}$ to a minimum value $\eta_{\min}$ following a cosine curve:

$$\eta_e = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left[1 + \cos \left(\frac{e}{E - 1} \pi \right) \right]$$

²⁴This schedule has been studied as part of the SGDR family (Loshchilov and Hutter, 2017)

which reaches $\eta_{\max}$ at $e = 0$ and $\eta_{\min}$ at $e = E - 1$;

- in the warm-up schedule, the training begins with a short linear "warm-up" phase, where the learning rate ramps up to a peak value, before following a cosine decay for the remainder of the training:

$$\eta_e = \begin{cases} \eta_{\text{start}} + \frac{e}{W}(\eta_{\text{peak}} - \eta_{\text{start}}), & 0 \leq e < W, \\ \eta_{\min} + \frac{1}{2}(\eta_{\text{peak}} - \eta_{\min}) \left[1 + \cos \left(\frac{e - W}{E - W - 1} \pi \right) \right], & W \leq e < E, \end{cases}$$

where W is the number of warm-up epochs. In the recorded grid, $E = 100$, $W = 5$, $\eta_{\text{start}} = \eta_{\min} = 10^{-5}$, and $\eta_{\text{peak}} = 10^{-1}$. Thus, the warm-up branch returns 10^{-5} at the first completed epoch and reaches 10^{-1} after 5 completed epochs (at the start of epoch 6).

In the software architecture, each of these mathematical policies is implemented as a specialized subclass of the abstract `LRScheduler` base class (`ConstantLR`, `StepLR`, `CosineAnnealingLR` and `WarmupCosineLR`) and instantiated through an `LRSchedulerRecipe`. At the beginning of each epoch, the `Trainer` queries the active scheduler to adjust the optimizer's learning rate η_e in a deterministic manner, ensuring full reproducibility across parallel MPI workers.

The grid search evaluated eleven candidates, which are summarized in Table 12, combining the schedule types above with three different initial learning rate magnitudes: a low rate (10^{-5}), a medium rate (10^{-3}) and a high rate (10^{-1}). All experiments were performed using the winning activation function, LeakyReLU with $\alpha = 0.05$, and the same fixed topology `conv5x5-dense-1024-512-256-128`. The other hyperparameters were kept constant, including the physics weight 0.1, L1 and L2 weights 0, dropout rate 0, global batch size 64, gradient clipping at 1, and no early stopping. Each candidate was evaluated using the same five-fold cross-validation procedure as in the activation search, with 100 epochs per fold.

Candidate	Configuration	Mean validation MSE $\pm$ SD	Fold range
<code>const_1e-3</code>	Constant, $\eta_0 = 10^{-3}$	0.003609 ± 0.001335	0.002092 to 0.006094
<code>step_s15_1e-3</code>	Step, $S = 15$, $\gamma = 0.5$, $\eta_0 = 10^{-3}$	0.004005 ± 0.000704	0.002826 to 0.004779
<code>step_s30_1e-3</code>	Step, $S = 30$, $\gamma = 0.1$, $\eta_0 = 10^{-3}$	0.004072 ± 0.000785	0.002865 to 0.004955
<code>const_1e-5</code>	Constant, $\eta_0 = 10^{-5}$	0.004419 ± 0.000790	0.002880 to 0.005001
<code>step_s30_1e-5</code>	Step, $S = 30$, $\gamma = 0.1$, $\eta_0 = 10^{-5}$	0.005034 ± 0.000901	0.003434 to 0.005786
<code>step_s15_1e-5</code>	Step, $S = 15$, $\gamma = 0.5$, $\eta_0 = 10^{-5}$	0.005094 ± 0.000905	0.003484 to 0.005861
<code>cosine_1e-1</code>	Cosine, $\eta_{\max} = 10^{-1}$, $\eta_{\min} = 10^{-5}$	$5.577 \times 10^8 \pm 2.251 \times 10^8$	2.087×10^8 to 7.949×10^8
<code>warmup_cosine_1e-1</code>	Warm-up and cosine, $\eta_{\text{peak}} = 10^{-1}$	$9.047 \times 10^8 \pm 8.866 \times 10^8$	1.207×10^8 to 2.507×10^9
<code>step_s15_1e-1</code>	Step, $S = 15$, $\gamma = 0.5$, $\eta_0 = 10^{-1}$	$5.010 \times 10^9 \pm 2.956 \times 10^9$	1.530×10^9 to 9.316×10^9
<code>step_s30_1e-1</code>	Step, $S = 30$, $\gamma = 0.1$, $\eta_0 = 10^{-1}$	$9.449 \times 10^9 \pm 4.726 \times 10^9$	3.197×10^9 to 1.631×10^{10}
<code>const_1e-1</code>	Constant, $\eta_0 = 10^{-1}$	$2.131 \times 10^{17} \pm 3.116 \times 10^{17}$	1.843×10^{16} to 8.313×10^{17}

Table 12: Complete ranking of the eleven learning-rate candidates. Means and deviations are the sample-weighted physical validation MSE and its weighted population standard deviation over the five folds, as recorded in `cv_summary.csv`; the final column gives the minimum and maximum fold scores.

The results of the learning rate sweep, shown also in Figure 28, reveal a clear separation between the low-rate candidates and the high-rate candidates that reach 10^{-1} .

The best performance was achieved by the constant learning rate of 10^{-3} , which secured a mean validation MSE of 0.003609 ± 0.001335 . This medium learning rate was consistently effective, winning four of the five cross-validation folds and delivering an 18% error reduction compared to the constant 10^{-5} baseline. The nearest alternative is `step_s15_1e-3`, whose mean is 0.004005474214; its lower dispersion does not compensate for

the higher aggregate validation error under the stated selection rule.

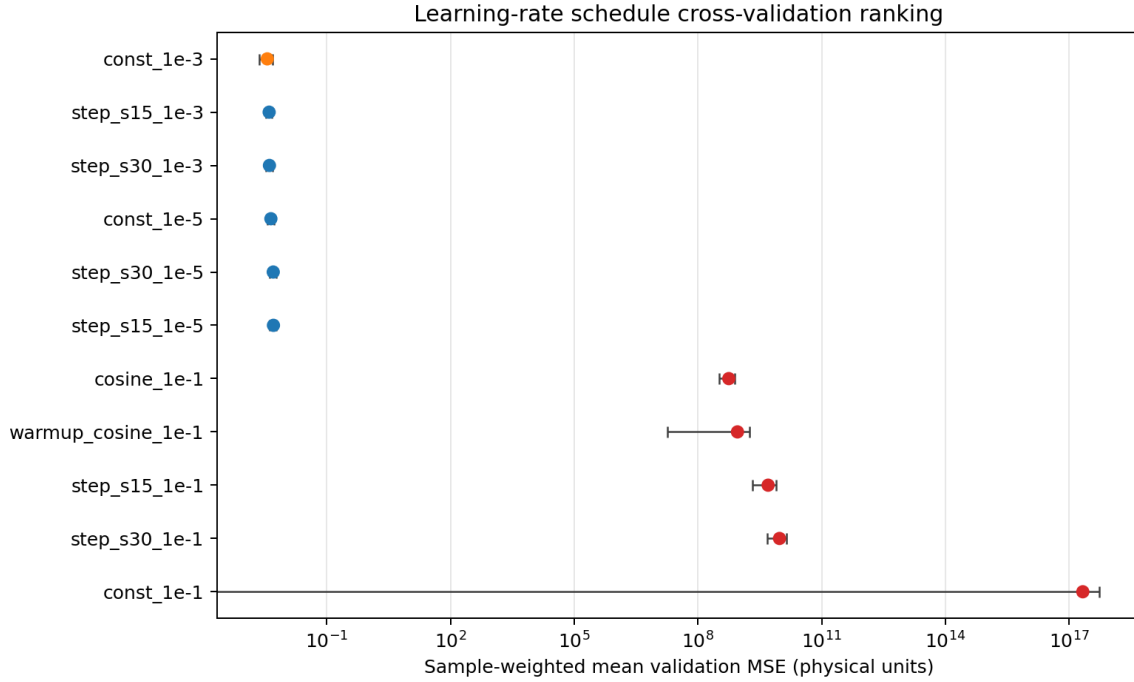


Figure 28: Complete learning-rate schedule ranking. Points show the sample-weighted mean physical validation MSE and horizontal bars show the weighted population standard deviation over five folds. The horizontal axis is logarithmic; orange identifies the selected constant 10^{-3} candidate and red identifies candidates whose rates reach 10^{-1} .

In contrast, all candidates that used a high learning rate of 10^{-1} failed catastrophically. As shown by the red markers in Figure 28, their validation errors were orders of magnitude higher, ranging from 10^8 to 10^{17} . This indicates that the optimization process diverged, likely due to an excessive step size in the parameter updates. Even the warm-up schedule, which initially starts at a safe low rate and ramps up to 10^{-1} , suffered the same fate, demonstrating that even a brief exposure to this aggressive rate permanently destabilized training.

The convergence histories in Figure 29 provide a dynamic view of the training process and visually confirm the final rankings. The plot is split into two panels, distinguishing the behaviour of the stable and unstable learning rates:

- the top panel shows that all candidates with initial learning rates of 10^{-5} and 10^{-3} remain well-behaved with the 10^{-3} reaching the lowest overall validation errors. Among them, the constant 10^{-3} candidate achieves the lowest final validation error at epoch 100, despite noticeable mid-training fluctuations (e.g., around epochs 50 and 80). In contrast, the stepped 10^{-3} schedules exhibit smoother, monotonic trajectories, while the 10^{-5} candidates converge much more slowly and plateau early, confirming that their step size is insufficient for the 100-epoch budget;
- the bottom panel shows that the validation error for all the candidates that reach 10^{-1} immediately explodes to extreme values and remains many orders of magnitude above the stable group for the entire duration of training. Even for schedules that eventually decay the learning rate (like cosine and step decay), the initial instability was so severe that the models could never recover.

This analysis reinforces the conclusion that a learning rate of 10^{-3} represents an optimal trade-off, offering a balance between rapid convergence and numerical stability.

Fold-weighted validation histories

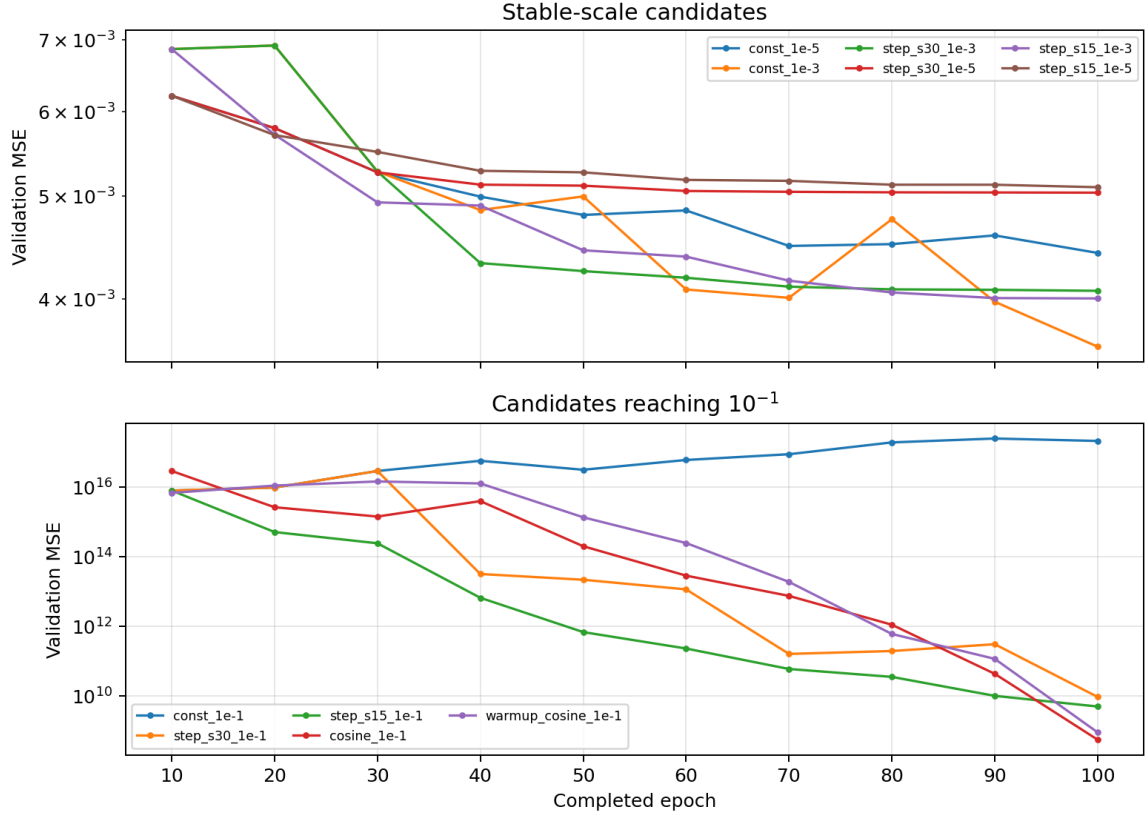


Figure 29: Sample-weighted validation histories at the ten-epoch checkpoints recorded for cross-validation. The upper panel contains the six candidates whose rates remain in the low-error scale, while the lower panel contains the five candidates that reach 10^{-1} . The upper panel uses a linear vertical axis, whereas the lower panel uses a logarithmic scale to accomodate the large divergence in physical-unit MSE.

The divergence at $\eta = 10^{-1}$ stems from a fundamental mismatch between Adam’s update dynamics and the network’s weight scale:

- under Kaiming initialization, weights in the wide first dense layer ($D_{\text{in}} = 7202$) have a standard deviation of $\sigma = \sqrt{2/7202} \approx 0.0166$. Since Adam’s coordinate update $\Delta\theta_j = -\eta \frac{\hat{m}_j}{\sqrt{\hat{v}_j + \epsilon}}$ produces a step of magnitude $\approx \eta$ along active directions, setting $\eta = 0.1$ forces each parameter update to be approximately six times larger than the entire initial parameter scale in a single step. This immediately destroys the initial structural features, causing pre-activations and downstream predictions to blow up exponentially;
- although component-wise gradient clipping ($\tilde{g}_{t,j} = \text{clip}(g_{t,j}, -1, 1)$) prevents gradient norms from reaching infinity, Adam is scale-invariant. When gradients consistently saturate at the clipping threshold ± 1 , the ratio $\hat{m}/\sqrt{\hat{v}}$ remains $\approx \pm 1$, leaving the effective update size directly determined by $\eta = 0.1$;
- in schedules such as **warmup_cosine_1e-1**, the network trains stably during the first few epochs at low rates. However, as soon as η hits 0.1 at epoch 6, the parameters enter a high-loss, chaotic plateau. The moving moments m_t and v_t become corrupted by massive gradient noise, making it impossible for the optimizer to recover even when the cosine decay cools η back down to 10^{-5} in later epochs.

Figure 30 tracks the effective learning rate profiles across completed epochs, highlighting how the transient peak at epoch 6 in the warm-up branch permanently derails optimization.

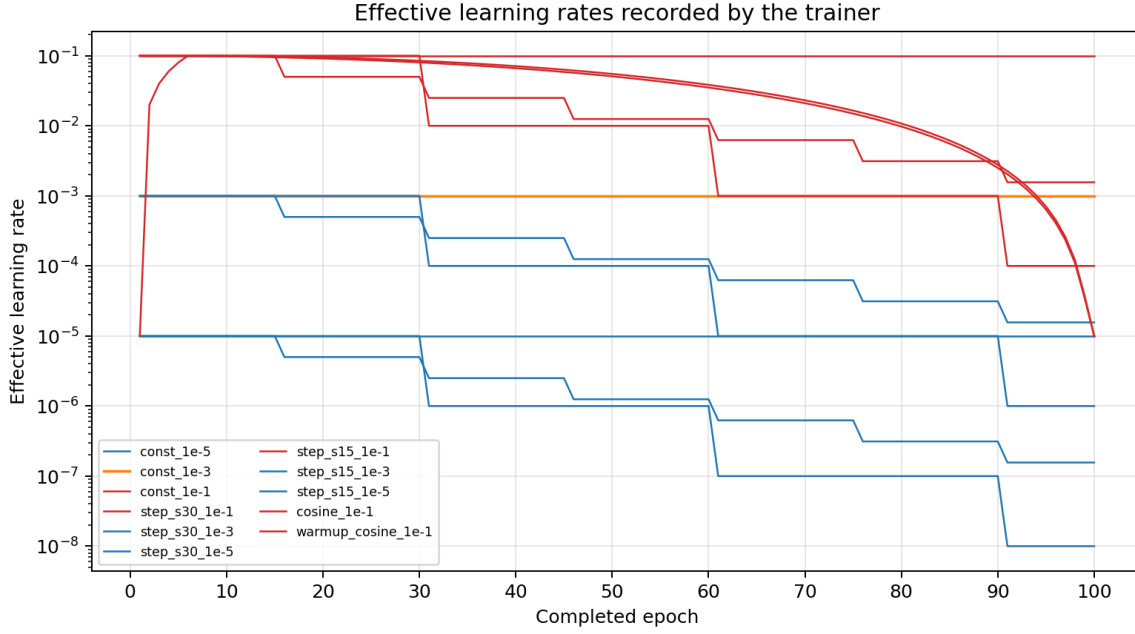


Figure 30: Effective learning rates from `epoch_metrics.csv` for zero-based fold 0, plotted against completed epoch. The scheduler settings are deterministic and identical across folds. The logarithmic vertical axis makes the constant, step, cosine, and warm-up branches comparable.

The high-rate candidates should be described as numerically unstable rather than as failed jobs. All 55 candidate-fold fits completed with finite diagnostic values and were included in the official ranking, but the five candidates that reach 10^{-1} have validation means from 5.577×10^8 to 2.131×10^{17} . The detailed diagnostics, presented in Figure 31, provide an illustration of this.

In the top left panel, the bar chart represents the "worst-case scenario" for gradients in each run. For every candidate, it searches through all 100 epochs, all five folds and all layers of the network to find the single largest gradient value. We can see a clear separation between the stable and unstable candidates:

- for the stable rates (10^{-5} and 10^{-3}), the peak gradient norms are controlled, with the winning candidate maintaining a peak gradient of approximately 183;
- for the unstable rates (10^{-1}), the peak gradient norms grow uncontrollably, leading to a complete breakdown of the optimization trajectory.

Moving to the right panel, we observe the direct consequence on parameter movement: this plot measures the largest relative change made to the network's weights in a single optimization step. While the stable cases show well-behaved updates (around 0.017 for the 10^{-3} candidate), the unstable 10^{-1} candidates exhibit update ratios approaching 100%. This means the optimizer attempts to displace weights by an amount equal to their entire existing magnitude in a single step, wiping out any learned representation.

The lower panels of Figure 31 provide a further check on the selected run.

The first plot shows the average gradient magnitude for each trainable layer at the very beginning of training (epoch 1) and at the end of training (epoch 100). The RMS gradient is a measure of how much the weights are being updated during training. A high RMS gradient indicates that the model is learning quickly, while a low RMS gradient suggests that learning has slowed down. The plot shows that the RMS gradient decreases over time, which is expected as the model converges to a solution. Indeed, we can see that the learning process starts strongly, with a peak norm of approximately 37 at epoch 1; by epoch 100, the gradients have naturally decreased, settling in a stable range below 1.0. This indicates that the model has learned a solid representation of the data and is performing refined parameter adjustments.

Regarding the post-activation variance, the lower-right panel shows the pooled population variance of the activations after the non-linear transformation at each layer. This is a measure of how much the activations vary across different inputs. In this case, we can see that at the beginning the signal variance grows as information passes deeper into the network, but at epoch 50 and epoch 100, the overall variance decreases and stabilizes. Crucially, the variance never collapses to zero and never explodes.

These two plots confirm that information is flowing effectively through all layers of the deep network for the

entire duration of training and that the network does not suffer from vanishing or exploding activations.

Learning-rate diagnostics from the five-fold runs

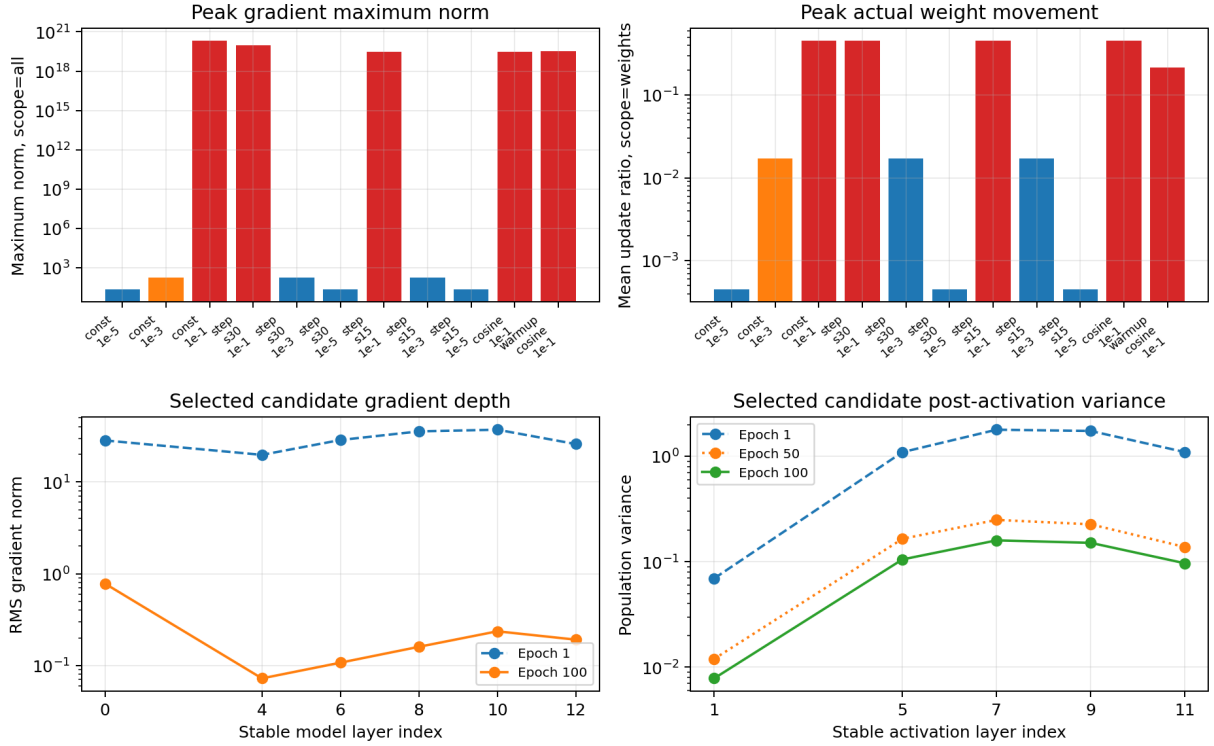


Figure 31: Learning-rate diagnostics from the complete five-fold run. The upper panels show the maximum over folds, layers, and epochs of `maximum_norm` with `parameter_scope=all` and of the actual `mean_ratio` with `parameter_scope=weights`. The lower-left panel shows the selected candidate’s fold-averaged RMS gradient by stable model layer at epochs 1 and 100. The lower-right panel shows the selected candidate’s pooled post-activation population variance at epochs 1, 50, and 100. All vertical axes are logarithmic.

The cross-validation experiment identifies the constant 10^{-3} learning rate as the clear winner, achieving a mean validation MSE of 0.003609 ± 0.001335 . The analysis confirms that this learning rate represents a robust "sweet spot", offering rapid convergence without triggering the catastrophic numerical instability observed at higher rates like 10^{-1} . After being refit on the full training set, this winning experimental candidate achieved a final test MSE of 0.003254.

4.9. Final production training

The preceding tuning experiments identified a set of optimal hyperparameters for the network architecture and optimizer. This final section details the development of the production training run, which combines these findings with a refined training protocol to achieve the best possible performance. Unlike the cross-validation used for tuning, this final training uses a single, fixed 90/10 split of the data for training and validation, with a normalization fitted only on the 1542 training samples and applied unchanged to the 171 validation samples and the subsequent test evaluation.

The final ordinary training was developed through a sequence of training trials rather than launched as a single fixed experiment. The first trial used a global batch size of 64 with the original range-based batching. Since the ordinary training split contains 1542 samples, this produced 24 batches of 64 samples and a final batch of only 6 samples, leading to wasted computational resources in our parallel MPI environment.

To solve this, we implemented a balanced batching strategy. Instead of creating as many full batches as possible, this strategy first determines the optimal number of batches, K , needed to keep the batch size as close to the target B as possible. It then distributes the samples as evenly as possible among these K batches. The exact

size of each batch, n_j , is calculated as follows:

$$K = \left\lceil \frac{n}{B} \right\rceil, \quad q = \left\lfloor \frac{n}{K} \right\rfloor, \quad r = n - Kq, \quad n_j = \begin{cases} q+1, & j < r, \\ q, & j \geq r, \end{cases} \quad j = 0, \dots, K-1, \quad (60)$$

where n is the total number of training samples. Equation (60) gives 17 batches of 62 and 8 batches of 61 for the balanced 64-sample experiment, while $B = 257$ produces six exact batches of 257. The former change removed the small tail without changing the 25 optimizer steps per epoch; the latter also reduced the number of updates per epoch to six. This distinction is important because the global batch size is not multiplied by the MPI rank count, and because the gradient synchronization in `CNN/src/model/CNNModel.cpp` weights each local gradient by its local sample count before dividing by the global count.

Early stopping is crucial for selecting the best model and preventing overfitting. However, our initial policy, which stopped training at the first sign of overfitting, proved to be too sensitive to normal training noise and often terminated runs prematurely. To address this, we implemented a more robust "20/20" policy. This new rule only stops training after the validation error has failed to improve for 20 consecutive epochs. This ensures that the model has truly converged and is not just experiencing a temporary fluctuation, leading to a much more reliable and stable model selection process.

The impact of these iterative improvements to the training protocol is documented in Table 13. The table chronologically presents the results of several training trials, clearly showing how the refinement of the batching and early stopping policies led to progressively better and more stable outcomes. All errors in the Table are physical-unit MSE values, where "budget" is the maximum configured epoch count, "completed" is the length of the diagnostic history and "selected" identifies the restored validation-minimum checkpoint.

Stage	B	Budget	Completed	Selected	Train MSE	Val. MSE	Test MSE
Initial tail, trial 1	64	200	200	162	0.001957	0.001714	0.001647
Balanced, trial 2	64	200	191	139	0.002081	0.001719	0.001894
Exact batches, trial 3	257	200	200	183	0.002614	0.002325	0.001998
First ratio, trial 4	257	834	319	318	0.001518	0.001425	0.001512
First ratio, seed 0	257	834	5	5	0.011170	0.013795	0.009084
First ratio, seed 1	257	834	6	5	0.012426	0.013655	0.010525
20/20 policy, seed 0	257	834	834	820	0.000389	0.000552	0.000420
20/20 policy, seed 1	257	834	159	139	0.002924	0.004752	0.002381
20/20 policy, seed 42	257	834	834	797	0.000395	0.000435	0.000487
Final production (trial 5), seed 42	257	1200	860	797	0.000395	0.000435	0.000487

Table 13: Ordinary training results from the ten run directories under `results/ordinary-training`. Restored metrics come from `final_metrics.csv`; budgets, batch construction and stopping policies come from metadata, and completed epochs from diagnostic histories. The final two seed-42 rows retain the same checkpoint; the increased budget is not an accuracy gain.

The first three trials demonstrate the effect of refining the batching strategy. While the initial move from a tailed to a balanced batch (**trial 2**) improved parallel efficiency, increasing the global batch size to a perfectly divisible 257 (**trial 3**) provided a more stable foundation for subsequent, longer training runs.

The most dramatic improvement came from refining the early stopping rule. The trials labeled "First ratio" show that the original, sensitive policy often terminated training far too early (e.g., at epoch 5 or 6), resulting in very poor test performance ($\text{MSE} > 0.009$). In contrast, the robust "20/20" policy allowed the models to train for much longer, navigating through normal fluctuations in the validation loss to find a much better solution. This is best seen in the seed 0 and seed 42 runs, where the new policy allowed training to continue for over 800 epochs, leading to a more than 20-fold reduction in test MSE compared to the premature stops.

The final run (**trial 5**) combines all of these best practices: the optimal $B = 257$ batch size, the robust "20/20" stopping policy, and an extended epoch budget of 1200. This culminated in the best performance of the sequence, identifying a model checkpoint with a test MSE of 0.000487.

The physical-MSE histories show recurrent excursions in the unbalanced run that affect training and validation together. Figure 32 uses the `training_physical_mse` and `validation_physical_mse` columns of `epoch_metrics.csv`, evaluated after the updates at every epoch. In trial 1, the excursion at epoch 84 reaches training MSE 0.0210883 and validation MSE 0.0218529, rising from 0.0042626 and 0.0039933 at epoch 83 and returning to 0.0075658 and 0.0069386 at epoch 85. Balancing reduces these recurrent excursions but does not eliminate all fluctuations: trial 2 still reaches 0.0088347 and 0.0090290 at epoch 11, and 0.0086777 and 0.0086519 at epoch 12. Both curves remain below 0.008 thereafter. All 200 and 191 recorded epochs are finite and use a constant effective learning rate of 10^{-3} . The simultaneous movement of training and validation is consistent with a noisy update affecting the shared model rather than a validation-only increase in error.

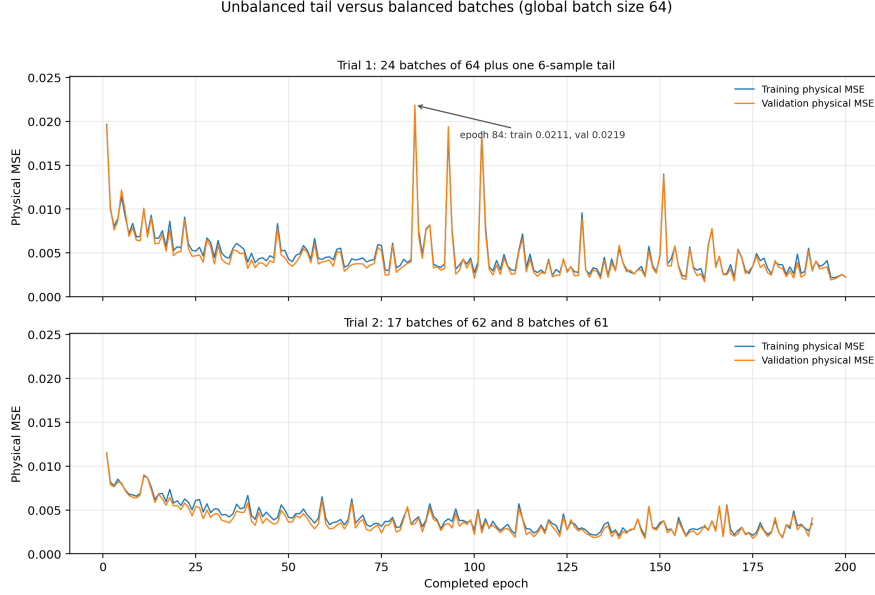


Figure 32: Physical training and validation MSE for the two seed-42 runs at global batch size 64. The upper panel is trial 1 with 24 batches of 64 plus a 6-sample tail; the lower panel is trial 2 with 17 batches of 62 and 8 batches of 61. The common linear scale includes the full histories of 200 and 191 completed epochs. The annotation marks the epoch-84 excursion in trial 1. These are single-run histories, not fold averages; the standardized SIMM objective is not shown.

The gradient and parameter-update diagnostics give a more detailed view of the same excursion. Figure 33 uses `parameter_scope=all` from `gradient_norms.csv` and `parameter_update_ratios.csv`, combining weights and biases rather than isolating small bias denominators. Gradients are measured after weighted MPI synchronization and before clipping and the optimizer update; the actual update ratio in Equation (46) includes clipping, Adam moments and the current parameter scale. The left panel plots epoch-84 RMS gradient norms at stable indices 0, 4, 6, 8, 10 and 12, corresponding to the convolution and five dense layers. The convolution norm is 1.27083 in trial 1 against 0.53502 in trial 2, whereas the dense-layer profiles are much closer: the first dense values are 0.07483 and 0.06975. At epoch 85 the convolution norms rise to 2.96752 and 1.09226, respectively. The right panel follows the first dense layer’s mean update ratio over each run’s own completed history. At epoch 84 this ratio is actually lower in trial 1, 0.0012268 against 0.0016265, but at epoch 85 it reaches 0.0055988 against 0.0026526. The corresponding mean pre-update parameter norms lie between 46.60 and 46.85, with no near-zero denominator steps. Thus the diagnostics do not show a uniform increase in every layer at the MSE peak: the larger actual mean movement occurs in the following epoch. Moreover, the last convolution gradient norm at epoch 84 equals that epoch’s maximum, 4.45210, in the unbalanced run.

The reduction in recurrent excursions must be distinguished from an accuracy gain. Trial 1 restores epoch 162, with training MSE 0.0019568, validation MSE 0.0017142 and test MSE 0.0016466; trial 2 restores epoch 139, with 0.0020810, 0.0017195 and 0.0018941, respectively. Their validation minima are practically equal, and the balanced run does not improve the reported test error. We therefore retain balancing as a way of avoiding the very small final update, not as evidence of better predictive accuracy by itself. The comparison is limited to these two seed-42 trajectories and does not establish that all optimization noise has been removed.

The move to $B = 257$ gives six exact batches because $1542 = 6 \times 257$. However, the 200-epoch trial then performs only 1200 Adam updates, compared with the 5000-update budget at $B = 64$. Its selected valida-

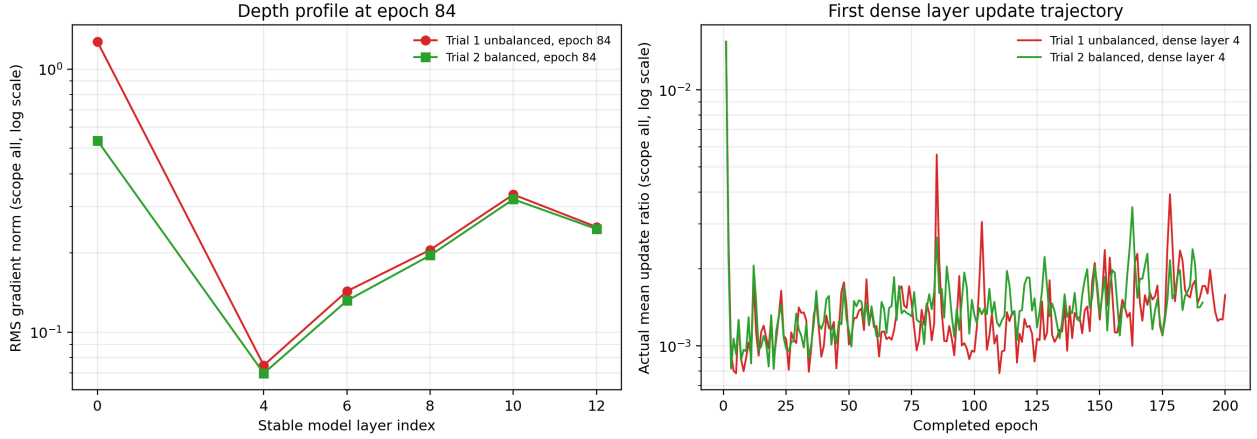


Figure 33: Diagnostics around the epoch-84 excursion. Left: synchronized, pre-clipping RMS gradient norms across all six trainable layers at epoch 84. Right: actual mean update ratio of the first dense layer, index 4, over 200 epochs for trial 1 and 191 for trial 2. Both vertical axes are logarithmic and use `parameter_scope=all`. Red denotes the unbalanced run and green the balanced run.

tion MSE is 0.0023254 and its test MSE is 0.0019978, so increasing the batch size alone does not improve the matched-epoch result. To allow a comparable number of updates, we increase the budget to 834 epochs, corresponding to $834 \times 6 = 5004$ planned updates. This is a budget comparison, not a claim that every run completes those updates. With the earlier stopping rule, the seed-42 trial stops at epoch 319 and retains epoch 318, with validation MSE 0.0014248 and test MSE 0.0015124. The early termination of this longer configuration motivates examining the stopping policy.

The overfitting check is based on the physical training and validation MSE evaluated after the optimizer updates at every epoch. An epoch satisfies the gap condition in Equation (61) when

$$\text{MSE}_{\text{val},e} > (1 + \rho) \text{MSE}_{\text{train},e}, \quad \rho = 0.15, \quad (61)$$

but this condition is not sufficient by itself in the current implementation. The earlier `first_ratio` rule stopped at the first qualifying epoch. With the 834-epoch budget, it terminates the seed-0 and seed-1 runs at epochs 5 and 6, respectively, and both retain epoch 5. Their selected validation MSE values are still 0.0137949 and 0.0136545, with test MSE 0.0090836 and 0.0105255. These short histories show why a gap during the initial optimizer transient should not automatically be interpreted as sustained overfitting.

The revised policy retains the 15% ratio but uses a minimum of 20 epochs and a patience of 20 consecutive qualifying epochs. In `Trainer`, the streak resets before epoch 20, whenever validation reaches a new minimum, or whenever validation MSE is at most 1.15 times training MSE. Otherwise it increases by one. A parameter snapshot is stored at every new validation minimum, and the best parameters are restored before the final metrics are recomputed. With this policy and the 834-epoch budget, seed 0 completes the budget and selects epoch 820, seed 1 stops at epoch 159 and selects epoch 139, and seed 42 completes the budget and selects epoch 797. Their validation MSE values are 0.0005521, 0.0047519 and 0.0004354, respectively. The policy permits substantially longer optimization than the first-ratio rule, particularly for seeds 0 and 1, but the remaining variation shows that it does not remove sensitivity to initialization and the holdout split.

The final production configuration synthesizes the findings of the tuning experiments. The model architecture consists of a 5×5 convolutional layer (stride 5, 8 channels) and a four-layer dense head of sizes (1024, 512, 256, 128, 1), using a LeakyReLU activation ($\alpha = 0.05$) throughout. This configuration is trained using the Adam optimizer with a constant learning rate of 10^{-3} . Based on our analysis, a small physics weight of $\lambda = 0.10$ is included, while explicit regularizers like dropout, L1 and L2 penalties are omitted. The training is stabilized with gradient clipping at ± 1 and executed with a balanced global batch size of 257 across 64 MPI ranks, governed by a "20/20" early stopping policy within a 1200-epoch budget.

Increasing the seed-42 budget from 834 to 1200 epochs allows the stopping rule to finish at epoch 860 in trial 5, the final production run. The 834-epoch run already selects epoch 797, and trial 5 restores that same checkpoint.

Its history through epoch 834 matches the seed-42 834-epoch run, and both restore identical parameters. The extra budget therefore allows termination under the stopping policy, but does not lower the retained validation minimum or constitute an accuracy improvement.

The final history begins with a finite initial transient: training and validation MSE are 0.2116009 and 0.2113380 at epoch 1, fall to 0.0390008 and 0.0383861 at epoch 2, and reach 0.0089447 and 0.0087936 at epoch 3. At the retained epoch 797 they are 0.000394531 and 0.000435441. Training continues until epoch 860, whose un-restored training and validation MSE are 0.000648353 and 0.000960955. The final qualifying streak is epochs 841–860, not the 20 epochs immediately following the selected checkpoint, because intervening returns below the ratio reset the counter. Production therefore performs 5160 optimizer updates, while the selected parameters come from update 4782. Figure 34 shows the complete physical-MSE history in linear and logarithmic views and distinguishes selection from termination. The standardized SIMM objective, which is 2.42682 at epoch 1, is a different quantity and is not plotted on the physical-MSE axes.

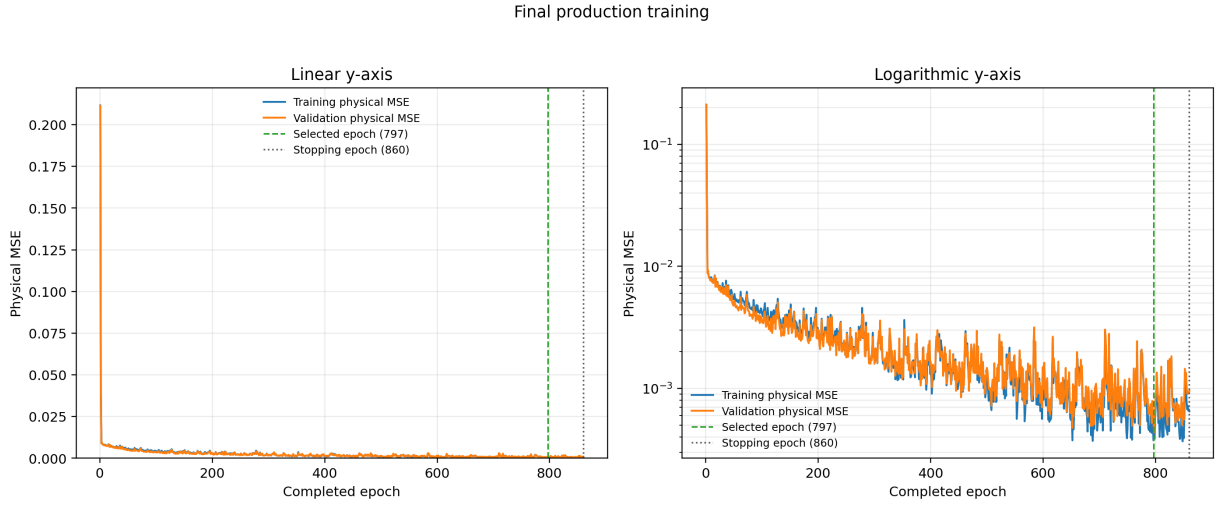


Figure 34: Physical training and validation MSE for the final production run, `trial-5_slurm-56688207`. The left panel uses a linear y-axis and the right panel a logarithmic y-axis; both contain all 860 completed epochs. The dashed line marks selected epoch 797 and the dotted line stopping epoch 860. The standardized SIMM objective is not shown.

The production diagnostics provide a complementary check on this convergence. At epoch 1, the all-parameter RMS gradient norms range from 20.0543 to 41.1596 across the six trainable layers. At epoch 797, the convolution norm is 0.59468, while the five dense-layer norms are 0.02963, 0.03558, 0.05370, 0.09576 and 0.08463 in model order. The actual mean update ratios at the selected epoch range from 0.000210 to 0.000940; the largest over the production history is 0.03581 at the first dense layer in epoch 1, with no near-zero denominator events in the all-parameter rows. At activation indices 1, 5, 7, 9 and 11, the globally aggregated post-activation population variances change from (0.10460, 0.97827, 1.69423, 1.50668, 0.96488) to (0.00346, 0.06938, 0.14887, 0.15338, 0.10306). The smaller gradients, parameter movements and activation variances accompany lower physical error, rather than a late numerical runaway. Across the ten runs, the recorded diagnostics are finite and the activation histogram populations are consistent with the recorded tensor sizes.

The restored production model is then evaluated on the 429-sample test set, separated according to the same 10° angle threshold used by the SIMM mask. A sample belongs to the high-angle group only when $|\alpha| > 10^\circ$, so samples exactly at $\pm 10^\circ$ remain in the low-angle group. Table 14 reports the complete split. The high-angle MSE is 0.000552233 larger than the low-angle MSE and 2.374 times as large. Taking square roots gives RMSE values 0.02005 and 0.03089 for the low- and high-angle groups. The high-angle samples represent $66/429 = 15.38\%$ of the test set but account for approximately 30.15% of the total squared error. This larger error occurs in the regime where the SIMM relation is inactive, although the split alone cannot identify the mask as its cause; the two groups also differ in sample count and input regime.

Figure 35 gives the same comparison graphically and includes the overall test MSE as a reference. It is a comparison of two physically defined test subsets, not a comparison of two separately trained models. The test set was not used to select the stopping epoch, and the figure therefore describes the aggregate subset error of the one restored production model rather than a new tuning criterion.

Test subset	Samples	Physical MSE	Difference vs. low	High / low MSE
Overall	429	0.000486795	—	—
$ \alpha \leq 10^\circ$	363	0.000401836	—	—
$ \alpha > 10^\circ$	66	0.000954069	+0.000552233	2.374

Table 14: Angle-of-attack split of the final production test metrics read from `test_metrics.csv`. The high-angle difference and ratio are computed from the two stored subset MSE values; no per-sample residual spread was recorded.

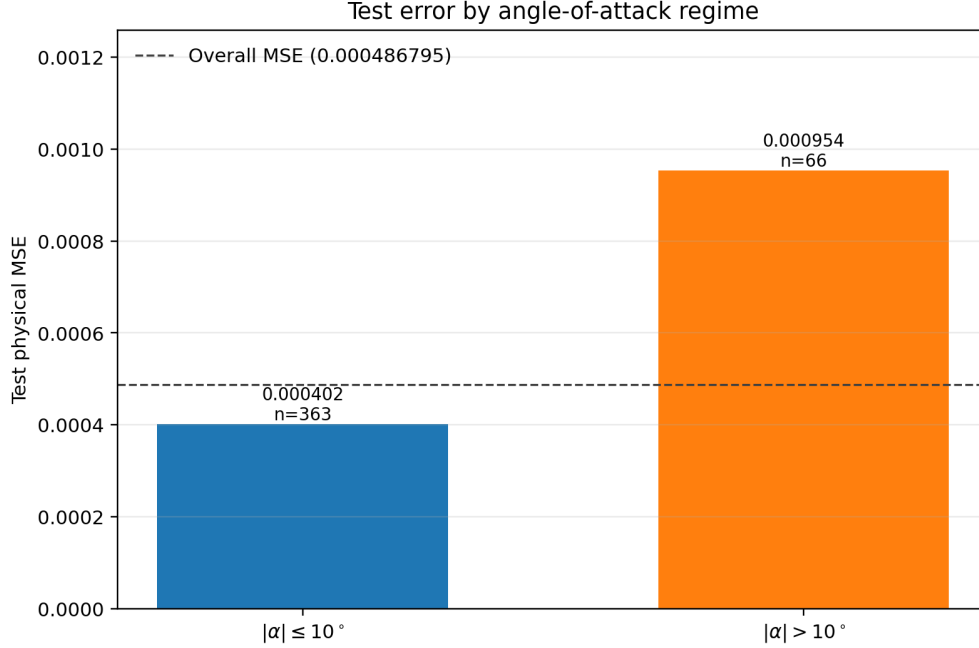


Figure 35: Post-training test physical MSE for the restored seed-42 production model, split into the low-angle ($|\alpha| \leq 10^\circ$) and high-angle ($|\alpha| > 10^\circ$) regimes. The bars are based on 363 and 66 samples, respectively, and the dashed line marks the overall MSE over all 429 test samples.

The final values should not be compared with every cross-validation value without qualification. The tuning experiments use fold-specific normalization and approximately 80 percent of the 1713 training samples in each fold, whereas ordinary training uses a 90/10 split and a different update budget. The final numbers describe the recorded large-topology production configuration with validation-based checkpoint restoration. Although each run evaluates the test set only after training, test metrics were recorded for all development trials; these artifacts do not establish that the test set was never inspected during development. We therefore report the production result as a post-training test evaluation, not as an independently untouched estimate for the entire tuning campaign.

4.10. Tuning summary and conclusions

The tuning campaign detailed in this chapter systematically refined the surrogate model, providing several key insights into the architecture and training protocols required for high-fidelity aerodynamic prediction.

The largest improvement among the structural choices came from replacing the initial (128,64) dense head with the selected topology `conv5x5-dense-1024-512-256-128`. Its cross-validated physical MSE was 0.004560, compared with 0.006010 for the topology baseline, and the improvement was observed in all five folds. In addition to this, the tuning experiments identified Adam as the best optimizer at the tested rate 10^{-5} , with a mean validation MSE of 0.004427. The activation comparison selected LeakyReLU with $\alpha = 0.05$ and the learning-rate search selected a constant Adam rate of 10^{-3} , which reduced the cross-validated error to 0.003609 for that experiment. These choices are supported not only by the mean scores, but also by finite diagnostics, small actual update ratios, and the absence of late-epoch divergence in the selected candidates.

The regularization results were different from what might be expected for a large network. Both dropout and weight penalties increased the recorded validation error, and the physics-weight sweep also selected the zero-weight reference. Nevertheless, to embed domain-specific physical grounding into the network, the moderate physics weight $\lambda = 0.10$ was chosen for the final production model: its validation error increase over the unconstrained baseline was negligible (+0.000028) relative to fold variability, while preserving structural consistency with Thin-Airfoil Theory in the linear regime.

Ultimately, the final production model, combining the large topology with the refined training protocol, Adam at 10^{-3} , and the $\lambda = 0.10$ SIMM prior, produced an excellent final test MSE of 0.000487. The 1200-epoch maximum was not exhausted: the run stopped at epoch 860, and the longer budget did not improve on the checkpoint already obtained with the 834-epoch budget. Crucially, when evaluating the final test predictions across aerodynamic regimes, the price of the physical constraint is clearly quantified: test samples with an angle of attack greater than 10° ($|\alpha| > 10^\circ$, $n = 66$) produced a physical MSE of 0.000954, which is nearly $2.4\times$ higher (almost double) than the physical MSE of 0.000402 achieved on samples in the linear regime ($|\alpha| \leq 10^\circ$, $n = 363$). This disparity occurs because the analytical thin-airfoil relation ($C_L \approx 2\pi\alpha$) is strictly valid and active only for $|\alpha| \leq 10^\circ$; beyond this threshold, flow separation and stall introduce strong aerodynamic nonlinearities where the surrogate model must rely entirely on data-driven learning without an analytical prior.

Overall, the experiments show that surrogate accuracy depends on the joint choice of topology, optimizer, activation, learning rate, batch construction and stopping policy rather than on adding regularization indiscriminately. They also show why numerical diagnostics and explicit provenance are necessary: a low validation value is meaningful only when the topology, split, metric and training configuration that produced it are stated alongside the result.

5. Conclusions

This work successfully demonstrates an integrated approach to aerodynamic analysis, combining high-fidelity numerical simulation with a data-driven surrogate model. The primary achievement was the extension of the **LifeX** library to handle turbulent external aerodynamic flows through the integration of the **Spalart-Allmaras RANS model**. This new capability was validated on the **Leonardo HPC cluster**, establishing a robust pipeline for generating reference-quality simulation data. Additionally, a **Convolutional Neural Network** was developed and trained on this data, creating a surrogate model capable of near-instantaneous aerodynamic prediction.

The CFD simulations of the NACA airfoil produced lift coefficients that were physically consistent, though moderately underestimated compared to some experimental data. This behavior is physically consistent with the modeling choices made: the VMS-LES formulation employed in the Navier-Stokes step acts as an implicit filter that dissipates small-scale turbulent structures, suppressing the contribution of fine-grained vortical activity to the overall lift generation. Nonetheless, the results remain well within the range of published numerical and experimental values, confirming the physical validity of the simulation framework by Ladson (1988) and McCroskey (1988).

A further modeling assumption concerns the compressibility of the flow. The simulations were performed at a Mach number significantly below 0.3, a regime in which the relative density variations induced by pressure fluctuations are negligible, making the incompressible Navier-Stokes equations an appropriate model.

Regarding the deep learning implementation, our initial baseline model confirmed that a Convolutional Neural Network could approximate the lift coefficient directly from geometry and flow conditions, achieving an initial Mean Squared Error of 0.0104. The primary advantage of this approach lies in its computational efficiency: while a high-fidelity CFD simulation requires approximately 3 hours on an HPC node, the trained CNN executes inference in less than a millisecond, delivering an online evaluation speedup of over four orders of magnitude ($\sim 10^4\times$) compared to conventional numerical simulation.

Building upon this proof of concept, we conducted a systematic tuning campaign across network topology, optimization algorithms, regularization, activation functions and training protocols. This empirical search established the optimal production configuration:

- a 5×5 convolution (stride 5, 8 feature channels) paired with an expanded 4-layer regression head (1024, 512, 256, 128, 1) totaling $\approx 8.065 \times 10^6$ parameters;
- Adam optimizer with coordinate-wise gradient clipping $[-1, 1]$, a tuned learning rate of 10^{-3} , and LeakyReLU activations ($\alpha = 0.05$);

- deterministic exclusion of dropout ($p = 0$) and weight decay, combined with a calibrated SIMM loss weight $\lambda = 0.10$ to ground the linear pre-stall regime ($|\alpha| \leq 10^\circ$);
- a balanced batch size $B = 257$ across 64 MPI ranks and a robust "20/20" early stopping policy.

This optimized surrogate achieved an exceptional test Mean Squared Error of 0.000487, representing a more than 21-fold accuracy improvement over the baseline.

A deeper inspection of the test predictions under the physics-informed SIMM loss function revealed that the surrogate model's accuracy varies across aerodynamic regimes. Specifically, test samples with an angle of attack exceeding 10° ($|\alpha| > 10^\circ$, $n = 66$) exhibited a physical MSE of 0.000954, that is nearly $2.4\times$ higher than the MSE of 0.000402 obtained on samples in the linear regime ($|\alpha| \leq 10^\circ$, $n = 363$). This discrepancy illustrates the localized scope of the analytical thin-airfoil prior ($C_L \approx 2\pi\alpha$): while the analytical constraint stabilizes predictions in the pre-stall linear attached-flow regime, high-angle flow separation and post-stall phenomena remain inherently non-linear, requiring the surrogate to rely entirely on data-driven learning in the absence of an active analytical prior.

Nevertheless, certain limitations inherent to each method must be acknowledged. These include the computational demand of the RANS simulations and, for the data-driven model, the absence of an *a priori* error estimate, the strict dependency on a training dataset that accurately resembles the target inference domain²⁵, and the general lack of interpretability characteristic of black-box models.

Looking forward, the framework developed here opens up promising avenues for engineering design and analysis. The speed of the trained CNN makes it ideal for real-time applications and rapid design iterations on consumer-grade hardware, completely bypassing the need for constant HPC access. This capability is particularly relevant for the agile aerodynamic design and field-testing of components for Unmanned Aerial Vehicles and quadcopters, where on-the-fly performance evaluation can accelerate development cycles. The simulation pipeline itself is now poised to serve as a data engine for training future surrogate models on more complex geometries, such as the iced airfoils that motivated this study.

6. Contributors

Name	NS Sim.	CNN API	Data Struct.	Topology	Dropout	Physics W.	Optimizer	Regulariz.	Activation	Lr. Rate	Training	Report
Giulio Enzo Donninelli		✓	✓	✓				✓	✓		✓	✓
Alessia Rigoni		✓			✓		✓			✓		✓
Vittorio Sironi	✓				✓	✓						
Alessandro Verrengia	✓			✓				✓				

Table 15: Contributions to the work presented in this report. A checkmark indicates participation in the corresponding task.

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²⁵As noted, since the training data from the CFD was slightly biased, we expect this bias persists in the surrogate model.

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